VCU220

VCU220 SYS.1 / SYS.2, HARA, FSC, DFMEA & ICD

System Definition, Safety Analysis, Cybersecurity and Interface Baseline

Document ID	VCU-EV220-DOC2-SYS1-SYS2-SAFETY-001
Revision	Rev A2 - Final Technical Hardening
Technical Freeze / Authority	JAN-2026 Program Baseline / Rev A2 hardening review 14-Aug-2026
Vehicle	220 kW peak / 110 kW continuous EV platform
Architecture	Central ECU with independent safety MCU + high-performance cockpit/application SoC
Process Scope	Automotive SPICE 4.0 SYS.1 / SYS.2 / SYS.3 with HWE.1 / SWE.1 / SYS.4 / SYS.5 / VAL.1 handoff
Safety / Cybersecurity	ISO 26262 vehicle safety lifecycle; ISO/SAE 21434 TARA; final ASIL from vehicle HARA/FSC/TSC
Evidence Status	Engineering baseline. Planned requirements/tests are not executed release or compliance evidence.

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1. Purpose and Systems-Engineering Position

This document converts the concept and RA baseline into stakeholder/system requirements, preliminary safety goals and functional safety requirements, failure analysis, cybersecurity threats, project risks and interface contracts. It is the primary design-review bridge between the product concept and the block-level SYS.3 architecture.

Authority model: The central safety MCU remains the authoritative motion/HV controller. Cockpit/infotainment software can request modes and present state but cannot directly publish unvalidated torque, regen or pack-contactor commands.

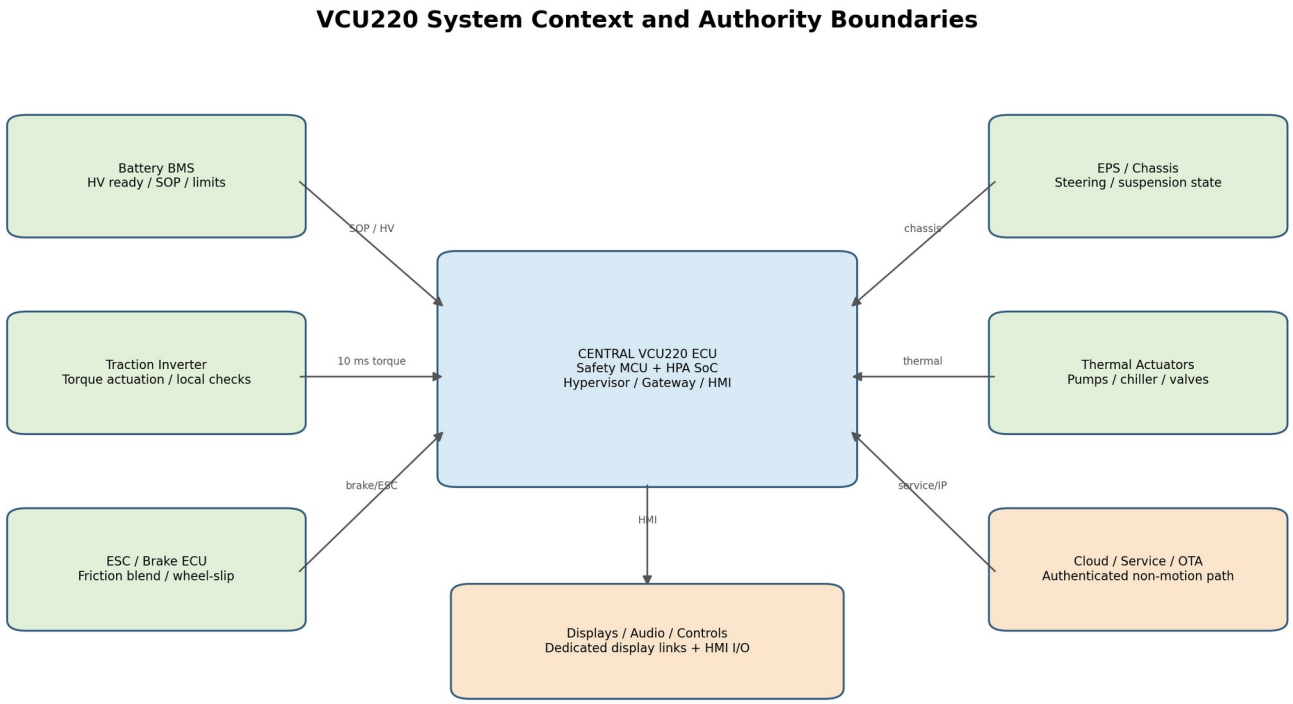


Figure 1 - System boundary and external ECU ownership

2. SYS.1 Stakeholder Requirements

SYS1 ID	Source RA	Area	Stakeholder Requirement	Priority	Verification	Role
SYS1-VCU-001	RA-VCU-001	Safety & Motion	Stakeholders require the central ECU to prevent a QM cockpit/infotainment fault or compromise from creating an unvalidated positive traction torque request.	Must	Requirement review + system acceptance	Driver / Vehicle / OEM / Service / Manufacturing
SYS1-VCU-002	RA-VCU-002	Driver Modes	Driver-mode changes shall be validated against vehicle state, thermal/SOP limits, market configuration and chassis permissions before they affect torque/regen/chassis requests.	Must	Requirement review + system acceptance	Driver / Vehicle / OEM / Service / Manufacturing
SYS1-VCU-003	RA-VCU-003	HV Coordination	VCU HV orchestration shall coordinate BMS, inverter and charger states while pack contactor actuation and pack protection remain owned by the BMS unless a vehicle	Must	Requirement review + system acceptance	Driver / Vehicle / OEM / Service / Manufacturing

SYS1 ID	Source RA	Area	Stakeholder Requirement	Priority	Verification	Role
			item definition explicitly reallocates ownership.			
SYS1-VCU-004	RA-VCU-004	Thermal & Energy	No driver mode shall increase torque or regen above the minimum of BMS SOP, inverter/motor capability, thermal limits, traction limits and legal/market caps.	Must	Requirement review + system acceptance	Driver / Vehicle / OEM / Service / Manufacturing
SYS1-VCU-005	RA-VCU-005	Chassis & Regen	Regenerative braking coordination shall remain subordinate to ESC/ABS wheel-slip and friction-brake safety authority.	Must	Requirement review + system acceptance	Driver / Vehicle / OEM / Service / Manufacturing
SYS1-VCU-006	RA-VCU-006	Central Compute	The architecture shall use a physically independent safety MCU or equivalently independent safety island for final motion-authority supervision even when IVI and vehicle services share a central SoC/hypervisor.	Must	Requirement review + system acceptance	Driver / Vehicle / OEM / Service / Manufacturing
SYS1-VCU-007	RA-VCU-007	Networking	Stakeholders require the VCU220 system to provide deterministic CAN FD motion networks and scalable Ethernet/TSN service backbone; requirement instance 1 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Requirement review + system acceptance	Driver / Vehicle / OEM / Service / Manufacturing
SYS1-VCU-008	RA-VCU-008	Cockpit/HMI	Stakeholders require the VCU220 system to provide responsive driver interaction and safety-telltale independence from rich infotainment; requirement instance 1 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Requirement review + system acceptance	Driver / Vehicle / OEM / Service / Manufacturing
SYS1-VCU-009	RA-VCU-009	Diagnostics/Service	Stakeholders require the VCU220 system to support UDS/DoIP, event recording, calibration, EOL and service workflows; requirement instance 1 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Requirement review + system acceptance	Driver / Vehicle / OEM / Service / Manufacturing
SYS1-VCU-010	RA-VCU-010	Cybersecurity	Stakeholders require the VCU220 system to prevent unauthorized command/update/configuration paths and support lifecycle TARA/monitoring; requirement instance 1 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Requirement review + system acceptance	Driver / Vehicle / OEM / Service / Manufacturing
SYS1-VCU-011	RA-VCU-011	Electrical/EMC	Stakeholders require the VCU220 system to meet LV electrical, EMC, thermal,	Must	Requirement review + system acceptance	Driver / Vehicle / OEM / Service / Manufacturing

SYS1 ID	Source RA	Area	Stakeholder Requirement	Priority	Verification	Role
			mechanical and environmental design targets; requirement instance 1 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.			
SYS1-VCU-012	RA-VCU-012	Lifecycle/Quality	Stakeholders require the VCU220 system to support ASPICE traceability, supplier evidence, OTA, obsolescence, FRACAS and production quality; requirement instance 1 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Requirement review + system acceptance	Driver / Vehicle / OEM / Service / Manufacturing
SYS1-VCU-013	RA-VCU-013	Safety & Motion	Stakeholders require the VCU220 system to prevent unintended torque, retain deterministic safe/degraded behavior and independent actuation safety; requirement instance 2 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Requirement review + system acceptance	Driver / Vehicle / OEM / Service / Manufacturing
SYS1-VCU-014	RA-VCU-014	Driver Modes	Stakeholders require the VCU220 system to support controlled Eco/Normal/Sport/Track/Valet profiles with jerk-bounded transition and safety-clamped calibration; requirement instance 2 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Requirement review + system acceptance	Driver / Vehicle / OEM / Service / Manufacturing
SYS1-VCU-015	RA-VCU-015	HV Coordination	Stakeholders require the VCU220 system to coordinate vehicle HV enable/precharge/drive/charge states without replacing BMS pack protection; requirement instance 2 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Requirement review + system acceptance	Driver / Vehicle / OEM / Service / Manufacturing
SYS1-VCU-016	RA-VCU-016	Thermal & Energy	Stakeholders require the VCU220 system to enforce BMS/inverter/motor/charger power limits and prioritize cooling without violating component limits; requirement instance 2 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Requirement review + system acceptance	Driver / Vehicle / OEM / Service / Manufacturing
SYS1-VCU-017	RA-VCU-017	Chassis & Regen	Stakeholders require the VCU220 system to coordinate regenerative braking and chassis requests while preserving ESC/ABS/EPS	Must	Requirement review + system acceptance	Driver / Vehicle / OEM / Service / Manufacturing

SYS1 ID	Source RA	Area	Stakeholder Requirement	Priority	Verification	Role
			authority; requirement instance 2 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.			
SYS1-VCU-018	RA-VCU-018	Central Compute	Stakeholders require the VCU220 system to consolidate cockpit and vehicle services with independent safety MCU and hardware-enforced isolation; requirement instance 2 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Requirement review + system acceptance	Driver / Vehicle / OEM / Service / Manufacturing
SYS1-VCU-019	RA-VCU-019	Networking	Stakeholders require the VCU220 system to provide deterministic CAN FD motion networks and scalable Ethernet/TSN service backbone; requirement instance 2 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Requirement review + system acceptance	Driver / Vehicle / OEM / Service / Manufacturing
SYS1-VCU-020	RA-VCU-020	Cockpit/HMI	Stakeholders require the VCU220 system to provide responsive driver interaction and safety-telltale independence from rich infotainment; requirement instance 2 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Requirement review + system acceptance	Driver / Vehicle / OEM / Service / Manufacturing
SYS1-VCU-021	RA-VCU-021	Diagnostics/Service	Stakeholders require the VCU220 system to support UDS/DoIP, event recording, calibration, EOL and service workflows; requirement instance 2 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Requirement review + system acceptance	Driver / Vehicle / OEM / Service / Manufacturing
SYS1-VCU-022	RA-VCU-022	Cybersecurity	Stakeholders require the VCU220 system to prevent unauthorized command/update/configuration paths and support lifecycle TARA/monitoring; requirement instance 2 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Requirement review + system acceptance	Driver / Vehicle / OEM / Service / Manufacturing
SYS1-VCU-023	RA-VCU-023	Electrical/EMC	Stakeholders require the VCU220 system to meet LV electrical, EMC, thermal, mechanical and environmental design targets; requirement instance 2 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Requirement review + system acceptance	Driver / Vehicle / OEM / Service / Manufacturing
SYS1-VCU-024	RA-VCU-024	Lifecycle/Quality	Stakeholders require	Must	Requirement review +	Driver / Vehicle / OEM

SYS1 ID	Source RA	Area	Stakeholder Requirement	Priority	Verification	Role
			the VCU220 system to support ASPICE traceability, supplier evidence, OTA, obsolescence, FRACAS and production quality; requirement instance 2 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.		system acceptance	/ Service / Manufacturing
SYS1-VCU-025	RA-VCU-025	Safety & Motion	Stakeholders require the VCU220 system to prevent unintended torque, retain deterministic safe/degraded behavior and independent actuation safety; requirement instance 3 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Requirement review + system acceptance	Driver / Vehicle / OEM / Service / Manufacturing
SYS1-VCU-026	RA-VCU-026	Driver Modes	Stakeholders require the VCU220 system to support controlled Eco/Normal/Sport/Track/Valet profiles with jerk-bounded transition and safety-clamped calibration; requirement instance 3 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Requirement review + system acceptance	Driver / Vehicle / OEM / Service / Manufacturing
SYS1-VCU-027	RA-VCU-027	HV Coordination	Stakeholders require the VCU220 system to coordinate vehicle HV enable/precharge/drive/charge states without replacing BMS pack protection; requirement instance 3 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Requirement review + system acceptance	Driver / Vehicle / OEM / Service / Manufacturing
SYS1-VCU-028	RA-VCU-028	Thermal & Energy	Stakeholders require the VCU220 system to enforce BMS/inverter/motor/charger power limits and prioritize cooling without violating component limits; requirement instance 3 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Requirement review + system acceptance	Driver / Vehicle / OEM / Service / Manufacturing
SYS1-VCU-029	RA-VCU-029	Chassis & Regen	Stakeholders require the VCU220 system to coordinate regenerative braking and chassis requests while preserving ESC/ABS/EPS authority; requirement instance 3 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Requirement review + system acceptance	Driver / Vehicle / OEM / Service / Manufacturing
SYS1-VCU-030	RA-VCU-030	Central Compute	Stakeholders require the VCU220 system to consolidate cockpit	Must	Requirement review + system acceptance	Driver / Vehicle / OEM / Service / Manufacturing

SYS1 ID	Source RA	Area	Stakeholder Requirement	Priority	Verification	Role
			and vehicle services with independent safety MCU and hardware-enforced isolation; requirement instance 3 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.			
SYS1-VCU-031	RA-VCU-031	Networking	Stakeholders require the VCU220 system to provide deterministic CAN FD motion networks and scalable Ethernet/TSN service backbone; requirement instance 3 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Requirement review + system acceptance	Driver / Vehicle / OEM / Service / Manufacturing
SYS1-VCU-032	RA-VCU-032	Cockpit/HMI	Stakeholders require the VCU220 system to provide responsive driver interaction and safety-telltale independence from rich infotainment; requirement instance 3 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Requirement review + system acceptance	Driver / Vehicle / OEM / Service / Manufacturing
SYS1-VCU-033	RA-VCU-033	Diagnostics/Service	Stakeholders require the VCU220 system to support UDS/DoIP, event recording, calibration, EOL and service workflows; requirement instance 3 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Requirement review + system acceptance	Driver / Vehicle / OEM / Service / Manufacturing
SYS1-VCU-034	RA-VCU-034	Cybersecurity	Stakeholders require the VCU220 system to prevent unauthorized command/update/configuration paths and support lifecycle TARA/monitoring; requirement instance 3 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Requirement review + system acceptance	Driver / Vehicle / OEM / Service / Manufacturing
SYS1-VCU-035	RA-VCU-035	Electrical/EMC	Stakeholders require the VCU220 system to meet LV electrical, EMC, thermal, mechanical and environmental design targets; requirement instance 3 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Requirement review + system acceptance	Driver / Vehicle / OEM / Service / Manufacturing
SYS1-VCU-036	RA-VCU-036	Lifecycle/Quality	Stakeholders require the VCU220 system to support ASPICE traceability, supplier evidence, OTA, obsolescence, FRACAS and production quality; requirement instance 3 shall be verified at the appropriate model,	Must	Requirement review + system acceptance	Driver / Vehicle / OEM / Service / Manufacturing

SYS1 ID	Source RA	Area	Stakeholder Requirement	Priority	Verification	Role
			bench, HIL, vehicle or production gate.			
SYS1-VCU-037	RA-VCU-037	Safety & Motion	Stakeholders require the VCU220 system to prevent unintended torque, retain deterministic safe/degraded behavior and independent actuation safety; requirement instance 4 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Requirement review + system acceptance	Driver / Vehicle / OEM / Service / Manufacturing
SYS1-VCU-038	RA-VCU-038	Driver Modes	Stakeholders require the VCU220 system to support controlled Eco/Normal/Sport/Track/Valet profiles with jerk-bounded transition and safety-clamped calibration; requirement instance 4 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Requirement review + system acceptance	Driver / Vehicle / OEM / Service / Manufacturing
SYS1-VCU-039	RA-VCU-039	HV Coordination	Stakeholders require the VCU220 system to coordinate vehicle HV enable/precharge/drive/charge states without replacing BMS pack protection; requirement instance 4 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Requirement review + system acceptance	Driver / Vehicle / OEM / Service / Manufacturing
SYS1-VCU-040	RA-VCU-040	Thermal & Energy	Stakeholders require the VCU220 system to enforce BMS/inverter/motor/charger power limits and prioritize cooling without violating component limits; requirement instance 4 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Requirement review + system acceptance	Driver / Vehicle / OEM / Service / Manufacturing
SYS1-VCU-041	RA-VCU-041	Chassis & Regen	Stakeholders require the VCU220 system to coordinate regenerative braking and chassis requests while preserving ESC/ABS/EPS authority; requirement instance 4 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Requirement review + system acceptance	Driver / Vehicle / OEM / Service / Manufacturing
SYS1-VCU-042	RA-VCU-042	Central Compute	Stakeholders require the VCU220 system to consolidate cockpit and vehicle services with independent safety MCU and hardware-enforced isolation; requirement instance 4 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Requirement review + system acceptance	Driver / Vehicle / OEM / Service / Manufacturing

SYS1 ID	Source RA	Area	Stakeholder Requirement	Priority	Verification	Role
SYS1-VCU-043	RA-VCU-043	Networking	Stakeholders require the VCU220 system to provide deterministic CAN FD motion networks and scalable Ethernet/TSN service backbone; requirement instance 4 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Requirement review + system acceptance	Driver / Vehicle / OEM / Service / Manufacturing
SYS1-VCU-044	RA-VCU-044	Cockpit/HMI	Stakeholders require the VCU220 system to provide responsive driver interaction and safety-telltale independence from rich infotainment; requirement instance 4 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Requirement review + system acceptance	Driver / Vehicle / OEM / Service / Manufacturing
SYS1-VCU-045	RA-VCU-045	Diagnostics/Service	Stakeholders require the VCU220 system to support UDS/DoIP, event recording, calibration, EOL and service workflows; requirement instance 4 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Requirement review + system acceptance	Driver / Vehicle / OEM / Service / Manufacturing
SYS1-VCU-046	RA-VCU-046	Cybersecurity	Stakeholders require the VCU220 system to prevent unauthorized command/update/configuration paths and support lifecycle TARA/monitoring; requirement instance 4 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Requirement review + system acceptance	Driver / Vehicle / OEM / Service / Manufacturing
SYS1-VCU-047	RA-VCU-047	Electrical/EMC	Stakeholders require the VCU220 system to meet LV electrical, EMC, thermal, mechanical and environmental design targets; requirement instance 4 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Requirement review + system acceptance	Driver / Vehicle / OEM / Service / Manufacturing
SYS1-VCU-048	RA-VCU-048	Lifecycle/Quality	Stakeholders require the VCU220 system to support ASPICE traceability, supplier evidence, OTA, obsolescence, FRACAS and production quality; requirement instance 4 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Requirement review + system acceptance	Driver / Vehicle / OEM / Service / Manufacturing
SYS1-VCU-049	RA-VCU-049	Safety & Motion	Stakeholders require the VCU220 system to prevent unintended torque, retain deterministic safe/degraded behavior and independent actuation	Must	Requirement review + system acceptance	Driver / Vehicle / OEM / Service / Manufacturing

SYS1 ID	Source RA	Area	Stakeholder Requirement	Priority	Verification	Role
			safety; requirement instance 5 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.			
SYS1-VCU-050	RA-VCU-050	Driver Modes	Stakeholders require the VCU220 system to support controlled Eco/Normal/Sport/Track/Valet profiles with jerk-bounded transition and safety-clamped calibration; requirement instance 5 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Requirement review + system acceptance	Driver / Vehicle / OEM / Service / Manufacturing

3. SYS.2 System Requirements

SYS.2 requirements define the externally observable system behavior and system-level constraints before selecting detailed implementation. Where numeric values remain dependent on BMS/inverter/ESC suppliers or vehicle calibration, the requirement explicitly points to the controlled ICD rather than embedding an arbitrary universal threshold.

SYS2 ID	Domain	Parent	System Requirement	Acceptance / Boundary	Allocation	Safety	Status
SYS2-VCU-001	Safety & Motion	SYS1-VCU-001	Only the safety-controlled motion domain may emit the authoritative propulsion torque request.	No unsafe authority path; HIL fault injection closes timing.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-002	Safety & Motion	SYS1-VCU-002	Torque shall be checked against pedal/brake/gear/speed/vehicle state and local actuator feedback.	No unsafe authority path; HIL fault injection closes timing.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-003	Safety & Motion	SYS1-VCU-003	Unsafe torque deviation shall trigger torque inhibit/degrade inside the allocated internal reaction budget.	No unsafe authority path; HIL fault injection closes timing.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-004	Safety & Motion	SYS1-VCU-004	Independent supervision shall detect task/clock/rail faults without relying on infotainment execution.	No unsafe authority path; HIL fault injection closes timing.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-005	Safety & Motion	SYS1-VCU-005	Faults shall map to controlled torque-zero, limited drive, controlled stop or HV isolation according to safety analysis.	No unsafe authority path; HIL fault injection closes timing.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-006	Driver Modes	SYS1-VCU-006	Mode requests shall be validated by the safety domain before calibration activation.	Transition/state tests meet accepted-mode and jerk/slew criteria.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-007	Driver Modes	SYS1-VCU-007	Mode request status shall update within 50 ms reference.	Transition/state tests meet accepted-mode and jerk/slew criteria.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-008	Driver Modes	SYS1-VCU-008	Torque/regen maps shall cross-fade with jerk-bounded slew rather than instantaneous full-scale changes.	Transition/state tests meet accepted-mode and jerk/slew criteria.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-009	Driver Modes	SYS1-VCU-009	Track mode shall be gated by market/vehicle/chassis/thermal permissions.	Transition/state tests meet accepted-mode and jerk/slew criteria.	HW/SW/System	Safety-related	Proposed Rev A1

SYS2 ID	Domain	Parent	System Requirement	Acceptance / Boundary	Allocation	Safety	Status
SYS2-VCU-010	Driver Modes	SYS1-VCU-010	Valet/custom restrictions shall be authenticated and shall never relax hard safety/thermal limits.	Transition/state tests meet accepted-mode and jerk/slew criteria.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-011	HV Coordination	SYS1-VCU-011	VCU shall coordinate OFF/ACCESSORY/ READY/PRECHARGE/DRIVE/CHARGE/ FAULT states with BMS and inverter.	State HIL demonstrates no illegal HV/drive transition.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-012	HV Coordination	SYS1-VCU-012	VCU shall verify BMS-reported precharge completion and inverter DC-link readiness before enabling propulsion.	State HIL demonstrates no illegal HV/drive transition.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-013	HV Coordination	SYS1-VCU-013	Pack contactor drive remains BMS-owned unless item definition allocates otherwise.	State HIL demonstrates no illegal HV/drive transition.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-014	HV Coordination	SYS1-VCU-014	Crash/HVIL faults shall cause a coordinated non-permissive propulsion/HV sequence.	State HIL demonstrates no illegal HV/drive transition.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-015	HV Coordination	SYS1-VCU-015	Drive torque and charging states shall be mutually constrained according to vehicle architecture.	State HIL demonstrates no illegal HV/drive transition.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-016	Thermal & Energy	SYS1-VCU-016	Available torque shall use the minimum of BMS, inverter, motor and thermal capability limits.	Boundary sweeps never exceed minimum capability limit.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-017	Thermal & Energy	SYS1-VCU-017	Thermal derating shall be monotonic with hysteresis/recovery and supplier-calibrated thresholds.	Boundary sweeps never exceed minimum capability limit.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-018	Thermal & Energy	SYS1-VCU-018	Sport/Track may request pre-chilling but may not commandeer unavailable thermal capacity.	Boundary sweeps never exceed minimum capability limit.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-019	Thermal & Energy	SYS1-VCU-019	220 kW peak and 110 kW continuous are vehicle references subject to time/temperature/SOP.	Boundary sweeps never exceed minimum capability limit.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-020	Thermal & Energy	SYS1-VCU-020	Accessory/cabin/ traction thermal priorities shall be arbitrated by vehicle mode and safety state.	Boundary sweeps never exceed minimum capability limit.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-021	Chassis & Regen	SYS1-VCU-021	VCU shall compute desired regenerative contribution and provide it to the brake/ESC authority.	Brake/ESC overrides remain authoritative.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-022	Chassis & Regen	SYS1-VCU-022	Brake pedal/ABS/ESC intervention shall immediately reduce or revoke positive propulsion request.	Brake/ESC overrides remain authoritative.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-023	Chassis & Regen	SYS1-VCU-023	Regen shall be limited by BMS charge SOP, inverter capability, motor speed and wheel-slip state.	Brake/ESC overrides remain authoritative.	HW/SW/System	Safety-related	Proposed Rev A1

SYS2 ID	Domain	Parent	System Requirement	Acceptance / Boundary	Allocation	Safety	Status
SYS2-VCU-024	Chassis & Regen	SYS1-VCU-024	VCU may request EPS/ESC/suspension mode; local ECUs retain hard safety limits.	Brake/ESC overrides remain authoritative.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-025	Chassis & Regen	SYS1-VCU-025	Mode/brake blending shall target longitudinal jerk <=2.0 m/s3 reference in normal road transitions.	Brake/ESC overrides remain authoritative.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-026	Central Compute	SYS1-VCU-026	Central ECU shall include independent safety MCU plus application SoC/hypervisor.	Isolation, boot and resource stress evidence passes.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-027	Central Compute	SYS1-VCU-027	DMA, IOMMU/MPU, interrupt and peripheral access shall prevent QM guests from writing safety actuators.	Isolation, boot and resource stress evidence passes.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-028	Central Compute	SYS1-VCU-028	Safety processes shall use bounded/static memory; application partitions shall use bounded pools and health limits.	Isolation, boot and resource stress evidence passes.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-029	Central Compute	SYS1-VCU-029	Safety MCU shall reach safe supervision before rich IVI boot completes.	Isolation, boot and resource stress evidence passes.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-030	Central Compute	SYS1-VCU-030	Release shall preserve >=30% CPU/memory/network reserve in worst released mode where practicable.	Isolation, boot and resource stress evidence passes.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-031	Networking	SYS1-VCU-031	Torque/brake/HV safety PDUs shall use deterministic CAN FD with age/counter/E2E protection.	Worst-case timing/bus-load/packet-loss criteria pass.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-032	Networking	SYS1-VCU-032	1000BASE-T1 shall support DoIP/SOME-IP/cockpit services; 2.5/10G upgrade path shall be reserved.	Worst-case timing/bus-load/packet-loss criteria pass.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-033	Networking	SYS1-VCU-033	Critical Ethernet traffic shall use time sync and traffic-class controls when sharing links with infotainment/OTA.	Worst-case timing/bus-load/packet-loss criteria pass.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-034	Networking	SYS1-VCU-034	Safety CAN nominal sustained utilization shall remain <=40% reference and fault-burst <=60%.	Worst-case timing/bus-load/packet-loss criteria pass.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-035	Networking	SYS1-VCU-035	CAN partial networking and Ethernet TC10-class sleep/wake shall support low-power states.	Worst-case timing/bus-load/packet-loss criteria pass.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-036	Cockpit/HMI	SYS1-VCU-036	Android/IVI mode selection is a request only; safety MCU publishes authoritative accepted mode.	HMI fault cannot change authoritative safety state.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-037	Cockpit/HMI	SYS1-VCU-037	Critical telltales shall have an independent monitored path capable of indicating safety state if Android stalls.	HMI fault cannot change authoritative safety state.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-038	Cockpit/HMI	SYS1-VCU-038	Driver shall receive	HMI fault cannot	HW/SW/System	QM/Functional	Proposed Rev A1

SYS2 ID	Domain	Parent	System Requirement	Acceptance / Boundary	Allocation	Safety	Status
			accepted/rejected mode and reason within 100 ms reference.	change authoritative safety state.			
SYS2-VCU-039	Cockpit/HMI	SYS1-VCU-039	High-rate display pixels shall use dedicated display interface/SerDes rather than consuming vehicle 1G Ethernet.	HMI fault cannot change authoritative safety state.	HW/SW/System	QM/Functional	Proposed Rev A1
SYS2-VCU-040	Cockpit/HMI	SYS1-VCU-040	Track/custom features shall follow market HMI/distraction and homologation rules.	HMI fault cannot change authoritative safety state.	HW/SW/System	QM/Functional	Proposed Rev A1
SYS2-VCU-041	Diagnostics/Service	SYS1-VCU-041	VCU shall support UDS diagnostic sessions, DTC, routines and data services per released diagnostic specification.	Security/session/ state constraints verified.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-042	Diagnostics/Service	SYS1-VCU-042	Ethernet diagnostic access shall use DoIP and security controls.	Security/session/ state constraints verified.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-043	Diagnostics/Service	SYS1-VCU-043	Safety-relevant torque/mode/HV faults shall record pre/post-event context with synchronized time.	Security/session/ state constraints verified.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-044	Diagnostics/Service	SYS1-VCU-044	Service functions that can affect torque/HV shall require explicit vehicle-safe conditions and authorization.	Security/session/ state constraints verified.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-045	Diagnostics/Service	SYS1-VCU-045	Calibration download shall be authenticated, range-checked and variant-bound.	Security/session/ state constraints verified.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-046	Cybersecurity	SYS1-VCU-046	Safety MCU and SoC boot chains shall verify authenticated software and prevent rollback outside approved policy.	TARA controls and negative tests pass.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-047	Cybersecurity	SYS1-VCU-047	Safety command PDUs shall use SecOC authenticity/freshness where allocated by TARA.	TARA controls and negative tests pass.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-048	Cybersecurity	SYS1-VCU-048	AUTOSAR E2E shall be configured independently from cryptographic SecOC.	TARA controls and negative tests pass.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-049	Cybersecurity	SYS1-VCU-049	IVI/telematics shall not route arbitrary CAN commands into safety networks.	TARA controls and negative tests pass.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-050	Cybersecurity	SYS1-VCU-050	Update manifests, dependencies, vehicle state and recovery shall be verified before activation.	TARA controls and negative tests pass.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-051	Electrical/EMC	SYS1-VCU-001	ECU shall tolerate the released ISO 16750-2/OEM LV envelope without unsafe output.	Electrical/EMC/ thermal DVP passes with safe outputs.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-052	Electrical/EMC	SYS1-VCU-002	Central ECU shall satisfy CISPR 25/OEM emission targets in worst	Electrical/EMC/ thermal DVP passes with safe outputs.	HW/SW/System	Safety-related	Proposed Rev A1

SYS2 ID	Domain	Parent	System Requirement	Acceptance / Boundary	Allocation	Safety	Status
			CPU/GPU/network activity.				
SYS2-VCU-053	Electrical/EMC	SYS1-VCU-003	Central ECU shall preserve safe outputs during ISO 11452/OEM immunity tests.	Electrical/EMC/ thermal DVP passes with safe outputs.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-054	Electrical/EMC	SYS1-VCU-004	130 W sustained-peak electronics thermal design shall retain supplier junction/case margin at released ambient.	Electrical/EMC/ thermal DVP passes with safe outputs.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-055	Electrical/EMC	SYS1-VCU-005	High-speed DDR/PCIe/Ethernet/ display interfaces and multi-rail PDN shall close SI/PI before layout release.	Electrical/EMC/ thermal DVP passes with safe outputs.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-056	Lifecycle/Quality	SYS1-VCU-006	RA-SYS1-SYS2-SYS3-HWE/SWE-test traceability shall be bidirectional and change controlled.	Audit and production evidence complete.	HW/SW/System	QM/Functional	Proposed Rev A1
SYS2-VCU-057	Lifecycle/Quality	SYS1-VCU-007	Safety MCU/SoC/hypervisor /PMIC suppliers shall provide safety manuals, assumptions and defect/change notification.	Audit and production evidence complete.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-058	Lifecycle/Quality	SYS1-VCU-008	OTA/cyber process shall support market UNECE R155/R156 obligations where applicable.	Audit and production evidence complete.	HW/SW/System	QM/Functional	Proposed Rev A1
SYS2-VCU-059	Lifecycle/Quality	SYS1-VCU-009	Production shall verify secure provisioning, calibration, network IDs, power rails, safety I/O and communication.	Audit and production evidence complete.	HW/SW/System	QM/Functional	Proposed Rev A1
SYS2-VCU-060	Lifecycle/Quality	SYS1-VCU-010	Field safety/cyber/quality events shall feed closed-loop corrective action and calibration/software updates.	Audit and production evidence complete.	HW/SW/System	QM/Functional	Proposed Rev A1
SYS2-VCU-061	Safety & Motion	SYS1-VCU-011	Only the safety-controlled motion domain may emit the authoritative propulsion torque request.	No unsafe authority path; HIL fault injection closes timing.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-062	Safety & Motion	SYS1-VCU-012	Torque shall be checked against pedal/brake/gear/speed/vehicle state and local actuator feedback.	No unsafe authority path; HIL fault injection closes timing.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-063	Safety & Motion	SYS1-VCU-013	Unsafe torque deviation shall trigger torque inhibit/degrade inside the allocated internal reaction budget.	No unsafe authority path; HIL fault injection closes timing.	HW/SW/System	Safety-related	Proposed Rev A1

SYS2 ID	Domain	Parent	System Requirement	Acceptance / Boundary	Allocation	Safety	Status
SYS2-VCU-064	Safety & Motion	SYS1-VCU-014	Independent supervision shall detect task/clock/rail faults without relying on infotainment execution.	No unsafe authority path; HIL fault injection closes timing.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-065	Safety & Motion	SYS1-VCU-015	Faults shall map to controlled torque-zero, limited drive, controlled stop or HV isolation according to safety analysis.	No unsafe authority path; HIL fault injection closes timing.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-066	Driver Modes	SYS1-VCU-016	Mode requests shall be validated by the safety domain before calibration activation.	Transition/state tests meet accepted-mode and jerk/slew criteria.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-067	Driver Modes	SYS1-VCU-017	Mode request status shall update within 50 ms reference.	Transition/state tests meet accepted-mode and jerk/slew criteria.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-068	Driver Modes	SYS1-VCU-018	Torque/regen maps shall cross-fade with jerk-bounded slew rather than instantaneous full-scale changes.	Transition/state tests meet accepted-mode and jerk/slew criteria.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-069	Driver Modes	SYS1-VCU-019	Track mode shall be gated by market/vehicle/chassis/thermal permissions.	Transition/state tests meet accepted-mode and jerk/slew criteria.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-070	Driver Modes	SYS1-VCU-020	Valet/custom restrictions shall be authenticated and shall never relax hard safety/thermal limits.	Transition/state tests meet accepted-mode and jerk/slew criteria.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-071	HV Coordination	SYS1-VCU-021	VCU shall coordinate OFF/ACCESSORY/READY/PRECHARGE/DRIVE/CHARGE/FAULT states with BMS and inverter.	State HIL demonstrates no illegal HV/drive transition.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-072	HV Coordination	SYS1-VCU-022	VCU shall verify BMS-reported precharge completion and inverter DC-link readiness before enabling propulsion.	State HIL demonstrates no illegal HV/drive transition.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-073	HV Coordination	SYS1-VCU-023	Pack contactor drive remains BMS-owned unless item definition allocates otherwise.	State HIL demonstrates no illegal HV/drive transition.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-074	HV Coordination	SYS1-VCU-024	Crash/HVIL faults shall cause a coordinated non-permissive propulsion/HV sequence.	State HIL demonstrates no illegal HV/drive transition.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-075	HV Coordination	SYS1-VCU-025	Drive torque and charging states shall be mutually constrained according to vehicle architecture.	State HIL demonstrates no illegal HV/drive transition.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-076	Thermal & Energy	SYS1-VCU-026	Available torque shall use the minimum of BMS, inverter, motor and thermal capability limits.	Boundary sweeps never exceed minimum capability limit.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-077	Thermal & Energy	SYS1-VCU-027	Thermal derating shall be monotonic with hysteresis/recovery and supplier-calibrated thresholds.	Boundary sweeps never exceed minimum capability limit.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-078	Thermal & Energy	SYS1-VCU-028	Sport/Track may request pre-chilling but may not commandeer	Boundary sweeps never exceed minimum capability limit.	HW/SW/System	Safety-related	Proposed Rev A1

SYS2 ID	Domain	Parent	System Requirement	Acceptance / Boundary	Allocation	Safety	Status
			unavailable thermal capacity.				
SYS2-VCU-079	Thermal & Energy	SYS1-VCU-029	220 kW peak and 110 kW continuous are vehicle references subject to time/temperature/SOP.	Boundary sweeps never exceed minimum capability limit.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-080	Thermal & Energy	SYS1-VCU-030	Accessory/cabin/traction thermal priorities shall be arbitrated by vehicle mode and safety state.	Boundary sweeps never exceed minimum capability limit.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-081	Chassis & Regen	SYS1-VCU-031	VCU shall compute desired regenerative contribution and provide it to the brake/ESC authority.	Brake/ESC overrides remain authoritative.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-082	Chassis & Regen	SYS1-VCU-032	Brake pedal/ABS/ESC intervention shall immediately reduce or revoke positive propulsion request.	Brake/ESC overrides remain authoritative.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-083	Chassis & Regen	SYS1-VCU-033	Regen shall be limited by BMS charge SOP, inverter capability, motor speed and wheel-slip state.	Brake/ESC overrides remain authoritative.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-084	Chassis & Regen	SYS1-VCU-034	VCU may request EPS/ESC/suspension mode; local ECUs retain hard safety limits.	Brake/ESC overrides remain authoritative.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-085	Chassis & Regen	SYS1-VCU-035	Mode/brake blending shall target longitudinal jerk <=2.0 m/s3 reference in normal road transitions.	Brake/ESC overrides remain authoritative.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-086	Central Compute	SYS1-VCU-036	Central ECU shall include independent safety MCU plus application SoC/hypervisor.	Isolation, boot and resource stress evidence passes.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-087	Central Compute	SYS1-VCU-037	DMA, IOMMU/MPU, interrupt and peripheral access shall prevent QM guests from writing safety actuators.	Isolation, boot and resource stress evidence passes.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-088	Central Compute	SYS1-VCU-038	Safety processes shall use bounded/static memory; application partitions shall use bounded pools and health limits.	Isolation, boot and resource stress evidence passes.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-089	Central Compute	SYS1-VCU-039	Safety MCU shall reach safe supervision before rich IVI boot completes.	Isolation, boot and resource stress evidence passes.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-090	Central Compute	SYS1-VCU-040	Release shall preserve >=30% CPU/memory/network reserve in worst released mode where practicable.	Isolation, boot and resource stress evidence passes.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-091	Networking	SYS1-VCU-041	Torque/brake/HV safety PDUs shall use deterministic CAN FD with age/counter/E2E protection.	Worst-case timing/bus-load/packet-loss criteria pass.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-092	Networking	SYS1-VCU-042	1000BASE-T1 shall support DoIP/SOME-IP/cockpit services; 2.5/10G upgrade path shall be	Worst-case timing/bus-load/packet-loss criteria pass.	HW/SW/System	Safety-related	Proposed Rev A1

SYS2 ID	Domain	Parent	System Requirement	Acceptance / Boundary	Allocation	Safety	Status
			reserved.				
SYS2-VCU-093	Networking	SYS1-VCU-043	Critical Ethernet traffic shall use time sync and traffic-class controls when sharing links with infotainment/OTA.	Worst-case timing/bus-load/packet-loss criteria pass.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-094	Networking	SYS1-VCU-044	Safety CAN nominal sustained utilization shall remain <=40% reference and fault-burst <=60%.	Worst-case timing/bus-load/packet-loss criteria pass.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-095	Networking	SYS1-VCU-045	CAN partial networking and Ethernet TC10-class sleep/wake shall support low-power states.	Worst-case timing/bus-load/packet-loss criteria pass.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-096	Cockpit/HMI	SYS1-VCU-046	Android/IVI mode selection is a request only; safety MCU publishes authoritative accepted mode.	HMI fault cannot change authoritative safety state.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-097	Cockpit/HMI	SYS1-VCU-047	Critical telltales shall have an independent monitored path capable of indicating safety state if Android stalls.	HMI fault cannot change authoritative safety state.	HW/SW/System	Safety-related	Proposed Rev A1
SYS2-VCU-098	Cockpit/HMI	SYS1-VCU-048	Driver shall receive accepted/rejected mode and reason within 100 ms reference.	HMI fault cannot change authoritative safety state.	HW/SW/System	QM/Functional	Proposed Rev A1
SYS2-VCU-099	Cockpit/HMI	SYS1-VCU-049	High-rate display pixels shall use dedicated display interface/SerDes rather than consuming vehicle 1G Ethernet.	HMI fault cannot change authoritative safety state.	HW/SW/System	QM/Functional	Proposed Rev A1
SYS2-VCU-100	Cockpit/HMI	SYS1-VCU-050	Track/custom features shall follow market HMI/distraction and homologation rules.	HMI fault cannot change authoritative safety state.	HW/SW/System	QM/Functional	Proposed Rev A1

4. Item Definition and Operational Safety Boundaries

- Motion authority ends at bounded torque/deceleration requests; inverter, ESC/brake and EPS retain local safety mechanisms.
- HV authority is coordinated by the VCU but pack contactor/precharge actuation is BMS-owned in the reference variant.
- Track/Sport mode may request different calibrations and local chassis profiles but cannot disable local ABS/ESC hard protections.
- Thermal limits are provided as normalized capability/SOP inputs from battery, inverter and motor subsystems rather than a universal 65 C threshold.
- Loss or compromise of Android/HPA software shall not create a valid motion/HV command path.

5. Preliminary HARA and Safety Goals

HARA ID	Malfunction	Potential Effect	Operational Situation	S	E	C	Prelim ASIL	SG
HARA-VCU-001	Positive torque without pedal	Unexpected acceleration	Vehicle driving / transition / service condition applicable to malfunction	S3	E4	C3	D	SG-01
HARA-VCU-002	Positive torque while brake applied	Failure to decelerate	Vehicle driving / transition / service condition applicable to malfunction	S3	E4	C3	D	SG-02
HARA-VCU-003	Torque sign	Unexpected	Vehicle driving /	S3	E4	C3	D	SG-03

HARA ID	Malfunction	Potential Effect	Operational Situation	S	E	C	Prelim ASIL	SG
	inversion	acceleration/deceleration	transition / service condition applicable to malfunction					
HARA-VCU-004	Torque request stuck high	Sustained unintended acceleration	Vehicle driving / transition / service condition applicable to malfunction	S3	E4	C3	D	SG-04
HARA-VCU-005	Torque request stale	Incorrect vehicle response	Vehicle driving / transition / service condition applicable to malfunction	S3	E4	C3	C	SG-05
HARA-VCU-006	Torque lost during overtake	Loss of propulsion	Vehicle driving / transition / service condition applicable to malfunction	S3	E4	C3	C	SG-06
HARA-VCU-007	Regen excessive	Unexpected deceleration	Vehicle driving / transition / service condition applicable to malfunction	S3	E4	C3	C	SG-07
HARA-VCU-008	Regen absent on one-pedal transition	Longer stopping distance	Vehicle driving / transition / service condition applicable to malfunction	S3	E4	C3	C	SG-08
HARA-VCU-009	Regen during ABS	Wheel instability	Vehicle driving / transition / service condition applicable to malfunction	S3	E4	C3	C	SG-09
HARA-VCU-010	Track mode falsely enabled	Higher-than-intended response	Vehicle driving / transition / service condition applicable to malfunction	S2	E4	C3	B	SG-10
HARA-VCU-011	Valet cap bypassed	Unauthorized performance	Vehicle driving / transition / service condition applicable to malfunction	S2	E4	C3	B	SG-11
HARA-VCU-012	Mode oscillation	Jerk/loss of control	Vehicle driving / transition / service condition applicable to malfunction	S3	E4	C3	C	SG-12
HARA-VCU-013	BMS SOP ignored	Battery overcurrent/voltage	Vehicle driving / transition / service condition applicable to malfunction	S3	E4	C3	C	SG-01
HARA-VCU-014	Inverter limit ignored	Power-stage protection trip	Vehicle driving / transition / service condition applicable to malfunction	S3	E4	C3	C	SG-02
HARA-VCU-015	Thermal derate delayed	Component overtemperature	Vehicle driving / transition / service condition applicable to malfunction	S3	E4	C3	C	SG-03
HARA-VCU-016	HV ready falsely asserted	Drive enabled before HV ready	Vehicle driving / transition / service condition applicable to malfunction	S3	E4	C3	D	SG-04
HARA-VCU-017	Precharge timeout ignored	Contactor/DC-link stress	Vehicle driving / transition / service condition applicable to malfunction	S3	E4	C3	C	SG-05
HARA-VCU-018	Crash/HVIL ignored	HV hazard after crash	Vehicle driving / transition / service condition applicable to malfunction	S3	E4	C3	D	SG-06
HARA-VCU-019	Android writes safety peripheral	Unsafe command	Vehicle driving / transition / service	S3	E4	C2	C	SG-07

HARA ID	Malfunction	Potential Effect	Operational Situation	S	E	C	Prelim ASIL	SG
			condition applicable to malfunction					
HARA-VCU-020	Hypervisor isolation failure	QM-to-safety interference	Vehicle driving / transition / service condition applicable to malfunction	S3	E4	C2	C	SG-08
HARA-VCU-021	Safety MCU watchdog failure	Latent dangerous control fault	Vehicle driving / transition / service condition applicable to malfunction	S3	E4	C2	C	SG-09
HARA-VCU-022	CAN torque spoof	Unauthorized torque	Vehicle driving / transition / service condition applicable to malfunction	S3	E4	C3	D	SG-10
HARA-VCU-023	CAN frame replay	Stale authority reused	Vehicle driving / transition / service condition applicable to malfunction	S3	E4	C3	C	SG-11
HARA-VCU-024	Ethernet gateway compromise	Lateral movement into safety network	Vehicle driving / transition / service condition applicable to malfunction	S3	E4	C3	C	SG-12
HARA-VCU-025	HMI mode accepted without confirmation	Driver-system state mismatch	Vehicle driving / transition / service condition applicable to malfunction	S2	E3	C2	B	SG-01
HARA-VCU-026	Safety telltale lost	Driver unaware of degraded safety state	Vehicle driving / transition / service condition applicable to malfunction	S2	E3	C2	B	SG-02
HARA-VCU-027	Cluster freeze during fault	Missing warning	Vehicle driving / transition / service condition applicable to malfunction	S2	E3	C2	B	SG-03
HARA-VCU-028	Calibration variant mismatch	Wrong torque/thermal limits	Vehicle driving / transition / service condition applicable to malfunction	S3	E3	C2	C	SG-04
HARA-VCU-029	OTA partial update	Incompatible control software	Vehicle driving / transition / service condition applicable to malfunction	S3	E3	C2	C	SG-05
HARA-VCU-030	Rollback to vulnerable image	Cyber-safety exposure	Vehicle driving / transition / service condition applicable to malfunction	S3	E3	C2	C	SG-06
HARA-VCU-031	12V brownout causes MCU reset	Loss of control authority	Vehicle driving / transition / service condition applicable to malfunction	S3	E3	C2	C	SG-07
HARA-VCU-032	PMIC false safe state	Unexpected torque inhibit/enable	Vehicle driving / transition / service condition applicable to malfunction	S3	E3	C2	C	SG-08
HARA-VCU-033	SoC thermal shutdown	Cockpit/service loss with potential shared-resource effect	Vehicle driving / transition / service condition applicable to malfunction	S3	E3	C2	C	SG-09
HARA-VCU-034	Memory corruption in torque tables	Incorrect torque	Vehicle driving / transition / service condition applicable to malfunction	S3	E3	C2	C	SG-10
HARA-VCU-035	Timebase drift	Stale/misaligned data	Vehicle driving / transition / service condition	S3	E3	C2	C	SG-11

HARA ID	Malfunction	Potential Effect	Operational Situation	S	E	C	Prelim ASIL	SG
			applicable to malfunction					
HARA-VCU-036	Sensor pedal plausibility fault	Incorrect driver demand	Vehicle driving / transition / service condition applicable to malfunction	S3	E3	C2	C	SG-12
HARA-VCU-037	ESC override not honored	Traction instability	Vehicle driving / transition / service condition applicable to malfunction	S3	E3	C3	D	SG-01
HARA-VCU-038	Brake ECU comm loss	Regen blend unsafe	Vehicle driving / transition / service condition applicable to malfunction	S3	E3	C3	C	SG-02
HARA-VCU-039	EOL secure provisioning error	Wrong keys/configuration	Vehicle driving / transition / service condition applicable to malfunction	S3	E3	C2	C	SG-03
HARA-VCU-040	Service routine activates in drive	Unexpected actuation	Vehicle driving / transition / service condition applicable to malfunction	S3	E3	C2	C	SG-04

5.1 Safety Goals

SG	Safety Goal	Prelim ASIL	Concept
SG-01	Prevent unintended or excessive positive propulsion torque caused by VCU malfunction or invalid mode/calibration.	Preliminary ASIL D	Independent torque supervision, plausibility, bounded communication and inverter local safety.
SG-02	Prevent hazardous loss or reversal of commanded propulsion torque that could destabilize the vehicle.	Preliminary ASIL C	Degraded drive, driver warning and controlled stop strategy.
SG-03	Prevent hazardous regenerative braking or deceleration request inconsistent with driver/brake/ESC state.	Preliminary ASIL C	Brake authority, regen clamp and independent plausibility.
SG-04	Prevent an unsafe HV enable/contact state request or propulsion enable before valid precharge/HV readiness.	Preliminary ASIL C	BMS-owned contactors, VCU coordination, dual permissives.
SG-05	Prevent driver-mode corruption from increasing torque/regen/chassis authority beyond released limits.	Preliminary ASIL B/C	Safety-domain validation and signed calibration.
SG-06	Prevent QM infotainment/Android faults from interfering with safety motion-control resources.	Preliminary ASIL C	Independent safety MCU, MPU/IOMMU, watchdog and gateway filtering.
SG-07	Prevent loss of critical driver warning/telltale information during safety-relevant VCU faults.	Preliminary ASIL B	Independent monitored telltale path and degraded indication.
SG-08	Prevent thermal/energy-limit violation that can lead to loss of propulsion, battery or inverter damage.	Preliminary ASIL B/C	Minimum capability arbitration and monotonic derating.
SG-09	Prevent corrupted/stale network data from increasing motion/HV capability.	Preliminary ASIL C	Age/E2E/SecOC/timeout non-permissive behavior.
SG-10	Prevent unauthorized software/calibration/update from modifying safety-relevant vehicle-control behavior.	Security-to-Safety	Secure boot/update, HSM, signed calibration and safety re-validation.
SG-11	Prevent unsafe central-ECU behavior due to LV power, thermal, clock or memory failure.	Preliminary ASIL C	Independent PMIC/watchdog, ECC, thermal and brownout safe state.
SG-12	Prevent manufacturing/service configuration mismatch from enabling incompatible vehicle functions.	Preliminary ASIL B	Secure EOL binding, topology/variant checks and commissioning tests.

6. Functional Safety Concept / FSR Allocation

FSR	SG	Functional Safety Requirement	FTTI / Timing	Allocation
FSR-VCU-001	SG-01	Authoritative propulsion torque shall be computed/validated in the safety domain and independently bounded before transmission.	<=10 ms internal reaction	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-002	SG-01	Brake application, invalid gear, crash or severe plausibility fault shall revoke positive torque authority.	<=10 ms	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-003	SG-02	Loss of primary control shall transition to deterministic degraded torque/controlled stop rather than arbitrary output.	FTTI-derived	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-004	SG-03	Regen command shall be subordinate to ESC/ABS/brake authority and charge-SOP constraints.	10 ms command cycle	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-005	SG-04	HV propulsion enable shall require valid BMS HV-ready/precharge and inverter readiness.	Before enable	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-006	SG-05	Driver-mode request shall be treated as untrusted until safety-domain validation succeeds.	<=50 ms acknowledge	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-007	SG-06	QM partitions shall have no direct write access to safety actuation peripherals or safety CAN controllers.	Continuous	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-008	SG-07	Critical safety telltales shall be independently monitorable from Android rich-HMI execution.	<=100 ms indication	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-009	SG-08	Torque availability shall be the minimum of BMS SOP, inverter/motor limits, thermal limits and safety caps.	Each 10 ms cycle	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-010	SG-09	Stale/invalid safety messages shall be non-permissive and shall not increase vehicle authority.	Message-specific timeout	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-011	SG-10	Boot/update/calibration shall authenticate code/data and reject unauthorized rollback/configuration.	Before activation	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-012	SG-11	Independent PMIC/watchdog shall force safe output on MCU rail/clock/task faults.	FTTI-derived	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-013	SG-12	Variant/topology/configuration mismatch shall block incompatible safety feature activation.	Before READY	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-014	SG-01	Authoritative propulsion torque shall be computed/validated in the safety domain and independently bounded before transmission.	<=10 ms internal reaction	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-015	SG-01	Brake application, invalid gear, crash or severe plausibility fault shall revoke positive torque authority.	<=10 ms	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-016	SG-02	Loss of primary control shall transition to deterministic degraded torque/controlled stop rather than arbitrary output.	FTTI-derived	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-017	SG-03	Regen command shall be subordinate to ESC/ABS/brake authority and charge-SOP constraints.	10 ms command cycle	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-018	SG-04	HV propulsion enable shall require valid BMS HV-ready/precharge and inverter readiness.	Before enable	Safety MCU / central compute / network / external ECU as allocated

FSR	SG	Functional Safety Requirement	FTTI / Timing	Allocation
FSR-VCU-019	SG-05	Driver-mode request shall be treated as untrusted until safety-domain validation succeeds.	<=50 ms acknowledge	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-020	SG-06	QM partitions shall have no direct write access to safety actuation peripherals or safety CAN controllers.	Continuous	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-021	SG-07	Critical safety telltales shall be independently monitorable from Android rich-HMI execution.	<=100 ms indication	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-022	SG-08	Torque availability shall be the minimum of BMS SOP, inverter/motor limits, thermal limits and safety caps.	Each 10 ms cycle	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-023	SG-09	Stale/invalid safety messages shall be non-permissive and shall not increase vehicle authority.	Message-specific timeout	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-024	SG-10	Boot/update/calibration shall authenticate code/data and reject unauthorized rollback/configuration.	Before activation	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-025	SG-11	Independent PMIC/watchdog shall force safe output on MCU rail/clock/task faults.	FTTI-derived	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-026	SG-12	Variant/topology/configuration mismatch shall block incompatible safety feature activation.	Before READY	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-027	SG-01	Authoritative propulsion torque shall be computed/validated in the safety domain and independently bounded before transmission.	<=10 ms internal reaction	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-028	SG-01	Brake application, invalid gear, crash or severe plausibility fault shall revoke positive torque authority.	<=10 ms	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-029	SG-02	Loss of primary control shall transition to deterministic degraded torque/controlled stop rather than arbitrary output.	FTTI-derived	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-030	SG-03	Regen command shall be subordinate to ESC/ABS/brake authority and charge-SOP constraints.	10 ms command cycle	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-031	SG-04	HV propulsion enable shall require valid BMS HV-ready/precharge and inverter readiness.	Before enable	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-032	SG-05	Driver-mode request shall be treated as untrusted until safety-domain validation succeeds.	<=50 ms acknowledge	Safety MCU / central compute / network / external ECU as allocated

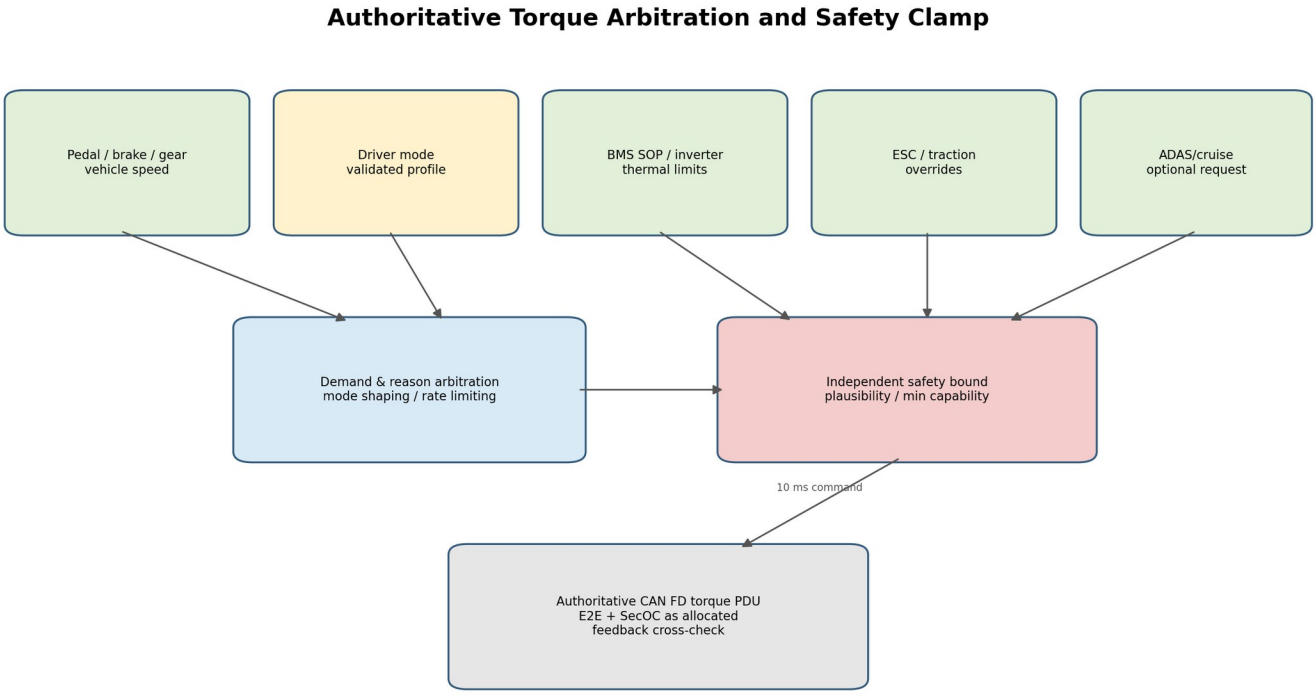


Figure 2 - Torque arbitration and independent safety clamp

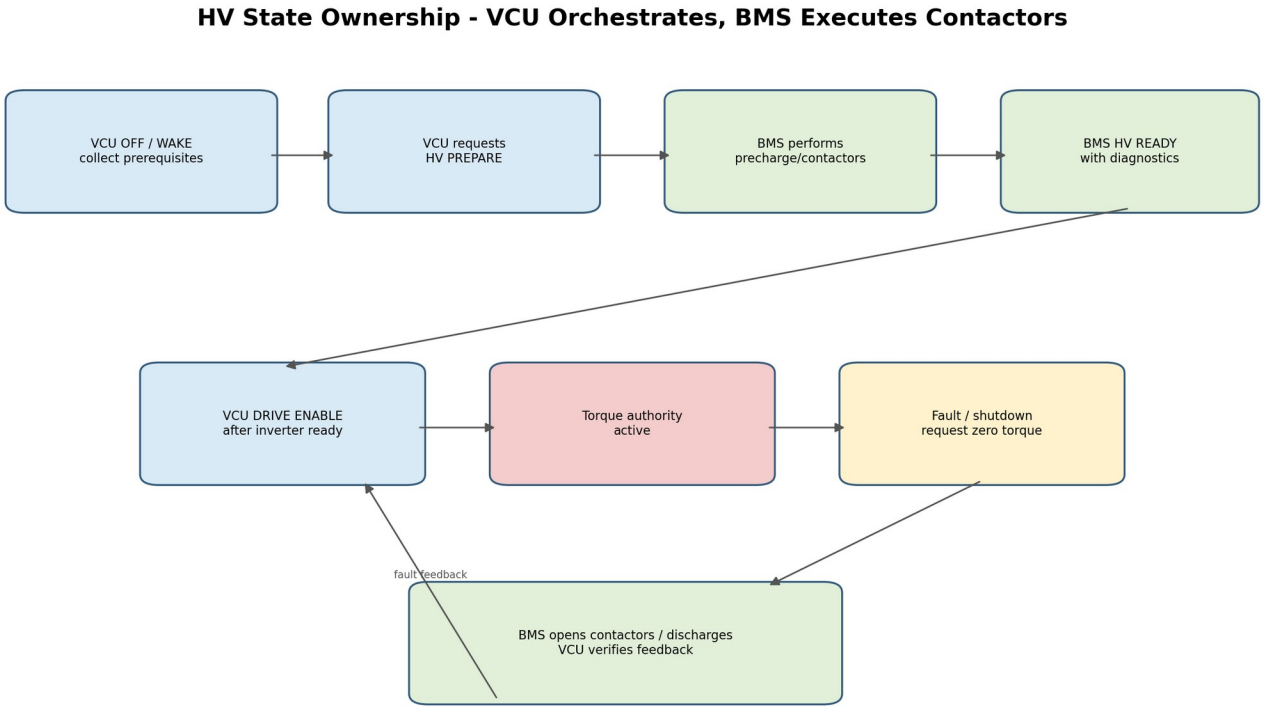


Figure 3 - HV-state ownership and BMS contactor boundary

7. Interface-Control Architecture

Each interface carries data value plus validity, age and diagnostic context. Safety communication uses deadline/timeout and E2E protection, with SecOC added where TARA requires authenticity/freshness. Service and cockpit Ethernet are not permitted to bypass safety-gateway policy.

ICD	Interface	Transport	Data	Rate / Latency	Integrity / Security	Owner	Verification
ICD-VCU-001	Accelerator pedal	Analog/SENT/PWM	2-channel demand +	1 ms acquisition	Plausibility / range /	VCU / External	Bench/HIL/network/

ICD	Interface	Transport	Data	Rate / Latency	Integrity / Security	Owner	Verification
			diagnostics		E2E internal	ECU / Cockpit as defined	vehicle verification
ICD-VCU-002	Brake/ESC -> VCU	CAN FD A	Brake pressure, ABS/ESC active, decel, wheel speeds	5-10 ms	E2E + timeout; SecOC if TARA	VCU / External ECU / Cockpit as defined	Bench/HIL/network/ vehicle verification
ICD-VCU-003	VCU -> Inverter	CAN FD A	Target torque, torque limits, direction, enable, mode, validity	10 ms	E2E + SecOC + age	VCU / External ECU / Cockpit as defined	Bench/HIL/network/ vehicle verification
ICD-VCU-004	Inverter -> VCU	CAN FD A	Delivered torque, speed, DC link, temps, limits, faults	10 ms	E2E + timeout	VCU / External ECU / Cockpit as defined	Bench/HIL/network/ vehicle verification
ICD-VCU-005	BMS -> VCU	CAN FD B	SOC/SOH/SOP, HV state, voltage/current, charge/discharge limits, faults	10-20 ms	E2E + timeout; SecOC if TARA	VCU / External ECU / Cockpit as defined	Bench/HIL/network/ vehicle verification
ICD-VCU-006	VCU -> BMS	CAN FD B	HV state request, vehicle mode, shutdown/charge request	20 ms	E2E + SecOC	VCU / External ECU / Cockpit as defined	Bench/HIL/network/ vehicle verification
ICD-VCU-007	ESC -> VCU	CAN FD A	Yaw/slip/traction/ chassis mode permission	10 ms	E2E + age	VCU / External ECU / Cockpit as defined	Bench/HIL/network/ vehicle verification
ICD-VCU-008	VCU -> ESC/Brake	CAN FD A	Regen capability/request, drive mode request, decel coordination	10 ms	E2E + SecOC if allocated	VCU / External ECU / Cockpit as defined	Bench/HIL/network/ vehicle verification
ICD-VCU-009	EPS <-> VCU	CAN FD chassis	Steering mode request, status, faults	10-20 ms	E2E + age	VCU / External ECU / Cockpit as defined	Bench/HIL/network/ vehicle verification
ICD-VCU-010	Suspension <-> VCU	CAN FD chassis	Damping/ride-height mode request/ack	20 ms	E2E	VCU / External ECU / Cockpit as defined	Bench/HIL/network/ vehicle verification
ICD-VCU-011	Thermal controller	CAN FD / Ethernet	Cooling capacity, requests, actuator status	20-100 ms	E2E/timeout	VCU / External ECU / Cockpit as defined	Bench/HIL/network/ vehicle verification
ICD-VCU-012	HVIL/crash	Hardwired + CAN redundant	Crash/HVIL/ emergency state	Immediate + CAN confirm	Independent hardwired safety path where allocated	VCU / External ECU / Cockpit as defined	Bench/HIL/network/ vehicle verification
ICD-VCU-013	HMI driver-mode request	IPC/Ethernet SOME/IP	Mode requested, user/profile ID, timestamp	Event	Authenticated local service + bounded enum	VCU / External ECU / Cockpit as defined	Bench/HIL/network/ vehicle verification
ICD-VCU-014	VCU authoritative mode -> HMI	IPC/Ethernet SOME/IP	Accepted mode, rejection reason, active limits	<=50 ms	Typed IPC/service version	VCU / External ECU / Cockpit as defined	Bench/HIL/network/ vehicle verification
ICD-VCU-015	Safety telltales	Safety MCU -> display safety path	HV/torque/critical fault/mode degradation	<=100 ms	Independent monitored channel	VCU / External ECU / Cockpit as defined	Bench/HIL/network/ vehicle verification
ICD-VCU-016	DoIP tester	1000BASE-T1 Ethernet	UDS diagnostics	On demand	DoIP + authenticated diagnostics	VCU / External ECU / Cockpit as defined	Bench/HIL/network/ vehicle verification
ICD-VCU-017	OTA client	Ethernet/cellular gateway	Manifest/image/ status	Background / parked activation	TLS/backend + signed package	VCU / External ECU / Cockpit as defined	Bench/HIL/network/ vehicle verification
ICD-VCU-018	Flight recorder	Internal IPC/storage	Torque/mode/HV/ network/security context	1-10 ms samples/event	CRC + monotonic time	VCU / External ECU / Cockpit as defined	Bench/HIL/network/ vehicle verification
ICD-VCU-019	PTP/gPTP	Ethernet 802.1AS	Time sync/quality	Continuous	Clock quality / domain control	VCU / External ECU / Cockpit as defined	Bench/HIL/network/ vehicle verification
ICD-VCU-020	Ethernet backbone	1000BASE-T1 baseline	SOME/IP, DoIP, cockpit services	Continuous	VLAN/QoS/MACsec optional	VCU / External ECU / Cockpit as defined	Bench/HIL/network/ vehicle verification
ICD-VCU-021	Ethernet high-speed uplink	2.5/10GBASE-T1 option	Future central compute/OTA/camer a/service traffic	Continuous	Traffic matrix / TSN	VCU / External ECU / Cockpit as defined	Bench/HIL/network/ vehicle verification
ICD-VCU-022	Display link	DP/eDP/automotive SerDes	Cluster/center/ passenger pixel streams	Frame rate	Link CRC/monitoring for safety display path	VCU / External ECU / Cockpit as defined	Bench/HIL/network/ vehicle verification
ICD-VCU-023	Audio/haptic alert	I2S/TDM/IPC + actuator	Safety warning request	Event	Ack/monitor where safety allocated	VCU / External ECU / Cockpit as defined	Bench/HIL/network/ vehicle verification
ICD-VCU-024	Physical mode switch	Discrete/LIN/CAN	Mode request/long press	Event	Debounce/ plausibility	VCU / External ECU / Cockpit as defined	Bench/HIL/network/ vehicle verification
ICD-VCU-025	Ignition/wake	Discrete/CAN	Wake/ignition/	Event	Authorized source +	VCU / External	Bench/HIL/network/

ICD	Interface	Transport	Data	Rate / Latency	Integrity / Security	Owner	Verification
		PN/Ethernet TC10	charger/service		debounce	ECU / Cockpit as defined	vehicle verification
ICD-VCU-026	PMIC/safety watchdog	SPI/hardwired	Rail/watchdog/ reset/safe-state	Sub-ms to ms	Independent safety path	VCU / External ECU / Cockpit as defined	Bench/HIL/network/ vehicle verification
ICD-VCU-027	Secure element/HSM	Internal bus	Keys, crypto requests, monotonic counters	On demand	Privilege/secure storage	VCU / External ECU / Cockpit as defined	Bench/HIL/network/ vehicle verification
ICD-VCU-028	Manufacturing tool	DoIP/debug manufacturing channel	Provisioning/ calibration/test	EOL only	Authenticated station + signed manifest	VCU / External ECU / Cockpit as defined	Bench/HIL/network/ vehicle verification
ICD-VCU-029	Cloud/fleet telemetry	Telematics gateway	Health/event data	Seconds/minutes	Privacy/ authentication; no direct torque	VCU / External ECU / Cockpit as defined	Bench/HIL/network/ vehicle verification
ICD-VCU-030	Charger/OBC state	CAN FD	Charge state, connector, power, wake	20-100 ms	E2E/timeout	VCU / External ECU / Cockpit as defined	Bench/HIL/network/ vehicle verification

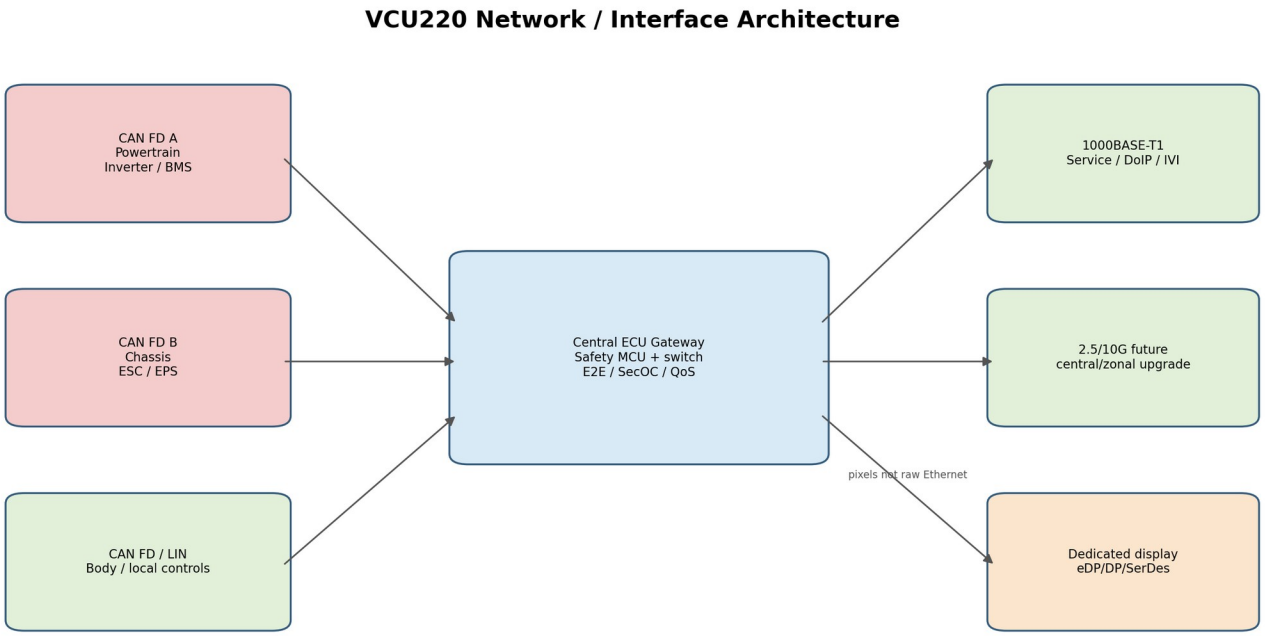


Figure 4 - Network and transport separation

8. Architecture DFMEA

The architecture DFMEA identifies failure mechanisms that must be decomposed at HWE.1/SWE.1 and validated by fault injection. Occurrence/detection values are program placeholders for prioritization, not production field statistics. AP and recommended actions are updated as supplier data and detailed schematics/software design become available.

ID	Block	Item	Failure Mode	Effect	S	Cause	O	Current Control	D	AP	Recommended Action	Owner
DFMEA-VCU-001	B01	LV Power Input, PMIC, Wake/Sleep & Power-State Control	PMIC output undervoltage	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	9	Component/ software/ configuration / environment/ common-cause failure	4	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	4	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-002	B01	LV Power Input, PMIC, Wake/Sleep & Power-State Control	PMIC output overvoltage	Loss of availability, incorrect motion/HV behavior, degraded	9	Component/ software/ configuration / environment/ common-	5	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and	5	H	Perform detailed FMEA/FMEDA; verify detection/containment	Subsystem Lead

ID	Block	Item	Failure Mode	Effect	S	Cause	O	Current Control	D	AP	Recommended Action	Owner
				cockpit or cybersecurity exposure depending on failure		cause failure		safe/degraded state			and maintenance action by allocated DVP	
DFMEA-VCU-003	B01	LV Power Input, PMIC, Wake/Sleep & Power-State Control	reverse polarity path failure	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	9	Component/ software/ configuration / environment/ common-cause failure	3	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	6	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-004	B01	LV Power Input, PMIC, Wake/Sleep & Power-State Control	brownout reset loop	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	9	Component/ software/ configuration / environment/ common-cause failure	4	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	3	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-005	B01	LV Power Input, PMIC, Wake/Sleep & Power-State Control	sleep/wake stuck	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	9	Component/ software/ configuration / environment/ common-cause failure	5	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	4	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-006	B02	Independent Safety MCU, Watchdog, Safe-State I/O & Actuation Gateway	lockstep core failure	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	10	Component/ software/ configuration / environment/ common-cause failure	3	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	5	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-007	B02	Independent Safety MCU, Watchdog, Safe-State I/O & Actuation Gateway	watchdog not serviced/detecting	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	10	Component/ software/ configuration / environment/ common-cause failure	4	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	6	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-008	B02	Independent Safety MCU, Watchdog, Safe-State I/O & Actuation Gateway	safe-state output stuck	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	10	Component/ software/ configuration / environment/ common-cause failure	5	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	3	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-009	B02	Independent Safety MCU, Watchdog, Safe-State I/O & Actuation Gateway	CAN controller failure	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	10	Component/ software/ configuration / environment/ common-cause failure	3	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	4	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-010	B02	Independent Safety MCU, Watchdog, Safe-State I/O & Actuation Gateway	safety MCU memory corruption	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	10	Component/ software/ configuration / environment/ common-cause failure	4	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	5	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead

ID	Block	Item	Failure Mode	Effect	S	Cause	O	Current Control	D	AP	Recommended Action	Owner
DFMEA-VCU-011	B03	Central Application SoC, Memory, Storage, Hypervisor & Hardware Virtualization	hypervisor isolation breach	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	7	Component/ software/ configuration / environment/ common-cause failure	5	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	6	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-012	B03	Central Application SoC, Memory, Storage, Hypervisor & Hardware Virtualization	LPDDR ECC storm	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	7	Component/ software/ configuration / environment/ common-cause failure	3	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	3	M	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-013	B03	Central Application SoC, Memory, Storage, Hypervisor & Hardware Virtualization	storage corruption	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	7	Component/ software/ configuration / environment/ common-cause failure	4	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	4	M	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-014	B03	Central Application SoC, Memory, Storage, Hypervisor & Hardware Virtualization	GPU hang	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	7	Component/ software/ configuration / environment/ common-cause failure	5	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	5	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-015	B03	Central Application SoC, Memory, Storage, Hypervisor & Hardware Virtualization	application SoC thermal shutdown	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	7	Component/ software/ configuration / environment/ common-cause failure	3	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	6	M	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-016	B04	Vehicle I/O Acquisition, Driver Controls & Discrete Safety Inputs	accelerator channel stuck	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	9	Component/ software/ configuration / environment/ common-cause failure	4	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	3	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-017	B04	Vehicle I/O Acquisition, Driver Controls & Discrete Safety Inputs	brake signal stuck	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	9	Component/ software/ configuration / environment/ common-cause failure	5	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	4	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-018	B04	Vehicle I/O Acquisition, Driver Controls & Discrete Safety Inputs	gear input invalid	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	9	Component/ software/ configuration / environment/ common-cause failure	3	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	5	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-019	B04	Vehicle I/O Acquisition, Driver Controls & Discrete	crash/HVIL input lost	Loss of availability, incorrect motion/HV behavior,	9	Component/ software/ configuration / environment/	4	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning	6	H	Perform detailed FMEA/FMEDA; verify detection/co	Subsystem Lead

ID	Block	Item	Failure Mode	Effect	S	Cause	O	Current Control	D	AP	Recommended Action	Owner
		Safety Inputs		degraded cockpit or cybersecurity exposure depending on failure		common-cause failure		and safe/degraded state			ntainment and maintenance action by allocated DVP	
DFMEA-VCU-020	B04	Vehicle I/O Acquisition, Driver Controls & Discrete Safety Inputs	vehicle-speed source stale	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	9	Component/ software/ configuration / environment/ common-cause failure	5	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	3	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-021	B05	Vehicle State Estimator, Torque Arbitration & Propulsion Request Manager	torque arbiter stuck high	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	10	Component/ software/ configuration / environment/ common-cause failure	3	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	4	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-022	B05	Vehicle State Estimator, Torque Arbitration & Propulsion Request Manager	wrong torque sign	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	10	Component/ software/ configuration / environment/ common-cause failure	4	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	5	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-023	B05	Vehicle State Estimator, Torque Arbitration & Propulsion Request Manager	limit omitted	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	10	Component/ software/ configuration / environment/ common-cause failure	5	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	6	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-024	B05	Vehicle State Estimator, Torque Arbitration & Propulsion Request Manager	feedback monitor disabled	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	10	Component/ software/ configuration / environment/ common-cause failure	3	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	3	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-025	B05	Vehicle State Estimator, Torque Arbitration & Propulsion Request Manager	timing overrun	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	10	Component/ software/ configuration / environment/ common-cause failure	4	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	4	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-026	B06	Regenerative Braking, Deceleration Coordination & Brake-Energy Manager	regen excess	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	10	Component/ software/ configuration / environment/ common-cause failure	5	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	5	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-027	B06	Regenerative Braking, Deceleration Coordination & Brake-Energy Manager	regen unavailable	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending	10	Component/ software/ configuration / environment/ common-cause failure	3	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	6	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated	Subsystem Lead

ID	Block	Item	Failure Mode	Effect	S	Cause	O	Current Control	D	AP	Recommended Action	Owner
				on failure							DVP	
DFMEA-VCU-028	B06	Regenerative Braking, Deceleration Coordination & Brake-Energy Manager	ABS override not honored	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	10	Component/ software/ configuration / environment/ common-cause failure	4	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	3	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-029	B06	Regenerative Braking, Deceleration Coordination & Brake-Energy Manager	low-speed fade error	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	10	Component/ software/ configuration / environment/ common-cause failure	5	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	4	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-030	B06	Regenerative Braking, Deceleration Coordination & Brake-Energy Manager	brake blend discontinuity	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	10	Component/ software/ configuration / environment/ common-cause failure	3	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	5	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-031	B07	Multi-Driver Mode Manager, Calibration, Profile Security & Transition Control	Track mode stuck	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	7	Component/ software/ configuration / environment/ common-cause failure	4	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	6	M	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-032	B07	Multi-Driver Mode Manager, Calibration, Profile Security & Transition Control	valet cap bypassed	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	7	Component/ software/ configuration / environment/ common-cause failure	5	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	3	M	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-033	B07	Multi-Driver Mode Manager, Calibration, Profile Security & Transition Control	mode transition step	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	7	Component/ software/ configuration / environment/ common-cause failure	3	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	4	M	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-034	B07	Multi-Driver Mode Manager, Calibration, Profile Security & Transition Control	wrong market calibration	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	7	Component/ software/ configuration / environment/ common-cause failure	4	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	5	M	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-035	B07	Multi-Driver Mode Manager, Calibration, Profile Security & Transition Control	mode state mismatch	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	7	Component/ software/ configuration / environment/ common-cause failure	5	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	6	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-036	B08	Vehicle Dynamics, EPS/ESC/Su	ESC override lost	Loss of availability, incorrect	7	Component/ software/ configuration	3	Diagnostics, watchdogs, E2E/SecOC,	3	M	Perform detailed FMEA/FMED	Subsystem Lead

ID	Block	Item	Failure Mode	Effect	S	Cause	O	Current Control	D	AP	Recommended Action	Owner
		spension Coordination & Chassis State Manager		motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure		/ environment/ common-cause failure		plausibility, partitioning and safe/degraded state			A; verify detection/containment and maintenance action by allocated DVP	
DFMEA-VCU-037	B08	Vehicle Dynamics, EPS/ESC/Suspension Coordination & Chassis State Manager	EPS mode mismatch	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	7	Component/ software/ configuration / environment/ common-cause failure	4	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	4	M	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-038	B08	Vehicle Dynamics, EPS/ESC/Suspension Coordination & Chassis State Manager	suspension request invalid	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	7	Component/ software/ configuration / environment/ common-cause failure	5	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	5	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-039	B08	Vehicle Dynamics, EPS/ESC/Suspension Coordination & Chassis State Manager	yaw/slip data stale	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	7	Component/ software/ configuration / environment/ common-cause failure	3	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	6	M	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-040	B08	Vehicle Dynamics, EPS/ESC/Suspension Coordination & Chassis State Manager	chassis mode comm loss	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	7	Component/ software/ configuration / environment/ common-cause failure	4	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	3	M	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-041	B09	HV State, Contactor/Precharge Coordination & Charge/Drive Interlocks	HV ready false	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	10	Component/ software/ configuration / environment/ common-cause failure	5	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	4	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-042	B09	HV State, Contactor/Precharge Coordination & Charge/Drive Interlocks	precharge timeout missed	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	10	Component/ software/ configuration / environment/ common-cause failure	3	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	5	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-043	B09	HV State, Contactor/Precharge Coordination & Charge/Drive Interlocks	charge-drive interlock missed	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	10	Component/ software/ configuration / environment/ common-cause failure	4	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	6	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-044	B09	HV State, Contactor/Precharge Coordination & Charge/Drive Interlocks	BMS weld status ignored	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity	10	Component/ software/ configuration / environment/ common-cause failure	5	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	3	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance	Subsystem Lead

ID	Block	Item	Failure Mode	Effect	S	Cause	O	Current Control	D	AP	Recommended Action	Owner
				exposure depending on failure							action by allocated DVP	
DFMEA-VCU-045	B09	HV State, Contactor/Precharge Coordination & Charge/Drive Interlocks	shutdown sequence incomplete	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	10	Component/ software/ configuration / environment/ common-cause failure	3	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	4	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-046	B10	Energy, Thermal, Performance Limit & Derating Supervisor	BMS SOP ignored	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	9	Component/ software/ configuration / environment/ common-cause failure	4	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	5	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-047	B10	Energy, Thermal, Performance Limit & Derating Supervisor	thermal limit stale	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	9	Component/ software/ configuration / environment/ common-cause failure	5	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	6	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-048	B10	Energy, Thermal, Performance Limit & Derating Supervisor	derate oscillation	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	9	Component/ software/ configuration / environment/ common-cause failure	3	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	3	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-049	B10	Energy, Thermal, Performance Limit & Derating Supervisor	peak timer failure	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	9	Component/ software/ configuration / environment/ common-cause failure	4	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	4	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-050	B10	Energy, Thermal, Performance Limit & Derating Supervisor	power recovery step	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	9	Component/ software/ configuration / environment/ common-cause failure	5	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	5	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-051	B11	CAN FD / Automotive Ethernet Gateway, TSN Switching & Network Management	CAN bus-off	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	9	Component/ software/ configuration / environment/ common-cause failure	3	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	6	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-052	B11	CAN FD / Automotive Ethernet Gateway, TSN Switching & Network Management	gateway route leak	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	9	Component/ software/ configuration / environment/ common-cause failure	4	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	3	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-	B11	CAN FD /	Ethernet	Loss of	9	Component/	5	Diagnostics,	4	H	Perform	Subsystem

ID	Block	Item	Failure Mode	Effect	S	Cause	O	Current Control	D	AP	Recommended Action	Owner
VCU-053		Automotive Ethernet Gateway, TSN Switching & Network Management	switch overload	availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure		software/ configuration / environment/ common-cause failure		watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state			detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Lead
DFMEA-VCU-054	B11	CAN FD / Automotive Ethernet Gateway, TSN Switching & Network Management	E2E configuration error	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	9	Component/ software/ configuration / environment/ common-cause failure	3	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	5	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-055	B11	CAN FD / Automotive Ethernet Gateway, TSN Switching & Network Management	SecOC freshness failure	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	9	Component/ software/ configuration / environment/ common-cause failure	4	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	6	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-056	B12	Time Synchronization, Service-Oriented Middleware, IPC & Data Freshness	time jump	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	7	Component/ software/ configuration / environment/ common-cause failure	5	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	3	M	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-057	B12	Time Synchronization, Service-Oriented Middleware, IPC & Data Freshness	PTP loss	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	7	Component/ software/ configuration / environment/ common-cause failure	3	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	4	M	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-058	B12	Time Synchronization, Service-Oriented Middleware, IPC & Data Freshness	IPC queue overflow	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	7	Component/ software/ configuration / environment/ common-cause failure	4	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	5	M	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-059	B12	Time Synchronization, Service-Oriented Middleware, IPC & Data Freshness	service schema mismatch	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	7	Component/ software/ configuration / environment/ common-cause failure	5	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	6	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-060	B12	Time Synchronization, Service-Oriented Middleware, IPC & Data Freshness	stale data accepted	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	7	Component/ software/ configuration / environment/ common-cause failure	3	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	3	M	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-061	B13	Cockpit/IVI/ HMI Domain, Safety Telltales & Driver-Interaction	Android sends unsafe request	Loss of availability, incorrect motion/HV behavior, degraded	7	Component/ software/ configuration / environment/ common-	4	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and	4	M	Perform detailed FMEA/FMEDA; verify detection/containment	Subsystem Lead

ID	Block	Item	Failure Mode	Effect	S	Cause	O	Current Control	D	AP	Recommended Action	Owner
		Boundary		cockpit or cybersecurity exposure depending on failure		cause failure		safe/degraded state			and maintenance action by allocated DVP	
DFMEA-VCU-062	B13	Cockpit/IVI/HMI Domain, Safety Telltales & Driver-Interaction Boundary	cluster telltale frozen	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	7	Component/software/configuration/environment/common-cause failure	5	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	5	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-063	B13	Cockpit/IVI/HMI Domain, Safety Telltales & Driver-Interaction Boundary	GPU compositor hang	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	7	Component/software/configuration/environment/common-cause failure	3	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	6	M	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-064	B13	Cockpit/IVI/HMI Domain, Safety Telltales & Driver-Interaction Boundary	mode feedback wrong	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	7	Component/software/configuration/environment/common-cause failure	4	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	3	M	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-065	B13	Cockpit/IVI/HMI Domain, Safety Telltales & Driver-Interaction Boundary	user profile corrupt	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	7	Component/software/configuration/environment/common-cause failure	5	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	4	M	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-066	B14	Diagnostics, UDS/DoIP, Event Recorder, Service & Fleet Logging	DTC mapping wrong	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	7	Component/software/configuration/environment/common-cause failure	3	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	5	M	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-067	B14	Diagnostics, UDS/DoIP, Event Recorder, Service & Fleet Logging	service routine in Drive	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	7	Component/software/configuration/environment/common-cause failure	4	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	6	M	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-068	B14	Diagnostics, UDS/DoIP, Event Recorder, Service & Fleet Logging	DoIP unauthorized access	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	7	Component/software/configuration/environment/common-cause failure	5	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	3	M	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-069	B14	Diagnostics, UDS/DoIP, Event Recorder, Service & Fleet Logging	flight recorder overflow	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	7	Component/software/configuration/environment/common-cause failure	3	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	4	M	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead

ID	Block	Item	Failure Mode	Effect	S	Cause	O	Current Control	D	AP	Recommended Action	Owner
DFMEA-VCU-070	B14	Diagnostics, UDS/DoIP, Event Recorder, Service & Fleet Logging	freeze-frame missing	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	7	Component/ software/ configuration / environment/ common-cause failure	4	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	5	M	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-071	B15	Boot, OTA Software Update, Configuration, Variant & Calibration Management	secure boot bypass	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	9	Component/ software/ configuration / environment/ common-cause failure	5	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	6	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-072	B15	Boot, OTA Software Update, Configuration, Variant & Calibration Management	A/B update incomplete	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	9	Component/ software/ configuration / environment/ common-cause failure	3	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	3	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-073	B15	Boot, OTA Software Update, Configuration, Variant & Calibration Management	dependency mismatch	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	9	Component/ software/ configuration / environment/ common-cause failure	4	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	4	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-074	B15	Boot, OTA Software Update, Configuration, Variant & Calibration Management	rollback vulnerability	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	9	Component/ software/ configuration / environment/ common-cause failure	5	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	5	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-075	B15	Boot, OTA Software Update, Configuration, Variant & Calibration Management	wrong variant coding	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	9	Component/ software/ configuration / environment/ common-cause failure	3	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	6	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-076	B16	Cybersecurity, HSM, SecOC/MACsec, IDS & Access-Control Services	HSM key fault	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	9	Component/ software/ configuration / environment/ common-cause failure	4	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	3	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-077	B16	Cybersecurity, HSM, SecOC/MACsec, IDS & Access-Control Services	SecOC key/freshness error	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	9	Component/ software/ configuration / environment/ common-cause failure	5	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	4	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-078	B16	Cybersecurity, HSM, SecOC/MACsec, IDS & Access-	firewall rule wrong	Loss of availability, incorrect motion/HV behavior,	9	Component/ software/ configuration / environment/	3	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning	5	H	Perform detailed FMEA/FMEDA; verify detection/co	Subsystem Lead

ID	Block	Item	Failure Mode	Effect	S	Cause	O	Current Control	D	AP	Recommended Action	Owner
		Control Services		degraded cockpit or cybersecurity exposure depending on failure		common-cause failure		and safe/degraded state			ntainment and maintenance action by allocated DVP	
DFMEA-VCU-079	B16	Cybersecurity, HSM, SecOC/MACsec, IDS & Access-Control Services	debug left enabled	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	9	Component/ software/ configuration / environment/ common-cause failure	4	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	6	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-080	B16	Cybersecurity, HSM, SecOC/MACsec, IDS & Access-Control Services	IDS overload	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	9	Component/ software/ configuration / environment/ common-cause failure	5	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	3	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-081	B17	Functional Safety Supervisor, Freedom-from-Interference & Health Management	safety supervisor blind	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	10	Component/ software/ configuration / environment/ common-cause failure	3	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	4	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-082	B17	Functional Safety Supervisor, Freedom-from-Interference & Health Management	FFI common-cause fault	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	10	Component/ software/ configuration / environment/ common-cause failure	4	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	5	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-083	B17	Functional Safety Supervisor, Freedom-from-Interference & Health Management	reaction timing overrun	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	10	Component/ software/ configuration / environment/ common-cause failure	5	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	6	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-084	B17	Functional Safety Supervisor, Freedom-from-Interference & Health Management	wrong safe-state selection	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	10	Component/ software/ configuration / environment/ common-cause failure	3	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	3	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-085	B17	Functional Safety Supervisor, Freedom-from-Interference & Health Management	latent diagnostic not run	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	10	Component/ software/ configuration / environment/ common-cause failure	4	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	4	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-086	B18	Central ECU Thermal, EMC, SI/PI, Mechanical & Environmental Design	PDN droop	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending	9	Component/ software/ configuration / environment/ common-cause failure	5	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	5	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated	Subsystem Lead

ID	Block	Item	Failure Mode	Effect	S	Cause	O	Current Control	D	AP	Recommended Action	Owner
				on failure							DVP	
DFMEA-VCU-087	B18	Central ECU Thermal, EMC, SI/PI, Mechanical & Environmental Design	DDR SI margin loss	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	9	Component/ software/ configuration / environment/ common-cause failure	3	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	6	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-088	B18	Central ECU Thermal, EMC, SI/PI, Mechanical & Environmental Design	Ethernet BER	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	9	Component/ software/ configuration / environment/ common-cause failure	4	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	3	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-089	B18	Central ECU Thermal, EMC, SI/PI, Mechanical & Environmental Design	EMC emission fail	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	9	Component/ software/ configuration / environment/ common-cause failure	5	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	4	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-090	B18	Central ECU Thermal, EMC, SI/PI, Mechanical & Environmental Design	thermal interface degradation	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	9	Component/ software/ configuration / environment/ common-cause failure	3	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	5	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-091	B19	Manufacturing, EOL Calibration, Secure Provisioning & Vehicle Commissioning	wrong key provisioning	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	9	Component/ software/ configuration / environment/ common-cause failure	4	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	6	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-092	B19	Manufacturing, EOL Calibration, Secure Provisioning & Vehicle Commissioning	wrong calibration	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	9	Component/ software/ configuration / environment/ common-cause failure	5	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	3	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-093	B19	Manufacturing, EOL Calibration, Secure Provisioning & Vehicle Commissioning	test fixture drift	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	9	Component/ software/ configuration / environment/ common-cause failure	3	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	4	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-094	B19	Manufacturing, EOL Calibration, Secure Provisioning & Vehicle Commissioning	network identity duplicate	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	9	Component/ software/ configuration / environment/ common-cause failure	4	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	5	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-095	B19	Manufacturing, EOL Calibration,	safety path EOL test skipped	Loss of availability, incorrect	9	Component/ software/ configuration	5	Diagnostics, watchdogs, E2E/SecOC,	6	H	Perform detailed FMEA/FMED	Subsystem Lead

ID	Block	Item	Failure Mode	Effect	S	Cause	O	Current Control	D	AP	Recommended Action	Owner
		Secure Provisioning & Vehicle Commissioning		motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure		/ environment/ common-cause failure		plausibility, partitioning and safe/degraded state			A; verify detection/containment and maintenance action by allocated DVP	
DFMEA-VCU-096	B20	Lifecycle Quality, Reliability, Obsolescence, Field Monitoring & FRACAS	supplier EOL without alternate	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	7	Component/ software/ configuration / environment/ common-cause failure	3	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	3	M	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-097	B20	Lifecycle Quality, Reliability, Obsolescence, Field Monitoring & FRACAS	storage wear-out	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	7	Component/ software/ configuration / environment/ common-cause failure	4	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	4	M	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-098	B20	Lifecycle Quality, Reliability, Obsolescence, Field Monitoring & FRACAS	thermal aging underestimated	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	7	Component/ software/ configuration / environment/ common-cause failure	5	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	5	H	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-099	B20	Lifecycle Quality, Reliability, Obsolescence, Field Monitoring & FRACAS	field defect trend ignored	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	7	Component/ software/ configuration / environment/ common-cause failure	3	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	6	M	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead
DFMEA-VCU-100	B20	Lifecycle Quality, Reliability, Obsolescence, Field Monitoring & FRACAS	service tool obsolete	Loss of availability, incorrect motion/HV behavior, degraded cockpit or cybersecurity exposure depending on failure	7	Component/ software/ configuration / environment/ common-cause failure	4	Diagnostics, watchdogs, E2E/SecOC, plausibility, partitioning and safe/degraded state	3	M	Perform detailed FMEA/FMEDA; verify detection/containment and maintenance action by allocated DVP	Subsystem Lead

9. Cybersecurity TARA Starter Baseline

TARA ID	Asset	Threat Scenario	Impact	Initial Risk	Key Controls	Owner
TARA-VCU-001	Android/IVI partition	Compromised app attempts to inject torque/mode/HV command	Unsafe motion authority	High	HSM/secure boot; least privilege; gateway filtering; E2E + SecOC where allocated; authenticated diagnostics/update; logging/IDS; physical debug control; lifecycle key/config management	Cybersecurity / System / Safety
TARA-VCU-002	CAN powertrain	Spoofed torque command on CAN FD	Unintended torque	High	HSM/secure boot; least privilege; gateway filtering; E2E + SecOC where allocated; authenticated diagnostics/update; logging/IDS; physical debug control; lifecycle key/config management	Cybersecurity / System / Safety
TARA-VCU-003	CAN powertrain	Replay of valid old torque/HV frame	Stale command accepted	High	HSM/secure boot; least privilege; gateway	Cybersecurity / System / Safety

TARA ID	Asset	Threat Scenario	Impact	Initial Risk	Key Controls	Owner
					filtering; E2E + SecOC where allocated; authenticated diagnostics/update; logging/IDS; physical debug control; lifecycle key/config management	
TARA-VCU-004	Ethernet gateway	Malicious service floods or pivots into safety domain	DoS / lateral movement	High	HSM/secure boot; least privilege; gateway filtering; E2E + SecOC where allocated; authenticated diagnostics/update; logging/IDS; physical debug control; lifecycle key/config management	Cybersecurity / System / Safety
TARA-VCU-005	OTA backend	Malicious or tampered update package	Unsafe software	High	HSM/secure boot; least privilege; gateway filtering; E2E + SecOC where allocated; authenticated diagnostics/update; logging/IDS; physical debug control; lifecycle key/config management	Cybersecurity / System / Safety
TARA-VCU-006	Update process	Rollback to vulnerable software	Known exploit restored	High	HSM/secure boot; least privilege; gateway filtering; E2E + SecOC where allocated; authenticated diagnostics/update; logging/IDS; physical debug control; lifecycle key/config management	Cybersecurity / System / Safety
TARA-VCU-007	Calibration	Unauthorized Track/torque calibration	Performance/safety limits bypassed	High	HSM/secure boot; least privilege; gateway filtering; E2E + SecOC where allocated; authenticated diagnostics/update; logging/IDS; physical debug control; lifecycle key/config management	Cybersecurity / System / Safety
TARA-VCU-008	Valet policy	Policy tampering raises valet cap/geofence	Unauthorized performance	High	HSM/secure boot; least privilege; gateway filtering; E2E + SecOC where allocated; authenticated diagnostics/update; logging/IDS; physical debug control; lifecycle key/config management	Cybersecurity / System / Safety
TARA-VCU-009	HSM/keys	Key extraction or cloning	Command/update impersonation	High	HSM/secure boot; least privilege; gateway filtering; E2E + SecOC where allocated; authenticated diagnostics/update; logging/IDS; physical debug control; lifecycle key/config management	Cybersecurity / System / Safety
TARA-VCU-010	Debug port	Unauthorized production JTAG/SWD access	Full ECU compromise	High	HSM/secure boot; least privilege; gateway filtering; E2E + SecOC where allocated; authenticated diagnostics/update; logging/IDS; physical debug control; lifecycle key/config management	Cybersecurity / System / Safety
TARA-VCU-011	DoIP diagnostics	Unauthorized routine/IO control	Unsafe service actuation	High	HSM/secure boot; least privilege; gateway filtering; E2E + SecOC where allocated; authenticated diagnostics/update; logging/IDS; physical debug control; lifecycle key/config management	Cybersecurity / System / Safety
TARA-VCU-012	Service tool	Compromised technician credentials	Malicious calibration/update	High	HSM/secure boot; least privilege; gateway filtering; E2E + SecOC where allocated; authenticated diagnostics/update; logging/IDS; physical	Cybersecurity / System / Safety

TARA ID	Asset	Threat Scenario	Impact	Initial Risk	Key Controls	Owner
					debug control; lifecycle key/config management	
TARA-VCU-013	Wi-Fi/Bluetooth	Remote infotainment exploit pivots to gateway	Remote compromise	Medium	HSM/secure boot; least privilege; gateway filtering; E2E + SecOC where allocated; authenticated diagnostics/update; logging/IDS; physical debug control; lifecycle key/config management	Cybersecurity / System / Safety
TARA-VCU-014	Telematics	Cloud token theft	Remote data/control exposure	Medium	HSM/secure boot; least privilege; gateway filtering; E2E + SecOC where allocated; authenticated diagnostics/update; logging/IDS; physical debug control; lifecycle key/config management	Cybersecurity / System / Safety
TARA-VCU-015	Ethernet TSN	PTP time manipulation	Event/freshness timing corruption	Medium	HSM/secure boot; least privilege; gateway filtering; E2E + SecOC where allocated; authenticated diagnostics/update; logging/IDS; physical debug control; lifecycle key/config management	Cybersecurity / System / Safety
TARA-VCU-016	SecOC freshness	Counter desynchronization attack	DoS or replay window	High	HSM/secure boot; least privilege; gateway filtering; E2E + SecOC where allocated; authenticated diagnostics/update; logging/IDS; physical debug control; lifecycle key/config management	Cybersecurity / System / Safety
TARA-VCU-017	Gateway ACL	Misconfiguration permits raw CAN access	Boundary bypass	High	HSM/secure boot; least privilege; gateway filtering; E2E + SecOC where allocated; authenticated diagnostics/update; logging/IDS; physical debug control; lifecycle key/config management	Cybersecurity / System / Safety
TARA-VCU-018	Storage	Offline modification of log/configuration	Forensics/config compromise	Medium	HSM/secure boot; least privilege; gateway filtering; E2E + SecOC where allocated; authenticated diagnostics/update; logging/IDS; physical debug control; lifecycle key/config management	Cybersecurity / System / Safety
TARA-VCU-019	Boot ROM/config	Fault injection bypasses secure boot	Untrusted code execution	Medium	HSM/secure boot; least privilege; gateway filtering; E2E + SecOC where allocated; authenticated diagnostics/update; logging/IDS; physical debug control; lifecycle key/config management	Cybersecurity / System / Safety
TARA-VCU-020	Hypervisor	Guest escape from Android/Linux	Cross-domain compromise	High	HSM/secure boot; least privilege; gateway filtering; E2E + SecOC where allocated; authenticated diagnostics/update; logging/IDS; physical debug control; lifecycle key/config management	Cybersecurity / System / Safety
TARA-VCU-021	DMA device	Malicious DMA into safety memory	Safety data corruption	High	HSM/secure boot; least privilege; gateway filtering; E2E + SecOC where allocated; authenticated diagnostics/update; logging/IDS; physical debug control; lifecycle key/config management	Cybersecurity / System / Safety
TARA-VCU-022	Supply chain	Counterfeit/reworked MCU/SoC	Security/safety assurance loss	Medium	HSM/secure boot; least privilege; gateway filtering; E2E + SecOC	Cybersecurity / System / Safety

TARA ID	Asset	Threat Scenario	Impact	Initial Risk	Key Controls	Owner
					where allocated; authenticated diagnostics/update; logging/IDS; physical debug control; lifecycle key/config management	
TARA-VCU-023	EOL provisioning	Duplicate certificate/device identity	Fleet identity collision	High	HSM/secure boot; least privilege; gateway filtering; E2E + SecOC where allocated; authenticated diagnostics/update; logging/IDS; physical debug control; lifecycle key/config management	Cybersecurity / System / Safety
TARA-VCU-024	Variant coding	Wrong market/track feature enabled	Homologation/safety issue	High	HSM/secure boot; least privilege; gateway filtering; E2E + SecOC where allocated; authenticated diagnostics/update; logging/IDS; physical debug control; lifecycle key/config management	Cybersecurity / System / Safety
TARA-VCU-025	Fleet logs	Personal data exfiltration	Privacy impact	Medium	HSM/secure boot; least privilege; gateway filtering; E2E + SecOC where allocated; authenticated diagnostics/update; logging/IDS; physical debug control; lifecycle key/config management	Cybersecurity / System / Safety
TARA-VCU-026	IDS	Log flooding hides real attack	Detection degradation	Medium	HSM/secure boot; least privilege; gateway filtering; E2E + SecOC where allocated; authenticated diagnostics/update; logging/IDS; physical debug control; lifecycle key/config management	Cybersecurity / System / Safety
TARA-VCU-027	Network wake	Wake-up abuse drains 12V battery	Availability impact	Medium	HSM/secure boot; least privilege; gateway filtering; E2E + SecOC where allocated; authenticated diagnostics/update; logging/IDS; physical debug control; lifecycle key/config management	Cybersecurity / System / Safety
TARA-VCU-028	Mode HMI	UI spoof shows Normal while Track active	Driver deception	High	HSM/secure boot; least privilege; gateway filtering; E2E + SecOC where allocated; authenticated diagnostics/update; logging/IDS; physical debug control; lifecycle key/config management	Cybersecurity / System / Safety
TARA-VCU-029	Certificate expiry	Expired credentials disable updates/diagnostics	Availability/safety maintenance impact	Medium	HSM/secure boot; least privilege; gateway filtering; E2E + SecOC where allocated; authenticated diagnostics/update; logging/IDS; physical debug control; lifecycle key/config management	Cybersecurity / System / Safety
TARA-VCU-030	Decommissioning	Credentials remain on retired ECU	Post-life compromise	Medium	HSM/secure boot; least privilege; gateway filtering; E2E + SecOC where allocated; authenticated diagnostics/update; logging/IDS; physical debug control; lifecycle key/config management	Cybersecurity / System / Safety

10. Project and Integration Risk Register

Risk ID	Category	Risk	Cause	P	I	Score	Level	Mitigation	Owner	Status
RISK-	Safety	ASIL C torque	Under-scoped	4	4	16	High	Allocate	Domain Lead	Open

Risk ID	Category	Risk	Cause	P	I	Score	Level	Mitigation	Owner	Status
VCU-001		target conflicts with HARA ASIL D unintended-torque goal	safety allocation					owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.		
RISK-VCU-002	Architecture	Single central ECU creates common-cause loss between cockpit and vehicle services	Excess consolidation	5	4	20	Critical	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-003	Compute	Chosen HPA SoC not mature enough for SOP timing	Sampling/ roadmap risk	3	4	12	Medium	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-004	Thermal	75 W peak budget insufficient for premium cockpit workload	GPU/AI/ display load growth	4	5	20	Critical	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-005	Network	1 Gbps Ethernet becomes bottleneck for future services	Centralization traffic growth	5	4	20	Critical	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-006	Software	Android/Linux restart affects shared services	Mixed-criticality integration	3	4	12	Medium	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-007	Hypervisor	Safety hypervisor/device assignment lacks required certification on chosen SoC	Platform support gap	4	4	16	High	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-	Domain Lead	Open

Risk ID	Category	Risk	Cause	P	I	Score	Level	Mitigation	Owner	Status
								controlled residual risk.		
RISK-VCU-008	Safety	Mode transition jerk target violated at large torque delta	50 ms interpretation too aggressive	5	5	25	Critical	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-009	Safety	Track mode calibration compromises road safety/homologation	Feature governance	3	4	12	Medium	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-010	Chassis	ESC/EPS supplier interfaces do not support requested mode semantics	Supplier mismatch	4	4	16	High	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-011	Brake	Regen/friction blending ownership unclear	VCU vs ESC boundary	5	4	20	Critical	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-012	BMS	HV contactor ownership duplicated	VCU/BMS boundary ambiguity	3	5	15	High	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-013	Thermal	Hard-coded 65C thresholds incompatible with supplier limits	Calibration oversimplification	4	4	16	High	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-014	Cyber	Infotainment compromise reaches safety CAN	Gateway isolation weakness	5	4	20	Critical	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and	Domain Lead	Open

Risk ID	Category	Risk	Cause	P	I	Score	Level	Mitigation	Owner	Status
								supplier evidence; maintain change-controlled residual risk.		
RISK-VCU-015	Cyber	SecOC overhead breaks 10ms CAN timing	Security/ performance co-design	3	4	12	Medium	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-016	OTA	Partial multi-domain update creates incompatible versions	Dependency management	4	5	20	Critical	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-017	Diagnostics	DoIP/service routine authorization insufficient	Service attack surface	5	4	20	Critical	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-018	Power	Low-voltage brownout resets central ECU while vehicle moving	LV resilience	3	4	12	Medium	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-019	EMC	High-speed SoC/DDR/SerDes noise fails CISPR25	Central compute EMC density	4	4	16	High	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-020	PI	HPA load-step causes DDR/SoC rail droop	Power integrity	5	5	25	Critical	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-021	SI	Display/ Ethernet/DDR SI fails on	High-speed layout	3	4	12	Medium	Allocate owner; close architecture	Domain Lead	Open

Risk ID	Category	Risk	Cause	P	I	Score	Level	Mitigation	Owner	Status
		compact PCB						decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.		
RISK-VCU-022	Manufacturing	Secure provisioning station compromised	Production cyber	4	4	16	High	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-023	Quality	Wrong variant/calibration at EOL	Configuration management	5	4	20	Critical	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-024	Lifecycle	Critical SoC/hypervisor supplier EOL before vehicle service life	Obsolescence	3	5	15	High	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-025	Toolchain	Model/code-generation/tool qualification evidence incomplete	ISO26262 tool confidence	4	4	16	High	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-026	ASPICE	Requirements duplicated across SYS2/SYS3 without ownership	Traceability quality	5	4	20	Critical	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-027	HIL	Plant models do not cover coupled brake/thermal/HV states	Verification coverage	3	4	12	Medium	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open

Risk ID	Category	Risk	Cause	P	I	Score	Level	Mitigation	Owner	Status
RISK-VCU-028	Timing	Worst-case CAN arbitration exceeds torque deadline	Network schedulability	4	5	20	Critical	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-029	Storage	Flight recorder/OTA writes exhaust storage endurance	Lifetime	5	4	20	Critical	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-030	Boot	Cockpit slow boot hides safety state	User/safety perception	3	4	12	Medium	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-031	HMI	Driver sees requested mode rather than authoritative accepted mode	State mismatch	4	4	16	High	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-032	Security	Key/certificate expiry disrupts service/update	Lifecycle PKI	5	5	25	Critical	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-033	Performance	220 kW peak requested when BMS SOP lower	Limit arbitration	3	4	12	Medium	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-034	Vehicle Dynamics	Torque rate limit conflicts with emergency cut	Comfort vs safety	4	4	16	High	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain	Domain Lead	Open

Risk ID	Category	Risk	Cause	P	I	Score	Level	Mitigation	Owner	Status
								change-controlled residual risk.		
RISK-VCU-035	Service	Actuator routine accidentally allowed in Drive	Diagnostic state gating	5	4	20	Critical	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-036	Supplier	Inverter/BMS limit signals have inconsistent units/age	ICD quality	3	5	15	High	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-037	Change	Track feature late change causes safety/cyber regression	Feature creep	4	4	16	High	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-038	Cost	Dual compute architecture exceeds BOM target	Cost pressure	5	4	20	Critical	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-039	Power	Dual SoC/MCU PMIC architecture exceeds 75 W	Power budget	3	4	12	Medium	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-040	Cooling	Cabin mounting cannot reject peak heat silently	Packaging	4	5	20	Critical	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-041	Privacy	Fleet logs contain personal driver data	Data governance	5	4	20	Critical	Allocate owner; close architecture decision early; add targeted analysis/HIL/D	Domain Lead	Open

Risk ID	Category	Risk	Cause	P	I	Score	Level	Mitigation	Owner	Status
								VP and supplier evidence; maintain change-controlled residual risk.		
RISK-VCU-042	Regulation	UNECE R155/R156 market scope addressed too late	Homologation	3	4	12	Medium	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-043	Testing	EMC test modes do not exercise worst GPU/network activity	False pass	4	4	16	High	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-044	Safety	Independent safety MCU shares reset/clock/power root with HPA	Common-cause	5	5	25	Critical	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-045	Calibration	User custom mode parameters exceed safe bounds	Customization control	3	4	12	Medium	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open

11. Technology/Interface Decisions Affecting SYS.2

ADR	Decision Area	Decision	Status	Rationale
ADR-001	Central compute topology	One central ECU with independent safety MCU + HPA application SoC	Selected	Maintains infotainment consolidation while keeping torque/HV authority independent from Android/Linux faults.
ADR-002	Torque execution	Safety MCU owns final torque arbitration and safety clamp; HPA SoC may compute advisory/comfort requests	Selected	Reduces hypervisor/Android qualification burden and gives deterministic WCET.
ADR-003	Cockpit SoC	R-Car X5H / Snapdragon Cockpit Elite class evaluated; no silicon frozen in concept baseline	Open RFQ	Future-proof multi-display/AI performance requires workload benchmark, supplier roadmap and power evidence.
ADR-004	Real-time central processor	AURIX TC4x primary safety-MCU reference; S32N55 class alternative for deeper real-time consolidation	Open RFQ	Both target centralized safety/real-time automotive control; final choice by timing, I/O, safety, security, ecosystem and lifecycle.
ADR-005	Hypervisor	Safety-qualified	Selected principle	QM Android/Linux guests must

ADR	Decision Area	Decision	Status	Rationale
		Type-1/partitioning hypervisor for HPA SoC; QNX Hypervisor for Safety class reference		be isolated from safety/cluster services and hardware resources.
ADR-006	Driver-mode source	HMI is a request source only; safety MCU validates mode against vehicle state, market, thermal and chassis permissions	Selected	Prevents infotainment compromise or UI fault from directly selecting unsafe torque/stability behavior.
ADR-007	Track mode	ESC final authority; VCU requests Track calibration but cannot defeat ABS/ESC hardware protections	Selected	Mode excitement cannot override safety limits or local actuator protections.
ADR-008	HV contactors	VCU owns high-level HV state orchestration; BMS owns pack contactor drive and pack-level protection	Selected	Avoids duplicated pack-safety ownership and keeps pack hardware protection local.
ADR-009	Display transport	Display pixels over DP/eDP/automotive SerDes; Ethernet for services/state/updates	Selected	1 Gbps Ethernet should not be consumed by raw multi-display pixels.
ADR-010	Safety comm	CAN FD A/B or independent safety channels for torque/brake/HV; Ethernet not sole safe motion path	Selected	Preserves actuation during Ethernet switch/SoC faults.
ADR-011	E2E vs SecOC	AUTOSAR E2E protects communication integrity; SecOC separately provides authenticity/freshness	Selected	Avoids treating CRC/counter as a cryptographic signature.
ADR-012	Thermal thresholds	VCU consumes normalized BMS/inverter thermal limits/SOP rather than universal hard-coded 65 C limits	Selected	Cell/module/inverter hardware have different valid thermal envelopes.

12. Regulatory, Safety and Process Mapping

Framework	Process / Area	Expected Evidence	Allocation
Automotive SPICE 4.0	SYS.1/SYS.2/SYS.3; SUP/MAN process interaction	Bidirectional requirements/architecture/test/change evidence	All DOCs + RTM
ISO 26262:2018	Item/HARA/FSC/TSC/system/HW/SW safety	Safety goals, FSR/TSR/HSR/SWSR, analyses, verification	DOC1/DOC2/DOC3
ISO/SAE 21434:2021	Cybersecurity lifecycle/TARA	Assets/threats/goals/controls/ evidence/post-development	DOC2/DOC3
ISO 24089:2023 + Amd1:2024	Software update engineering	Update plan, dependency, vehicle state, activation/recovery evidence	DOC1/DOC3
UNECE R155/R156	Market CSMS/SUMS type-approval obligations where applicable	Cyber/update management evidence and vehicle type trace	DOC1/DOC2
AUTOSAR Classic/Adaptive	MCU BSW, E2E, SecOC, service-oriented architecture	ARXML/BSW/service contracts, safety comm config	DOC3
ISO 11898 CAN/CAN FD Jan-2026 baseline	CAN data-link/physical layer	Network design, timing, conformance	DOC3
ISO 14229 / ISO 13400	UDS / DoIP	Diagnostic service/interface conformance	DOC2/DOC3
ISO 16750-2:2023 / ISO 7637-2:2011	Electrical loads/transients	LV power DVP	DOC3
CISPR 25:2021 / ISO 11452 series	EMC emissions/immunity	Component EMC evidence	DOC3

13. Verification Strategy at SYS.1 / SYS.2 Level

- Requirements review: completeness, singularity, verifiability, assumptions, safety/cyber classification and interface ownership.
- Model-in-the-loop / software-in-the-loop: torque, regen, driver mode, HV and thermal supervisory behavior including degraded states.
- Network simulation: worst-case CAN scheduling, E2E/SecOC timeouts, gateway routing and DoIP/diagnostic loading.
- HIL: sensor/network/CPU/PMIC fault injection with safety reaction timing and local ECU behavior.

- Dyno/vehicle: torque step, brake blend, mode transitions, HV startup/shutdown, thermal derating, limp/controlled-stop and driver warning.
- Cybersecurity: secure boot/update, privilege boundary, diagnostic authentication, replay/spoofing and compromised infotainment tests.

14. Source and Vendor Reference Register

Ref ID	Category	Document / Product	Source	Link	Use
REF-VCU-001	Process	Automotive SPICE 4.0 PRM/PAM	VDA QMC	Official source	SYS.1/SYS.2/SYS.3 process and assessment baseline
REF-VCU-002	Functional Safety	ISO 26262:2018 Road vehicles - Functional safety	ISO	Official source	Item/HARA/FSC/TSC/ system/HW/SW safety lifecycle
REF-VCU-003	Cybersecurity	ISO/SAE 21434:2021 Road vehicles - Cybersecurity engineering	ISO	Official source	TARA and cybersecurity lifecycle
REF-VCU-004	Software Update	ISO 24089:2023 + Amd 1:2024	ISO	Official source	Software update engineering
REF-VCU-005	Regulation	UN Regulation No.155 - Cyber security / CSMS	UNECE	Official source	Market cybersecurity type-approval comparator
REF-VCU-006	Regulation	UN Regulation No.156 - Software update / SUMS	UNECE	Official source	Market software-update type-approval comparator
REF-VCU-007	Diagnostics	ISO 14229-1:2020 UDS (Jan-2026 baseline)	ISO	Official source	UDS diagnostic service architecture
REF-VCU-008	Diagnostics	ISO 13400-2:2025 DoIP (Jan-2026 baseline)	ISO	Official source	IP diagnostic transport
REF-VCU-009	CAN	ISO 11898-1:2024 / ISO 11898-2:2024 Jan-2026 baseline	ISO	Official source	CAN/CAN FD data-link and high-speed physical layer
REF-VCU-010	Electrical	ISO 16750-2:2023 Electrical loads	ISO	Official source	LV electrical environmental loads
REF-VCU-011	Electrical	ISO 7637-2:2011 Conducted transients	ISO	Official source	12/24 V transient bench test baseline
REF-VCU-012	EMC	CISPR 25:2021	IEC	Official source	Component conducted/radiated emissions
REF-VCU-013	EMC	ISO 11452-2:2019 / ISO 11452-4:2020	ISO	Official source	Component radiated/BCI immunity
REF-VCU-014	AUTOSAR	AUTOSAR E2E Protocol Specification	AUTOSAR	Official source	Communication safety integrity profiles
REF-VCU-015	AUTOSAR	AUTOSAR Secure Onboard Communication (SecOC)	AUTOSAR	Official source	PDU authenticity and freshness
REF-VCU-016	Safety MCU	AURIX TC4x family	Infineon	Official source	Safety MCU / eMobility / vehicle motion control reference
REF-VCU-017	Safety MCU PMIC	OPTIREG TLF35585	Infineon	Official source	Safety PMIC/watchdog/rail/safe-state reference
REF-VCU-018	Central real-time compute	S32N55 / S32N family	NXP	Official source	Central real-time vehicle control / hardware isolation comparator
REF-VCU-019	HPA SoC	R-Car X5H / Gen5 central compute	Renesas	Official source	Multi-domain HPA / IVI / virtualization comparator
REF-VCU-020	Cockpit SoC	Snapdragon Cockpit Elite	Qualcomm	Official source	Premium virtualized cockpit/central compute comparator
REF-VCU-021	Hypervisor	QNX Hypervisor for Safety	QNX	Official source	Safety-qualified virtualization comparator
REF-VCU-022	Ethernet Switch	Brightlane Q622x central automotive Ethernet switch	Marvell	Official source	60/90-Gbps TSN/high-speed central switching comparator
REF-VCU-023	Ethernet PHY	Brightlane 88Q2110/88Q2112	Infineon/Marvell technology line	Official source	100/1000BASE-T1 PHY / TC10 / 802.1AS comparator
REF-VCU-024	CAN PHY	TCAN1043A/HG-Q1 CAN FD transceiver	Texas Instruments	Official source	CAN FD / wake / bus fault comparator
REF-VCU-025	Cockpit	Snapdragon Cockpit	Qualcomm	Official source	Cockpit virtualized

Ref ID	Category	Document / Product	Source	Link	Use
		Platform			processing / UX ecosystem
REF-VCU-026	Central Compute	R-Car X5H architecture overview	Renesas	Official source	Centralized multi-domain SoC capabilities
REF-VCU-027	Software	AUTOSAR Standards	AUTOSAR	Official source	Classic/Adaptive architecture baseline

Appendix A - Controlled Assumptions and Constraints

ID	Topic	Assumption	Closure Rule
A-001	HV Architecture	400 Vdc nominal traction battery; 220 kW peak / 110 kW continuous propulsion reference. An 800 V variant is supported by interface/calibration scaling rather than assuming identical current limits.	Freeze vehicle BMS/inverter voltage-current envelopes and connector ICD.
A-002	Centralization	The product is one central ECU/housing containing an independent safety MCU and a high-performance application SoC. Safety-critical torque authority does not depend on the Android/infotainment guest.	Close silicon/hypervisor safety concept and common-cause analysis.
A-003	Actuation Boundary	VCU sends bounded torque/deceleration/HV requests. Inverter, ESC/brake ECU, EPS and BMS retain local actuator-level safety checks and hardware protection.	Freeze each actuator ICD and safety manual assumptions.
A-004	Driver Modes	Eco, Normal, Sport, Track and Valet/Custom are calibration profiles; none may bypass battery/inverter/ESC/EPS thermal or safety hard limits.	Approve mode safety analysis and homologation/market rules.
A-005	Track Mode	Track mode may be market-, geofence-, service- or closed-course restricted. ESC retains the final stability envelope; VCU only requests a permitted mode.	Freeze legal/homologation and ESC feature contract.
A-006	Power Budget	70 W nominal and 130 W sustained peak with a 150 W short electrical input envelope apply to the central ECU electronics, excluding display backlights, speakers/amplifier and externally powered antennas/sensors.	Close power tree and vehicle LV budget.
A-007	Network	Safety motion commands use redundant or independent CAN FD channels; Ethernet carries service-oriented, diagnostic and cockpit traffic. Raw display pixels use dedicated display SerDes/DP links, not the 1-Gbps vehicle Ethernet baseline.	Freeze E/E architecture and traffic matrix.
A-008	Safety	Program target is ASIL C for torque arbitration; preliminary HARA may allocate ASIL D safety goals for unintended torque. Hardware/software architecture shall not prevent ASIL D-capable mitigation if HARA requires it.	Complete item definition/HARA/FSC/TSC.
A-009	Cockpit OS	Android Automotive/Linux/QNX guests are treated as QM unless specific functions receive a higher allocation. Safety-relevant telltales use an independent monitored path.	Freeze HMI safety concept.
A-010	Thermal Limits	Battery/inverter temperature thresholds are supplier-calibration inputs; the VCU shall not hard-code a	Freeze BMS/inverter thermal limit ICD.

ID	Topic	Assumption	Closure Rule
		universal 65 C protection threshold.	
A-011	Vehicle Dynamics	VCU coordinates high-level longitudinal/chassis requests; ESC/ABS retains wheel-slip and hydraulic brake authority and EPS retains steering hardware safety.	Close chassis arbitration ICD.
A-012	OTA	Vehicle supports secure OTA and workshop update with A/B or equivalent recovery. Update engineering follows ISO 24089 and applicable UNECE R156 market obligations.	Freeze SUMS/update architecture.

ID	Area	Constraint
C-001	Power	Central ECU <=45 W nominal and <=75 W peak; short transient may exceed only if vehicle LV and thermal design approve.
C-002	Control Latency	Pedal-to-valid-torque-request <=15 ms end-to-end reference; internal VCU compute/network allocation <=6 ms target.
C-003	Safety Reaction	Electronic safety monitor to propulsion inhibit/request <=10 ms internal capability target; final hazard FTTI follows FSC/TSC.
C-004	CAN FD	Powertrain safety networks: 500 kbps arbitration / 2 Mbps data baseline; nominal sustained utilization <=40%, fault burst <=60%.
C-005	Ethernet	1000BASE-T1 baseline for service/cockpit/gateway traffic; architecture shall support 2.5/10G uplink option if future central compute traffic requires it.
C-006	Thermal	130 W sustained-peak central ECU shall close Tj/case/junction margins at released ambient with >=20% thermal design margin.
C-007	Boot	Safety MCU motion-control availability <=150 ms after wake reference; cockpit fast-visible telltale <=500 ms; rich Android UX may boot later.
C-008	Security	Infotainment/telematics compromise shall not create a direct command path to torque/HV actuation.
C-009	Mode Transition	Mode request acknowledgement <=50 ms; torque-map blending is jerk-bounded and may take 200-500 ms or longer depending torque delta.
C-010	Calibration	All power/torque/regen/thermal/mode safety-relevant calibration is signed, variant-bound, range-checked and rollback controlled.
C-011	EMC	CISPR 25 and ISO 11452 component-level targets plus OEM vehicle EMC profile apply to central ECU and harness.
C-012	Electrical	ISO 16750-2:2023 and ISO 7637-2:2011 baseline for 12 V/24 V electrical loads/transients; vehicle-specific LV profile is controlling.

Appendix B - Traceability Handoff

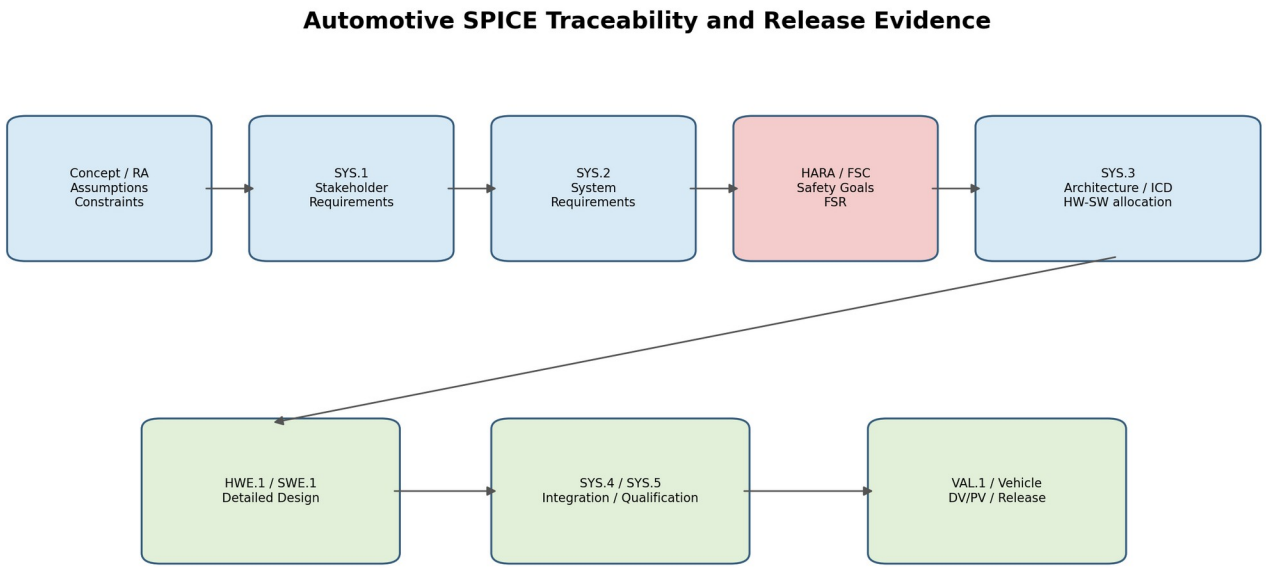


Figure 5 - Concept-to-validation traceability

Appendix C - SYS.2 Design Interpretation and Verification Matrix

The SYS.2 baseline is intended to survive supplier/SoC changes. This matrix makes the verification meaning explicit without prematurely constraining detailed SYS.3 implementation. Where a requirement depends on an external ECU, the final acceptance criterion is jointly frozen in the ICD and supplier DVP.

SYS2 ID	Domain	System Requirement	Design Interpretation	Verification / Evidence	Safety
SYS2-VCU-001	Safety & Motion	Only the safety-controlled motion domain may emit the authoritative propulsion torque request.	Treat the requirement as an authority/containment rule. The safety MCU and local actuator ECU shall remain capable of rejecting an unsafe command even if the HPA or HMI behaves incorrectly.	HIL fault injection, timing trace, local-inverter/ESC response and safety-case evidence.	Safety-related
SYS2-VCU-002	Safety & Motion	Torque shall be checked against pedal/brake/gear/speed/vehicle state and local actuator feedback.	Treat the requirement as an authority/containment rule. The safety MCU and local actuator ECU shall remain capable of rejecting an unsafe command even if the HPA or HMI behaves incorrectly.	HIL fault injection, timing trace, local-inverter/ESC response and safety-case evidence.	Safety-related
SYS2-VCU-003	Safety & Motion	Unsafe torque deviation shall trigger torque inhibit/degrade inside the allocated internal reaction budget.	Treat the requirement as an authority/containment rule. The safety MCU and local actuator ECU shall remain capable of rejecting an unsafe command even if the HPA or HMI behaves incorrectly.	HIL fault injection, timing trace, local-inverter/ESC response and safety-case evidence.	Safety-related
SYS2-VCU-004	Safety & Motion	Independent supervision shall detect task/clock/rail faults without relying on infotainment execution.	Treat the requirement as an authority/containment rule. The safety MCU and local actuator ECU shall remain capable of rejecting an unsafe command even if the HPA or HMI behaves incorrectly.	HIL fault injection, timing trace, local-inverter/ESC response and safety-case evidence.	Safety-related
SYS2-VCU-005	Safety & Motion	Faults shall map to controlled torque-zero, limited drive, controlled stop or HV isolation according to safety analysis.	Treat the requirement as an authority/containment rule. The safety MCU and local actuator ECU shall remain capable of rejecting an unsafe command even if the HPA or HMI behaves incorrectly.	HIL fault injection, timing trace, local-inverter/ESC response and safety-case evidence.	Safety-related
SYS2-VCU-006	Driver Modes	Mode requests shall be validated by the safety domain before calibration activation.	Convert the user-visible mode request into a safety-domain accepted profile. Separate request acknowledgement from physical torque/regen/chassis blending and enforce jerk/SOP/thermal/local-ECU limits.	Mode-transition HIL and vehicle measurements covering jerk, rejected/invalid mode, thermal/SOP clamps and Track/Valet restrictions.	Safety-related
SYS2-VCU-007	Driver Modes	Mode request status shall update within 50 ms reference.	Convert the user-visible mode request into a safety-domain accepted profile. Separate request acknowledgement from physical torque/regen/chassis blending and enforce jerk/SOP/thermal/local-ECU limits.	Mode-transition HIL and vehicle measurements covering jerk, rejected/invalid mode, thermal/SOP clamps and Track/Valet restrictions.	Safety-related
SYS2-VCU-008	Driver Modes	Torque/regen maps shall cross-fade with jerk-bounded slew rather than instantaneous full-scale changes.	Convert the user-visible mode request into a safety-domain accepted profile. Separate request acknowledgement from physical torque/regen/chassis blending and enforce jerk/SOP/thermal/local-ECU limits.	Mode-transition HIL and vehicle measurements covering jerk, rejected/invalid mode, thermal/SOP clamps and Track/Valet restrictions.	Safety-related
SYS2-VCU-009	Driver Modes	Track mode shall be gated by market/vehicle/chassis/thermal permissions.	Convert the user-visible mode request into a safety-domain accepted profile. Separate request acknowledgement from physical torque/regen/chassis blending and enforce jerk/SOP/thermal/local-	Mode-transition HIL and vehicle measurements covering jerk, rejected/invalid mode, thermal/SOP clamps and Track/Valet restrictions.	Safety-related

SYS2 ID	Domain	System Requirement	Design Interpretation	Verification / Evidence	Safety
			ECU limits.		
SYS2-VCU-010	Driver Modes	Valet/custom restrictions shall be authenticated and shall never relax hard safety/thermal limits.	Convert the user-visible mode request into a safety-domain accepted profile. Separate request acknowledgement from physical torque/regen/chassis blending and enforce jerk/SOP/thermal/local-ECU limits.	Mode-transition HIL and vehicle measurements covering jerk, rejected/invalid mode, thermal/SOP clamps and Track/Valet restrictions.	Safety-related
SYS2-VCU-011	HV Coordination	VCU shall coordinate OFF/ACCESSORY/READY/PRECHARGE/DRIVE/CHARGE/FAULT states with BMS and inverter.	Keep pack contactor/precharge actuation in the BMS reference allocation while the VCU orchestrates vehicle READY and inverter enable. Every transition uses explicit prerequisites and feedback.	BMS/inverter HIL, welded/failed contactor simulation, precharge timeout, crash/HVIL and shutdown sequence evidence.	Safety-related
SYS2-VCU-012	HV Coordination	VCU shall verify BMS-reported precharge completion and inverter DC-link readiness before enabling propulsion.	Keep pack contactor/precharge actuation in the BMS reference allocation while the VCU orchestrates vehicle READY and inverter enable. Every transition uses explicit prerequisites and feedback.	BMS/inverter HIL, welded/failed contactor simulation, precharge timeout, crash/HVIL and shutdown sequence evidence.	Safety-related
SYS2-VCU-013	HV Coordination	Pack contactor drive remains BMS-owned unless item definition allocates otherwise.	Keep pack contactor/precharge actuation in the BMS reference allocation while the VCU orchestrates vehicle READY and inverter enable. Every transition uses explicit prerequisites and feedback.	BMS/inverter HIL, welded/failed contactor simulation, precharge timeout, crash/HVIL and shutdown sequence evidence.	Safety-related
SYS2-VCU-014	HV Coordination	Crash/HVIL faults shall cause a coordinated non-permissive propulsion/HV sequence.	Keep pack contactor/precharge actuation in the BMS reference allocation while the VCU orchestrates vehicle READY and inverter enable. Every transition uses explicit prerequisites and feedback.	BMS/inverter HIL, welded/failed contactor simulation, precharge timeout, crash/HVIL and shutdown sequence evidence.	Safety-related
SYS2-VCU-015	HV Coordination	Drive torque and charging states shall be mutually constrained according to vehicle architecture.	Keep pack contactor/precharge actuation in the BMS reference allocation while the VCU orchestrates vehicle READY and inverter enable. Every transition uses explicit prerequisites and feedback.	BMS/inverter HIL, welded/failed contactor simulation, precharge timeout, crash/HVIL and shutdown sequence evidence.	Safety-related
SYS2-VCU-016	Thermal & Energy	Available torque shall use the minimum of BMS, inverter, motor and thermal capability limits.	Use normalized capability/SOP limits from the BMS, inverter, motor and thermal controller rather than a hard-coded universal temperature threshold. Capability may only decrease on invalid/stale input.	Thermal chamber/HIL and dyno testing at cold/hot/SOP boundaries, sensor faults and communication timeouts.	Safety-related
SYS2-VCU-017	Thermal & Energy	Thermal derating shall be monotonic with hysteresis/recovery and supplier-calibrated thresholds.	Use normalized capability/SOP limits from the BMS, inverter, motor and thermal controller rather than a hard-coded universal temperature threshold. Capability may only decrease on invalid/stale input.	Thermal chamber/HIL and dyno testing at cold/hot/SOP boundaries, sensor faults and communication timeouts.	Safety-related
SYS2-VCU-018	Thermal & Energy	Sport/Track may request pre-chilling but may not commandeer unavailable thermal capacity.	Use normalized capability/SOP limits from the BMS, inverter, motor and thermal controller rather than a hard-coded universal temperature threshold. Capability may only decrease on invalid/stale input.	Thermal chamber/HIL and dyno testing at cold/hot/SOP boundaries, sensor faults and communication timeouts.	Safety-related
SYS2-VCU-019	Thermal & Energy	220 kW peak and 110 kW continuous are vehicle references subject to time/temperature/SOP.	Use normalized capability/SOP limits from the BMS, inverter, motor and thermal controller	Thermal chamber/HIL and dyno testing at cold/hot/SOP boundaries, sensor faults and	Safety-related

SYS2 ID	Domain	System Requirement	Design Interpretation	Verification / Evidence	Safety
			rather than a hard-coded universal temperature threshold. Capability may only decrease on invalid/stale input.	communication timeouts.	
SYS2-VCU-020	Thermal & Energy	Accessory/cabin/traction thermal priorities shall be arbitrated by vehicle mode and safety state.	Use normalized capability/SOP limits from the BMS, inverter, motor and thermal controller rather than a hard-coded universal temperature threshold. Capability may only decrease on invalid/stale input.	Thermal chamber/HIL and dyno testing at cold/hot/SOP boundaries, sensor faults and communication timeouts.	Safety-related
SYS2-VCU-021	Chassis & Regen	VCU shall compute desired regenerative contribution and provide it to the brake/ESC authority.	The VCU calculates desired regen/chassis profile but the ESC/brake/EPS controllers retain local wheel-slip, hydraulic and steering safety authority.	Brake dyno/vehicle blend tests, ABS/ESC activation, low-friction split-mu, BMS charge-SOP loss and local-ECU refusal tests.	Safety-related
SYS2-VCU-022	Chassis & Regen	Brake pedal/ABS/ESC intervention shall immediately reduce or revoke positive propulsion request.	The VCU calculates desired regen/chassis profile but the ESC/brake/EPS controllers retain local wheel-slip, hydraulic and steering safety authority.	Brake dyno/vehicle blend tests, ABS/ESC activation, low-friction split-mu, BMS charge-SOP loss and local-ECU refusal tests.	Safety-related
SYS2-VCU-023	Chassis & Regen	Regen shall be limited by BMS charge SOP, inverter capability, motor speed and wheel-slip state.	The VCU calculates desired regen/chassis profile but the ESC/brake/EPS controllers retain local wheel-slip, hydraulic and steering safety authority.	Brake dyno/vehicle blend tests, ABS/ESC activation, low-friction split-mu, BMS charge-SOP loss and local-ECU refusal tests.	Safety-related
SYS2-VCU-024	Chassis & Regen	VCU may request EPS/ESC/suspension mode; local ECUs retain hard safety limits.	The VCU calculates desired regen/chassis profile but the ESC/brake/EPS controllers retain local wheel-slip, hydraulic and steering safety authority.	Brake dyno/vehicle blend tests, ABS/ESC activation, low-friction split-mu, BMS charge-SOP loss and local-ECU refusal tests.	Safety-related
SYS2-VCU-025	Chassis & Regen	Mode/brake blending shall target longitudinal jerk <=2.0 m/s3 reference in normal road transitions.	The VCU calculates desired regen/chassis profile but the ESC/brake/EPS controllers retain local wheel-slip, hydraulic and steering safety authority.	Brake dyno/vehicle blend tests, ABS/ESC activation, low-friction split-mu, BMS charge-SOP loss and local-ECU refusal tests.	Safety-related
SYS2-VCU-026	Central Compute	Central ECU shall include independent safety MCU plus application SoC/hypervisor.	Prove freedom from interference across safety MCU, hypervisor, HPA guests, memory, power and network resources. Authoritative motion control must survive HPA reset or overload.	WCET/load stress, memory/ECC faults, hypervisor/guest crash, PMIC/brownout and resource-exhaustion fault injection.	Safety-related
SYS2-VCU-027	Central Compute	DMA, IOMMU/MPU, interrupt and peripheral access shall prevent QM guests from writing safety actuators.	Prove freedom from interference across safety MCU, hypervisor, HPA guests, memory, power and network resources. Authoritative motion control must survive HPA reset or overload.	WCET/load stress, memory/ECC faults, hypervisor/guest crash, PMIC/brownout and resource-exhaustion fault injection.	Safety-related
SYS2-VCU-028	Central Compute	Safety processes shall use bounded/static memory; application partitions shall use bounded pools and health limits.	Prove freedom from interference across safety MCU, hypervisor, HPA guests, memory, power and network resources. Authoritative motion control must survive HPA reset or overload.	WCET/load stress, memory/ECC faults, hypervisor/guest crash, PMIC/brownout and resource-exhaustion fault injection.	Safety-related
SYS2-VCU-029	Central Compute	Safety MCU shall reach safe supervision before rich IVI boot completes.	Prove freedom from interference across safety MCU, hypervisor, HPA guests, memory, power and network resources. Authoritative motion control must survive HPA reset or overload.	WCET/load stress, memory/ECC faults, hypervisor/guest crash, PMIC/brownout and resource-exhaustion fault injection.	Safety-related
SYS2-VCU-030	Central Compute	Release shall preserve >=30% CPU/memory/network reserve in worst released mode where practicable.	Prove freedom from interference across safety MCU, hypervisor, HPA guests, memory, power and network resources. Authoritative motion control must survive HPA reset or overload.	WCET/load stress, memory/ECC faults, hypervisor/guest crash, PMIC/brownout and resource-exhaustion fault injection.	Safety-related
SYS2-VCU-031	Networking	Torque/brake/HV safety PDUs shall use deterministic CAN FD with	Specify timing, validity, age, routing and security separately for safety CAN	Bit-accurate CAN schedule, Ethernet VLAN/QoS/time-sync	Safety-related

SYS2 ID	Domain	System Requirement	Design Interpretation	Verification / Evidence	Safety
		age/counter/E2E protection.	and IP/Ethernet services. Ethernet must not be a single point of failure for motion safety.	tests, gateway overload, packet loss, stale/replay and bus-off scenarios.	
SYS2-VCU-032	Networking	1000BASE-T1 shall support DoIP/SOME-IP/cockpit services; 2.5/10G upgrade path shall be reserved.	Specify timing, validity, age, routing and security separately for safety CAN and IP/Ethernet services. Ethernet must not be a single point of failure for motion safety.	Bit-accurate CAN schedule, Ethernet VLAN/QoS/time-sync tests, gateway overload, packet loss, stale/replay and bus-off scenarios.	Safety-related
SYS2-VCU-033	Networking	Critical Ethernet traffic shall use time sync and traffic-class controls when sharing links with infotainment/OTA.	Specify timing, validity, age, routing and security separately for safety CAN and IP/Ethernet services. Ethernet must not be a single point of failure for motion safety.	Bit-accurate CAN schedule, Ethernet VLAN/QoS/time-sync tests, gateway overload, packet loss, stale/replay and bus-off scenarios.	Safety-related
SYS2-VCU-034	Networking	Safety CAN nominal sustained utilization shall remain <=40% reference and fault-burst <=60%.	Specify timing, validity, age, routing and security separately for safety CAN and IP/Ethernet services. Ethernet must not be a single point of failure for motion safety.	Bit-accurate CAN schedule, Ethernet VLAN/QoS/time-sync tests, gateway overload, packet loss, stale/replay and bus-off scenarios.	Safety-related
SYS2-VCU-035	Networking	CAN partial networking and Ethernet TC10-class sleep/wake shall support low-power states.	Specify timing, validity, age, routing and security separately for safety CAN and IP/Ethernet services. Ethernet must not be a single point of failure for motion safety.	Bit-accurate CAN schedule, Ethernet VLAN/QoS/time-sync tests, gateway overload, packet loss, stale/replay and bus-off scenarios.	Safety-related
SYS2-VCU-036	Cockpit/HMI	Android/IVI mode selection is a request only; safety MCU publishes authoritative accepted mode.	Treat rich HMI as a presentation/request layer. Safety-relevant telltales and mode status require monitored source integrity and a defined fallback if Android/display rendering fails.	HMI latency, display reset, lost acknowledgement, telltale fallback, physical control and driver-mode usability tests.	Safety-related
SYS2-VCU-037	Cockpit/HMI	Critical telltales shall have an independent monitored path capable of indicating safety state if Android stalls.	Treat rich HMI as a presentation/request layer. Safety-relevant telltales and mode status require monitored source integrity and a defined fallback if Android/display rendering fails.	HMI latency, display reset, lost acknowledgement, telltale fallback, physical control and driver-mode usability tests.	Safety-related
SYS2-VCU-038	Cockpit/HMI	Driver shall receive accepted/rejected mode and reason within 100 ms reference.	Treat rich HMI as a presentation/request layer. Safety-relevant telltales and mode status require monitored source integrity and a defined fallback if Android/display rendering fails.	HMI latency, display reset, lost acknowledgement, telltale fallback, physical control and driver-mode usability tests.	QM/Functional
SYS2-VCU-039	Cockpit/HMI	High-rate display pixels shall use dedicated display interface/SerDes rather than consuming vehicle 1G Ethernet.	Treat rich HMI as a presentation/request layer. Safety-relevant telltales and mode status require monitored source integrity and a defined fallback if Android/display rendering fails.	HMI latency, display reset, lost acknowledgement, telltale fallback, physical control and driver-mode usability tests.	QM/Functional
SYS2-VCU-040	Cockpit/HMI	Track/custom features shall follow market HMI/distraction and homologation rules.	Treat rich HMI as a presentation/request layer. Safety-relevant telltales and mode status require monitored source integrity and a defined fallback if Android/display rendering fails.	HMI latency, display reset, lost acknowledgement, telltale fallback, physical control and driver-mode usability tests.	QM/Functional
SYS2-VCU-041	Diagnostics/Service	VCU shall support UDS diagnostic sessions, DTC, routines and data services per released diagnostic specification.	Diagnostic and plant/service functions require vehicle-state interlocks, role authorization and complete audit/freeze-frame data. Service modes cannot bypass torque/HV safety.	UDS/DoIP conformance, session/security negative tests, interrupted routines and plant/service fault-injection tests.	Safety-related
SYS2-VCU-042	Diagnostics/Service	Ethernet diagnostic access shall use DoIP and security controls.	Diagnostic and plant/service functions require vehicle-state interlocks, role authorization and complete audit/freeze-frame data. Service modes cannot bypass torque/HV safety.	UDS/DoIP conformance, session/security negative tests, interrupted routines and plant/service fault-injection tests.	Safety-related
SYS2-VCU-043	Diagnostics/Service	Safety-relevant torque/mode/HV faults	Diagnostic and plant/service functions	UDS/DoIP conformance, session/security negative	Safety-related

SYS2 ID	Domain	System Requirement	Design Interpretation	Verification / Evidence	Safety
		shall record pre/post-event context with synchronized time.	require vehicle-state interlocks, role authorization and complete audit/freeze-frame data. Service modes cannot bypass torque/HV safety.	tests, interrupted routines and plant/service fault-injection tests.	
SYS2-VCU-044	Diagnostics/Service	Service functions that can affect torque/HV shall require explicit vehicle-safe conditions and authorization.	Diagnostic and plant/service functions require vehicle-state interlocks, role authorization and complete audit/freeze-frame data. Service modes cannot bypass torque/HV safety.	UDS/DoIP conformance, session/security negative tests, interrupted routines and plant/service fault-injection tests.	Safety-related
SYS2-VCU-045	Diagnostics/Service	Calibration download shall be authenticated, range-checked and variant-bound.	Diagnostic and plant/service functions require vehicle-state interlocks, role authorization and complete audit/freeze-frame data. Service modes cannot bypass torque/HV safety.	UDS/DoIP conformance, session/security negative tests, interrupted routines and plant/service fault-injection tests.	Safety-related
SYS2-VCU-046	Cybersecurity	Safety MCU and SoC boot chains shall verify authenticated software and prevent rollback outside approved policy.	Protect roots of trust, boot/update/calibration, diagnostics and network trust boundaries. E2E integrity and SecOC authenticity/freshness are treated as distinct mechanisms.	TARA-based penetration, secure boot/update rollback, key/certificate, replay/spoofing and compromised-cockpit tests.	Safety-related
SYS2-VCU-047	Cybersecurity	Safety command PDUs shall use SecOC authenticity/freshness where allocated by TARA.	Protect roots of trust, boot/update/calibration, diagnostics and network trust boundaries. E2E integrity and SecOC authenticity/freshness are treated as distinct mechanisms.	TARA-based penetration, secure boot/update rollback, key/certificate, replay/spoofing and compromised-cockpit tests.	Safety-related
SYS2-VCU-048	Cybersecurity	AUTOSAR E2E shall be configured independently from cryptographic SecOC.	Protect roots of trust, boot/update/calibration, diagnostics and network trust boundaries. E2E integrity and SecOC authenticity/freshness are treated as distinct mechanisms.	TARA-based penetration, secure boot/update rollback, key/certificate, replay/spoofing and compromised-cockpit tests.	Safety-related
SYS2-VCU-049	Cybersecurity	IVI/telematics shall not route arbitrary CAN commands into safety networks.	Protect roots of trust, boot/update/calibration, diagnostics and network trust boundaries. E2E integrity and SecOC authenticity/freshness are treated as distinct mechanisms.	TARA-based penetration, secure boot/update rollback, key/certificate, replay/spoofing and compromised-cockpit tests.	Safety-related
SYS2-VCU-050	Cybersecurity	Update manifests, dependencies, vehicle state and recovery shall be verified before activation.	Protect roots of trust, boot/update/calibration, diagnostics and network trust boundaries. E2E integrity and SecOC authenticity/freshness are treated as distinct mechanisms.	TARA-based penetration, secure boot/update rollback, key/certificate, replay/spoofing and compromised-cockpit tests.	Safety-related
SYS2-VCU-051	Electrical/EMC	ECU shall tolerate the released ISO 16750-2/OEM LV envelope without unsafe output.	Close power, transient, thermal, SI/PI and EMC margins on the complete central ECU. External display/audio power is separately budgeted unless intentionally included.	ISO 16750/7637 electrical tests, CISPR 25/ISO 11452 EMC, PI/SI, thermal chamber and load-step measurements.	Safety-related
SYS2-VCU-052	Electrical/EMC	Central ECU shall satisfy CISPR 25/OEM emission targets in worst CPU/GPU/network activity.	Close power, transient, thermal, SI/PI and EMC margins on the complete central ECU. External display/audio power is separately budgeted unless intentionally included.	ISO 16750/7637 electrical tests, CISPR 25/ISO 11452 EMC, PI/SI, thermal chamber and load-step measurements.	Safety-related
SYS2-VCU-053	Electrical/EMC	Central ECU shall preserve safe outputs during ISO 11452/OEM immunity tests.	Close power, transient, thermal, SI/PI and EMC margins on the complete central ECU. External display/audio power is separately budgeted unless intentionally included.	ISO 16750/7637 electrical tests, CISPR 25/ISO 11452 EMC, PI/SI, thermal chamber and load-step measurements.	Safety-related
SYS2-VCU-054	Electrical/EMC	130 W sustained-peak electronics	Close power, transient, thermal, SI/PI and EMC margins on the complete	ISO 16750/7637 electrical tests, CISPR 25/ISO 11452 EMC, PI/SI, thermal	Safety-related

SYS2 ID	Domain	System Requirement	Design Interpretation	Verification / Evidence	Safety
		thermal design shall retain supplier junction/case margin at released ambient.	central ECU. External display/audio power is separately budgeted unless intentionally included.	chamber and load-step measurements.	
SYS2-VCU-055	Electrical/EMC	High-speed DDR/PCIe/Ethernet/display interfaces and multi-rail PDN shall close SI/PI before layout release.	Close power, transient, thermal, SI/PI and EMC margins on the complete central ECU. External display/audio power is separately budgeted unless intentionally included.	ISO 16750/7637 electrical tests, CISPR 25/ISO 11452 EMC, PI/SI, thermal chamber and load-step measurements.	Safety-related
SYS2-VCU-056	Lifecycle/Quality	RA-SYS1-SYS2-SYS3-HWE/SWE-test traceability shall be bidirectional and change controlled.	Maintain serialized configuration, manufacturing evidence, PCN/EOL control, FRACAS, cybersecurity support and service/update compatibility through program life.	EOL station MSA, secure provisioning, PV audits, supplier PCN/EOL reviews, field FRACAS and post-SOP update evidence.	QM/Functional
SYS2-VCU-057	Lifecycle/Quality	Safety MCU/SoC/hypervisor/PMIC suppliers shall provide safety manuals, assumptions and defect/change notification.	Maintain serialized configuration, manufacturing evidence, PCN/EOL control, FRACAS, cybersecurity support and service/update compatibility through program life.	EOL station MSA, secure provisioning, PV audits, supplier PCN/EOL reviews, field FRACAS and post-SOP update evidence.	Safety-related
SYS2-VCU-058	Lifecycle/Quality	OTA/cyber process shall support market UNECE R155/R156 obligations where applicable.	Maintain serialized configuration, manufacturing evidence, PCN/EOL control, FRACAS, cybersecurity support and service/update compatibility through program life.	EOL station MSA, secure provisioning, PV audits, supplier PCN/EOL reviews, field FRACAS and post-SOP update evidence.	QM/Functional
SYS2-VCU-059	Lifecycle/Quality	Production shall verify secure provisioning, calibration, network IDs, power rails, safety I/O and communication.	Maintain serialized configuration, manufacturing evidence, PCN/EOL control, FRACAS, cybersecurity support and service/update compatibility through program life.	EOL station MSA, secure provisioning, PV audits, supplier PCN/EOL reviews, field FRACAS and post-SOP update evidence.	QM/Functional
SYS2-VCU-060	Lifecycle/Quality	Field safety/cyber/quality events shall feed closed-loop corrective action and calibration/software updates.	Maintain serialized configuration, manufacturing evidence, PCN/EOL control, FRACAS, cybersecurity support and service/update compatibility through program life.	EOL station MSA, secure provisioning, PV audits, supplier PCN/EOL reviews, field FRACAS and post-SOP update evidence.	QM/Functional
SYS2-VCU-061	Safety & Motion	Only the safety-controlled motion domain may emit the authoritative propulsion torque request.	Treat the requirement as an authority/containment rule. The safety MCU and local actuator ECU shall remain capable of rejecting an unsafe command even if the HPA or HMI behaves incorrectly.	HIL fault injection, timing trace, local-inverter/ESC response and safety-case evidence.	Safety-related
SYS2-VCU-062	Safety & Motion	Torque shall be checked against pedal/brake/gear/speed/vehicle state and local actuator feedback.	Treat the requirement as an authority/containment rule. The safety MCU and local actuator ECU shall remain capable of rejecting an unsafe command even if the HPA or HMI behaves incorrectly.	HIL fault injection, timing trace, local-inverter/ESC response and safety-case evidence.	Safety-related
SYS2-VCU-063	Safety & Motion	Unsafe torque deviation shall trigger torque inhibit/degrade inside the allocated internal reaction budget.	Treat the requirement as an authority/containment rule. The safety MCU and local actuator ECU shall remain capable of rejecting an unsafe command even if the HPA or HMI behaves incorrectly.	HIL fault injection, timing trace, local-inverter/ESC response and safety-case evidence.	Safety-related
SYS2-VCU-064	Safety & Motion	Independent supervision shall detect task/clock/rail	Treat the requirement as an authority/containment	HIL fault injection, timing trace, local-inverter/ESC	Safety-related

SYS2 ID	Domain	System Requirement	Design Interpretation	Verification / Evidence	Safety
		faults without relying on infotainment execution.	rule. The safety MCU and local actuator ECU shall remain capable of rejecting an unsafe command even if the HPA or HMI behaves incorrectly.	response and safety-case evidence.	
SYS2-VCU-065	Safety & Motion	Faults shall map to controlled torque-zero, limited drive, controlled stop or HV isolation according to safety analysis.	Treat the requirement as an authority/containment rule. The safety MCU and local actuator ECU shall remain capable of rejecting an unsafe command even if the HPA or HMI behaves incorrectly.	HIL fault injection, timing trace, local-inverter/ESC response and safety-case evidence.	Safety-related
SYS2-VCU-066	Driver Modes	Mode requests shall be validated by the safety domain before calibration activation.	Convert the user-visible mode request into a safety-domain accepted profile. Separate request acknowledgement from physical torque/regen/chassis blending and enforce jerk/SOP/thermal/local-ECU limits.	Mode-transition HIL and vehicle measurements covering jerk, rejected/invalid mode, thermal/SOP clamps and Track/Valet restrictions.	Safety-related
SYS2-VCU-067	Driver Modes	Mode request status shall update within 50 ms reference.	Convert the user-visible mode request into a safety-domain accepted profile. Separate request acknowledgement from physical torque/regen/chassis blending and enforce jerk/SOP/thermal/local-ECU limits.	Mode-transition HIL and vehicle measurements covering jerk, rejected/invalid mode, thermal/SOP clamps and Track/Valet restrictions.	Safety-related
SYS2-VCU-068	Driver Modes	Torque/regen maps shall cross-fade with jerk-bounded slew rather than instantaneous full-scale changes.	Convert the user-visible mode request into a safety-domain accepted profile. Separate request acknowledgement from physical torque/regen/chassis blending and enforce jerk/SOP/thermal/local-ECU limits.	Mode-transition HIL and vehicle measurements covering jerk, rejected/invalid mode, thermal/SOP clamps and Track/Valet restrictions.	Safety-related
SYS2-VCU-069	Driver Modes	Track mode shall be gated by market/vehicle/chassis/thermal permissions.	Convert the user-visible mode request into a safety-domain accepted profile. Separate request acknowledgement from physical torque/regen/chassis blending and enforce jerk/SOP/thermal/local-ECU limits.	Mode-transition HIL and vehicle measurements covering jerk, rejected/invalid mode, thermal/SOP clamps and Track/Valet restrictions.	Safety-related
SYS2-VCU-070	Driver Modes	Valet/custom restrictions shall be authenticated and shall never relax hard safety/thermal limits.	Convert the user-visible mode request into a safety-domain accepted profile. Separate request acknowledgement from physical torque/regen/chassis blending and enforce jerk/SOP/thermal/local-ECU limits.	Mode-transition HIL and vehicle measurements covering jerk, rejected/invalid mode, thermal/SOP clamps and Track/Valet restrictions.	Safety-related
SYS2-VCU-071	HV Coordination	VCU shall coordinate OFF/ACCESSORY/READY/PRECHARGE/DRIVE/CHARGE/FAULT states with BMS and inverter.	Keep pack contactor/precharge actuation in the BMS reference allocation while the VCU orchestrates vehicle READY and inverter enable. Every transition uses explicit prerequisites and feedback.	BMS/inverter HIL, welded/failed contactor simulation, precharge timeout, crash/HVIL and shutdown sequence evidence.	Safety-related
SYS2-VCU-072	HV Coordination	VCU shall verify BMS-reported precharge completion and inverter DC-link readiness before enabling propulsion.	Keep pack contactor/precharge actuation in the BMS reference allocation while the VCU orchestrates vehicle READY and inverter enable. Every transition uses explicit prerequisites and feedback.	BMS/inverter HIL, welded/failed contactor simulation, precharge timeout, crash/HVIL and shutdown sequence evidence.	Safety-related
SYS2-VCU-073	HV Coordination	Pack contactor drive remains BMS-owned unless item definition allocates otherwise.	Keep pack contactor/precharge actuation in the BMS reference allocation while the VCU orchestrates vehicle READY and	BMS/inverter HIL, welded/failed contactor simulation, precharge timeout, crash/HVIL and shutdown sequence evidence.	Safety-related

SYS2 ID	Domain	System Requirement	Design Interpretation	Verification / Evidence	Safety
			inverter enable. Every transition uses explicit prerequisites and feedback.		
SYS2-VCU-074	HV Coordination	Crash/HVIL faults shall cause a coordinated non-permissive propulsion/HV sequence.	Keep pack contactor/precharge actuation in the BMS reference allocation while the VCU orchestrates vehicle READY and inverter enable. Every transition uses explicit prerequisites and feedback.	BMS/inverter HIL, welded/failed contactor simulation, precharge timeout, crash/HVIL and shutdown sequence evidence.	Safety-related
SYS2-VCU-075	HV Coordination	Drive torque and charging states shall be mutually constrained according to vehicle architecture.	Keep pack contactor/precharge actuation in the BMS reference allocation while the VCU orchestrates vehicle READY and inverter enable. Every transition uses explicit prerequisites and feedback.	BMS/inverter HIL, welded/failed contactor simulation, precharge timeout, crash/HVIL and shutdown sequence evidence.	Safety-related
SYS2-VCU-076	Thermal & Energy	Available torque shall use the minimum of BMS, inverter, motor and thermal capability limits.	Use normalized capability/SOP limits from the BMS, inverter, motor and thermal controller rather than a hard-coded universal temperature threshold. Capability may only decrease on invalid/stale input.	Thermal chamber/HIL and dyno testing at cold/hot/SOP boundaries, sensor faults and communication timeouts.	Safety-related
SYS2-VCU-077	Thermal & Energy	Thermal derating shall be monotonic with hysteresis/recovery and supplier-calibrated thresholds.	Use normalized capability/SOP limits from the BMS, inverter, motor and thermal controller rather than a hard-coded universal temperature threshold. Capability may only decrease on invalid/stale input.	Thermal chamber/HIL and dyno testing at cold/hot/SOP boundaries, sensor faults and communication timeouts.	Safety-related
SYS2-VCU-078	Thermal & Energy	Sport/Track may request pre-chilling but may not commandeer unavailable thermal capacity.	Use normalized capability/SOP limits from the BMS, inverter, motor and thermal controller rather than a hard-coded universal temperature threshold. Capability may only decrease on invalid/stale input.	Thermal chamber/HIL and dyno testing at cold/hot/SOP boundaries, sensor faults and communication timeouts.	Safety-related
SYS2-VCU-079	Thermal & Energy	220 kW peak and 110 kW continuous are vehicle references subject to time/temperature/SOP.	Use normalized capability/SOP limits from the BMS, inverter, motor and thermal controller rather than a hard-coded universal temperature threshold. Capability may only decrease on invalid/stale input.	Thermal chamber/HIL and dyno testing at cold/hot/SOP boundaries, sensor faults and communication timeouts.	Safety-related
SYS2-VCU-080	Thermal & Energy	Accessory/cabin/traction thermal priorities shall be arbitrated by vehicle mode and safety state.	Use normalized capability/SOP limits from the BMS, inverter, motor and thermal controller rather than a hard-coded universal temperature threshold. Capability may only decrease on invalid/stale input.	Thermal chamber/HIL and dyno testing at cold/hot/SOP boundaries, sensor faults and communication timeouts.	Safety-related
SYS2-VCU-081	Chassis & Regen	VCU shall compute desired regenerative contribution and provide it to the brake/ESC authority.	The VCU calculates desired regen/chassis profile but the ESC/brake/EPS controllers retain local wheel-slip, hydraulic and steering safety authority.	Brake dyno/vehicle blend tests, ABS/ESC activation, low-friction split-mu, BMS charge-SOP loss and local-ECU refusal tests.	Safety-related
SYS2-VCU-082	Chassis & Regen	Brake pedal/ABS/ESC intervention shall immediately reduce or revoke positive propulsion request.	The VCU calculates desired regen/chassis profile but the ESC/brake/EPS controllers retain local wheel-slip, hydraulic and steering safety authority.	Brake dyno/vehicle blend tests, ABS/ESC activation, low-friction split-mu, BMS charge-SOP loss and local-ECU refusal tests.	Safety-related
SYS2-VCU-083	Chassis & Regen	Regen shall be limited by BMS charge SOP, inverter capability, motor speed and wheel-slip state.	The VCU calculates desired regen/chassis profile but the ESC/brake/EPS controllers retain local wheel-slip, hydraulic and steering safety authority.	Brake dyno/vehicle blend tests, ABS/ESC activation, low-friction split-mu, BMS charge-SOP loss and local-ECU refusal tests.	Safety-related
SYS2-VCU-084	Chassis & Regen	VCU may request EPS/ESC/suspension	The VCU calculates desired regen/chassis	Brake dyno/vehicle blend tests, ABS/ESC activation,	Safety-related

SYS2 ID	Domain	System Requirement	Design Interpretation	Verification / Evidence	Safety
		mode; local ECUs retain hard safety limits.	profile but the ESC/brake/EPS controllers retain local wheel-slip, hydraulic and steering safety authority.	low-friction split-mu, BMS charge-SOP loss and local-ECU refusal tests.	
SYS2-VCU-085	Chassis & Regen	Mode/brake blending shall target longitudinal jerk <=2.0 m/s3 reference in normal road transitions.	The VCU calculates desired regen/chassis profile but the ESC/brake/EPS controllers retain local wheel-slip, hydraulic and steering safety authority.	Brake dyno/vehicle blend tests, ABS/ESC activation, low-friction split-mu, BMS charge-SOP loss and local-ECU refusal tests.	Safety-related
SYS2-VCU-086	Central Compute	Central ECU shall include independent safety MCU plus application SoC/hypervisor.	Prove freedom from interference across safety MCU, hypervisor, HPA guests, memory, power and network resources. Authoritative motion control must survive HPA reset or overload.	WCET/load stress, memory/ECC faults, hypervisor/guest crash, PMIC/brownout and resource-exhaustion fault injection.	Safety-related
SYS2-VCU-087	Central Compute	DMA, IOMMU/MPU, interrupt and peripheral access shall prevent QM guests from writing safety actuators.	Prove freedom from interference across safety MCU, hypervisor, HPA guests, memory, power and network resources. Authoritative motion control must survive HPA reset or overload.	WCET/load stress, memory/ECC faults, hypervisor/guest crash, PMIC/brownout and resource-exhaustion fault injection.	Safety-related
SYS2-VCU-088	Central Compute	Safety processes shall use bounded/static memory; application partitions shall use bounded pools and health limits.	Prove freedom from interference across safety MCU, hypervisor, HPA guests, memory, power and network resources. Authoritative motion control must survive HPA reset or overload.	WCET/load stress, memory/ECC faults, hypervisor/guest crash, PMIC/brownout and resource-exhaustion fault injection.	Safety-related
SYS2-VCU-089	Central Compute	Safety MCU shall reach safe supervision before rich IVI boot completes.	Prove freedom from interference across safety MCU, hypervisor, HPA guests, memory, power and network resources. Authoritative motion control must survive HPA reset or overload.	WCET/load stress, memory/ECC faults, hypervisor/guest crash, PMIC/brownout and resource-exhaustion fault injection.	Safety-related
SYS2-VCU-090	Central Compute	Release shall preserve >=30% CPU/memory/network reserve in worst released mode where practicable.	Prove freedom from interference across safety MCU, hypervisor, HPA guests, memory, power and network resources. Authoritative motion control must survive HPA reset or overload.	WCET/load stress, memory/ECC faults, hypervisor/guest crash, PMIC/brownout and resource-exhaustion fault injection.	Safety-related
SYS2-VCU-091	Networking	Torque/brake/HV safety PDUs shall use deterministic CAN FD with age/counter/E2E protection.	Specify timing, validity, age, routing and security separately for safety CAN and IP/Ethernet services. Ethernet must not be a single point of failure for motion safety.	Bit-accurate CAN schedule, Ethernet VLAN/QoS/time-sync tests, gateway overload, packet loss, stale/replay and bus-off scenarios.	Safety-related
SYS2-VCU-092	Networking	1000BASE-T1 shall support DoIP/SOME-IP/cockpit services; 2.5/10G upgrade path shall be reserved.	Specify timing, validity, age, routing and security separately for safety CAN and IP/Ethernet services. Ethernet must not be a single point of failure for motion safety.	Bit-accurate CAN schedule, Ethernet VLAN/QoS/time-sync tests, gateway overload, packet loss, stale/replay and bus-off scenarios.	Safety-related
SYS2-VCU-093	Networking	Critical Ethernet traffic shall use time sync and traffic-class controls when sharing links with infotainment/OTA.	Specify timing, validity, age, routing and security separately for safety CAN and IP/Ethernet services. Ethernet must not be a single point of failure for motion safety.	Bit-accurate CAN schedule, Ethernet VLAN/QoS/time-sync tests, gateway overload, packet loss, stale/replay and bus-off scenarios.	Safety-related
SYS2-VCU-094	Networking	Safety CAN nominal sustained utilization shall remain <=40% reference and fault-burst <=60%.	Specify timing, validity, age, routing and security separately for safety CAN and IP/Ethernet services. Ethernet must not be a single point of failure for motion safety.	Bit-accurate CAN schedule, Ethernet VLAN/QoS/time-sync tests, gateway overload, packet loss, stale/replay and bus-off scenarios.	Safety-related
SYS2-VCU-095	Networking	CAN partial networking and Ethernet TC10-class sleep/wake shall support low-power states.	Specify timing, validity, age, routing and security separately for safety CAN and IP/Ethernet services. Ethernet must not be a single point of failure for motion safety.	Bit-accurate CAN schedule, Ethernet VLAN/QoS/time-sync tests, gateway overload, packet loss, stale/replay and bus-off scenarios.	Safety-related
SYS2-VCU-096	Cockpit/HMI	Android/IVI mode selection is a request only; safety MCU publishes authoritative accepted mode.	Treat rich HMI as a presentation/request layer. Safety-relevant telltales and mode status require monitored source integrity	HMI latency, display reset, lost acknowledgement, telltale fallback, physical control and driver-mode usability tests.	Safety-related

SYS2 ID	Domain	System Requirement	Design Interpretation	Verification / Evidence	Safety
			and a defined fallback if Android/display rendering fails.		
SYS2-VCU-097	Cockpit/HMI	Critical telltales shall have an independent monitored path capable of indicating safety state if Android stalls.	Treat rich HMI as a presentation/request layer. Safety-relevant telltales and mode status require monitored source integrity and a defined fallback if Android/display rendering fails.	HMI latency, display reset, lost acknowledgement, telltale fallback, physical control and driver-mode usability tests.	Safety-related
SYS2-VCU-098	Cockpit/HMI	Driver shall receive accepted/rejected mode and reason within 100 ms reference.	Treat rich HMI as a presentation/request layer. Safety-relevant telltales and mode status require monitored source integrity and a defined fallback if Android/display rendering fails.	HMI latency, display reset, lost acknowledgement, telltale fallback, physical control and driver-mode usability tests.	QM/Functional
SYS2-VCU-099	Cockpit/HMI	High-rate display pixels shall use dedicated display interface/SerDes rather than consuming vehicle 1G Ethernet.	Treat rich HMI as a presentation/request layer. Safety-relevant telltales and mode status require monitored source integrity and a defined fallback if Android/display rendering fails.	HMI latency, display reset, lost acknowledgement, telltale fallback, physical control and driver-mode usability tests.	QM/Functional
SYS2-VCU-100	Cockpit/HMI	Track/custom features shall follow market HMI/distraction and homologation rules.	Treat rich HMI as a presentation/request layer. Safety-relevant telltales and mode status require monitored source integrity and a defined fallback if Android/display rendering fails.	HMI latency, display reset, lost acknowledgement, telltale fallback, physical control and driver-mode usability tests.	QM/Functional

Appendix D - HARA Safeguard and Validation Rationale

HARA ID	Malfunction	Effect	Prelim ASIL	SG	Derived Safeguard	Validation Intent
HARA-VCU-001	Positive torque without pedal	Unexpected acceleration	D	SG-01	Independent torque supervision, plausibility, bounded communication and inverter local safety.	HIL/vehicle fault injection in the stated operational situation; confirm safe/degraded reaction and driver information.
HARA-VCU-002	Positive torque while brake applied	Failure to decelerate	D	SG-02	Degraded drive, driver warning and controlled stop strategy.	HIL/vehicle fault injection in the stated operational situation; confirm safe/degraded reaction and driver information.
HARA-VCU-003	Torque sign inversion	Unexpected acceleration/deceleration	D	SG-03	Brake authority, regen clamp and independent plausibility.	HIL/vehicle fault injection in the stated operational situation; confirm safe/degraded reaction and driver information.
HARA-VCU-004	Torque request stuck high	Sustained unintended acceleration	D	SG-04	BMS-owned contactors, VCU coordination, dual permissives.	HIL/vehicle fault injection in the stated operational situation; confirm safe/degraded reaction and driver information.
HARA-VCU-005	Torque request stale	Incorrect vehicle response	C	SG-05	Safety-domain validation and signed calibration.	HIL/vehicle fault injection in the stated operational situation; confirm safe/degraded reaction and driver information.
HARA-VCU-006	Torque lost during overtake	Loss of propulsion	C	SG-06	Independent safety MCU, MPU/IOMMU, watchdog and gateway filtering.	HIL/vehicle fault injection in the stated operational situation; confirm safe/degraded reaction and driver information.
HARA-VCU-007	Regen excessive	Unexpected deceleration	C	SG-07	Independent monitored telltale path and degraded indication.	HIL/vehicle fault injection in the stated operational situation; confirm safe/degraded reaction and driver information.
HARA-VCU-008	Regen absent on one-pedal transition	Longer stopping distance	C	SG-08	Minimum capability arbitration and monotonic derating.	HIL/vehicle fault injection in the stated operational situation; confirm safe/degraded reaction and driver information.
HARA-VCU-009	Regen during ABS	Wheel instability	C	SG-09	Age/E2E/SecOC/ timeout non-	HIL/vehicle fault injection in the stated

HARA ID	Malfunction	Effect	Prelim ASIL	SG	Derived Safeguard	Validation Intent
					permissive behavior.	operational situation; confirm safe/degraded reaction and driver information.
HARA-VCU-010	Track mode falsely enabled	Higher-than-intended response	B	SG-10	Secure boot/update, HSM, signed calibration and safety re-validation.	HIL/vehicle fault injection in the stated operational situation; confirm safe/degraded reaction and driver information.
HARA-VCU-011	Valet cap bypassed	Unauthorized performance	B	SG-11	Independent PMIC/watchdog, ECC, thermal and brownout safe state.	HIL/vehicle fault injection in the stated operational situation; confirm safe/degraded reaction and driver information.
HARA-VCU-012	Mode oscillation	Jerk/loss of control	C	SG-12	Secure EOL binding, topology/variant checks and commissioning tests.	HIL/vehicle fault injection in the stated operational situation; confirm safe/degraded reaction and driver information.
HARA-VCU-013	BMS SOP ignored	Battery overcurrent/voltage	C	SG-01	Independent torque supervision, plausibility, bounded communication and inverter local safety.	HIL/vehicle fault injection in the stated operational situation; confirm safe/degraded reaction and driver information.
HARA-VCU-014	Inverter limit ignored	Power-stage protection trip	C	SG-02	Degraded drive, driver warning and controlled stop strategy.	HIL/vehicle fault injection in the stated operational situation; confirm safe/degraded reaction and driver information.
HARA-VCU-015	Thermal derate delayed	Component overtemperature	C	SG-03	Brake authority, regen clamp and independent plausibility.	HIL/vehicle fault injection in the stated operational situation; confirm safe/degraded reaction and driver information.
HARA-VCU-016	HV ready falsely asserted	Drive enabled before HV ready	D	SG-04	BMS-owned contactors, VCU coordination, dual permissives.	HIL/vehicle fault injection in the stated operational situation; confirm safe/degraded reaction and driver information.
HARA-VCU-017	Precharge timeout ignored	Contactor/DC-link stress	C	SG-05	Safety-domain validation and signed calibration.	HIL/vehicle fault injection in the stated operational situation; confirm safe/degraded reaction and driver information.
HARA-VCU-018	Crash/HVIL ignored	HV hazard after crash	D	SG-06	Independent safety MCU, MPU/IOMMU, watchdog and gateway filtering.	HIL/vehicle fault injection in the stated operational situation; confirm safe/degraded reaction and driver information.
HARA-VCU-019	Android writes safety peripheral	Unsafe command	C	SG-07	Independent monitored telltale path and degraded indication.	HIL/vehicle fault injection in the stated operational situation; confirm safe/degraded reaction and driver information.
HARA-VCU-020	Hypervisor isolation failure	QM-to-safety interference	C	SG-08	Minimum capability arbitration and monotonic derating.	HIL/vehicle fault injection in the stated operational situation; confirm safe/degraded reaction and driver information.
HARA-VCU-021	Safety MCU watchdog failure	Latent dangerous control fault	C	SG-09	Age/E2E/SecOC/ timeout non-permissive behavior.	HIL/vehicle fault injection in the stated operational situation; confirm safe/degraded reaction and driver information.
HARA-VCU-022	CAN torque spoof	Unauthorized torque	D	SG-10	Secure boot/update, HSM, signed calibration and safety re-validation.	HIL/vehicle fault injection in the stated operational situation; confirm safe/degraded reaction and driver information.
HARA-VCU-023	CAN frame replay	Stale authority reused	C	SG-11	Independent PMIC/watchdog, ECC, thermal and brownout safe state.	HIL/vehicle fault injection in the stated operational situation; confirm safe/degraded reaction and driver information.
HARA-VCU-024	Ethernet gateway compromise	Lateral movement into safety network	C	SG-12	Secure EOL binding, topology/variant checks and commissioning tests.	HIL/vehicle fault injection in the stated operational situation; confirm safe/degraded

HARA ID	Malfunction	Effect	Prelim ASIL	SG	Derived Safeguard	Validation Intent
						reaction and driver information.
HARA-VCU-025	HMI mode accepted without confirmation	Driver-system state mismatch	B	SG-01	Independent torque supervision, plausibility, bounded communication and inverter local safety.	HIL/vehicle fault injection in the stated operational situation; confirm safe/degraded reaction and driver information.
HARA-VCU-026	Safety telltale lost	Driver unaware of degraded safety state	B	SG-02	Degraded drive, driver warning and controlled stop strategy.	HIL/vehicle fault injection in the stated operational situation; confirm safe/degraded reaction and driver information.
HARA-VCU-027	Cluster freeze during fault	Missing warning	B	SG-03	Brake authority, regen clamp and independent plausibility.	HIL/vehicle fault injection in the stated operational situation; confirm safe/degraded reaction and driver information.
HARA-VCU-028	Calibration variant mismatch	Wrong torque/thermal limits	C	SG-04	BMS-owned contactors, VCU coordination, dual permissives.	HIL/vehicle fault injection in the stated operational situation; confirm safe/degraded reaction and driver information.
HARA-VCU-029	OTA partial update	Incompatible control software	C	SG-05	Safety-domain validation and signed calibration.	HIL/vehicle fault injection in the stated operational situation; confirm safe/degraded reaction and driver information.
HARA-VCU-030	Rollback to vulnerable image	Cyber-safety exposure	C	SG-06	Independent safety MCU, MPU/IOMMU, watchdog and gateway filtering.	HIL/vehicle fault injection in the stated operational situation; confirm safe/degraded reaction and driver information.
HARA-VCU-031	12V brownout causes MCU reset	Loss of control authority	C	SG-07	Independent monitored telltale path and degraded indication.	HIL/vehicle fault injection in the stated operational situation; confirm safe/degraded reaction and driver information.
HARA-VCU-032	PMIC false safe state	Unexpected torque inhibit/enable	C	SG-08	Minimum capability arbitration and monotonic derating.	HIL/vehicle fault injection in the stated operational situation; confirm safe/degraded reaction and driver information.
HARA-VCU-033	SoC thermal shutdown	Cockpit/service loss with potential shared-resource effect	C	SG-09	Age/E2E/SecOC/ timeout non-permissive behavior.	HIL/vehicle fault injection in the stated operational situation; confirm safe/degraded reaction and driver information.
HARA-VCU-034	Memory corruption in torque tables	Incorrect torque	C	SG-10	Secure boot/update, HSM, signed calibration and safety re-validation.	HIL/vehicle fault injection in the stated operational situation; confirm safe/degraded reaction and driver information.
HARA-VCU-035	Timebase drift	Stale/misaligned data	C	SG-11	Independent PMIC/watchdog, ECC, thermal and brownout safe state.	HIL/vehicle fault injection in the stated operational situation; confirm safe/degraded reaction and driver information.
HARA-VCU-036	Sensor pedal plausibility fault	Incorrect driver demand	C	SG-12	Secure EOL binding, topology/variant checks and commissioning tests.	HIL/vehicle fault injection in the stated operational situation; confirm safe/degraded reaction and driver information.
HARA-VCU-037	ESC override not honored	Traction instability	D	SG-01	Independent torque supervision, plausibility, bounded communication and inverter local safety.	HIL/vehicle fault injection in the stated operational situation; confirm safe/degraded reaction and driver information.
HARA-VCU-038	Brake ECU comm loss	Regen blend unsafe	C	SG-02	Degraded drive, driver warning and controlled stop strategy.	HIL/vehicle fault injection in the stated operational situation; confirm safe/degraded reaction and driver information.
HARA-VCU-039	EOL secure provisioning error	Wrong keys/configuration	C	SG-03	Brake authority, regen clamp and independent plausibility.	HIL/vehicle fault injection in the stated operational situation; confirm safe/degraded reaction and driver

HARA ID	Malfunction	Effect	Prelim ASIL	SG	Derived Safeguard	Validation Intent
						information.
HARA-VCU-040	Service routine activates in drive	Unexpected actuation	C	SG-04	BMS-owned contactors, VCU coordination, dual permissives.	HIL/vehicle fault injection in the stated operational situation; confirm safe/degraded reaction and driver information.

Appendix E - Interface Failure Handling Matrix

ICD	Interface	Transport	Rate / Latency	Failure / Timeout Policy	Integrity / Security	Verification
ICD-VCU-001	Accelerator pedal	Analog/SENT/PWM	1 ms acquisition	On stale, invalid, missing or implausible safety data, the VCU shall enter the interface-specific non-permissive/degraded behavior; no communication failure may increase torque, regen, HV or Track-mode authority.	Plausibility / range / E2E internal	Bench/HIL/network/ vehicle verification
ICD-VCU-002	Brake/ESC -> VCU	CAN FD A	5-10 ms	On stale, invalid, missing or implausible safety data, the VCU shall enter the interface-specific non-permissive/degraded behavior; no communication failure may increase torque, regen, HV or Track-mode authority.	E2E + timeout; SecOC if TARA	Bench/HIL/network/ vehicle verification
ICD-VCU-003	VCU -> Inverter	CAN FD A	10 ms	On stale, invalid, missing or implausible safety data, the VCU shall enter the interface-specific non-permissive/degraded behavior; no communication failure may increase torque, regen, HV or Track-mode authority.	E2E + SecOC + age	Bench/HIL/network/ vehicle verification
ICD-VCU-004	Inverter -> VCU	CAN FD A	10 ms	On stale, invalid, missing or implausible safety data, the VCU shall enter the interface-specific non-permissive/degraded behavior; no communication failure may increase torque, regen, HV or Track-mode authority.	E2E + timeout	Bench/HIL/network/ vehicle verification
ICD-VCU-005	BMS -> VCU	CAN FD B	10-20 ms	On stale, invalid, missing or implausible safety data, the VCU shall enter the interface-specific non-permissive/degraded behavior; no communication failure may increase torque, regen, HV or Track-mode authority.	E2E + timeout; SecOC if TARA	Bench/HIL/network/ vehicle verification
ICD-VCU-006	VCU -> BMS	CAN FD B	20 ms	On stale, invalid, missing or implausible safety data, the VCU shall enter the interface-specific non-permissive/degraded behavior; no communication failure may increase torque, regen, HV or Track-mode authority.	E2E + SecOC	Bench/HIL/network/ vehicle verification
ICD-VCU-007	ESC -> VCU	CAN FD A	10 ms	On stale, invalid, missing or implausible safety data, the VCU shall enter the interface-specific non-permissive/degraded behavior; no communication failure may increase torque, regen, HV or Track-mode authority.	E2E + age	Bench/HIL/network/ vehicle verification

ICD	Interface	Transport	Rate / Latency	Failure / Timeout Policy	Integrity / Security	Verification
ICD-VCU-008	VCU -> ESC/Brake	CAN FD A	10 ms	On stale, invalid, missing or implausible safety data, the VCU shall enter the interface-specific non-permissive/degraded behavior; no communication failure may increase torque, regen, HV or Track-mode authority.	E2E + SecOC if allocated	Bench/HIL/network/ vehicle verification
ICD-VCU-009	EPS <-> VCU	CAN FD chassis	10-20 ms	On stale, invalid, missing or implausible safety data, the VCU shall enter the interface-specific non-permissive/degraded behavior; no communication failure may increase torque, regen, HV or Track-mode authority.	E2E + age	Bench/HIL/network/ vehicle verification
ICD-VCU-010	Suspension <-> VCU	CAN FD chassis	20 ms	On stale, invalid, missing or implausible safety data, the VCU shall enter the interface-specific non-permissive/degraded behavior; no communication failure may increase torque, regen, HV or Track-mode authority.	E2E	Bench/HIL/network/ vehicle verification
ICD-VCU-011	Thermal controller	CAN FD / Ethernet	20-100 ms	On stale, invalid, missing or implausible safety data, the VCU shall enter the interface-specific non-permissive/degraded behavior; no communication failure may increase torque, regen, HV or Track-mode authority.	E2E/timeout	Bench/HIL/network/ vehicle verification
ICD-VCU-012	HVIL/crash	Hardwired + CAN redundant	Immediate + CAN confirm	On stale, invalid, missing or implausible safety data, the VCU shall enter the interface-specific non-permissive/degraded behavior; no communication failure may increase torque, regen, HV or Track-mode authority.	Independent hardwired safety path where allocated	Bench/HIL/network/ vehicle verification
ICD-VCU-013	HMI driver-mode request	IPC/Ethernet SOME/IP	Event	On stale, invalid, missing or implausible safety data, the VCU shall enter the interface-specific non-permissive/degraded behavior; no communication failure may increase torque, regen, HV or Track-mode authority.	Authenticated local service + bounded enum	Bench/HIL/network/ vehicle verification
ICD-VCU-014	VCU authoritative mode -> HMI	IPC/Ethernet SOME/IP	<=50 ms	On stale, invalid, missing or implausible safety data, the VCU shall enter the interface-specific non-permissive/degraded behavior; no communication failure may increase torque, regen, HV or Track-mode authority.	Typed IPC/service version	Bench/HIL/network/ vehicle verification
ICD-VCU-015	Safety telltales	Safety MCU -> display safety path	<=100 ms	On stale, invalid, missing or implausible safety data, the VCU shall enter the interface-specific non-permissive/degraded behavior; no communication failure may increase torque, regen, HV or Track-mode authority.	Independent monitored channel	Bench/HIL/network/ vehicle verification
ICD-VCU-016	DoIP tester	1000BASE-T1 Ethernet	On demand	On stale, invalid, missing or implausible safety data, the VCU	DoIP + authenticated diagnostics	Bench/HIL/network/ vehicle verification

ICD	Interface	Transport	Rate / Latency	Failure / Timeout Policy	Integrity / Security	Verification
				shall enter the interface-specific non-permissive/degraded behavior; no communication failure may increase torque, regen, HV or Track-mode authority.		
ICD-VCU-017	OTA client	Ethernet/cellular gateway	Background / parked activation	On stale, invalid, missing or implausible safety data, the VCU shall enter the interface-specific non-permissive/degraded behavior; no communication failure may increase torque, regen, HV or Track-mode authority.	TLS/backend + signed package	Bench/HIL/network/ vehicle verification
ICD-VCU-018	Flight recorder	Internal IPC/storage	1-10 ms samples/event	On stale, invalid, missing or implausible safety data, the VCU shall enter the interface-specific non-permissive/degraded behavior; no communication failure may increase torque, regen, HV or Track-mode authority.	CRC + monotonic time	Bench/HIL/network/ vehicle verification
ICD-VCU-019	PTP/gPTP	Ethernet 802.1AS	Continuous	On stale, invalid, missing or implausible safety data, the VCU shall enter the interface-specific non-permissive/degraded behavior; no communication failure may increase torque, regen, HV or Track-mode authority.	Clock quality / domain control	Bench/HIL/network/ vehicle verification
ICD-VCU-020	Ethernet backbone	1000BASE-T1 baseline	Continuous	On stale, invalid, missing or implausible safety data, the VCU shall enter the interface-specific non-permissive/degraded behavior; no communication failure may increase torque, regen, HV or Track-mode authority.	VLAN/QoS/MACsec optional	Bench/HIL/network/ vehicle verification
ICD-VCU-021	Ethernet high-speed uplink	2.5/10GBASE-T1 option	Continuous	On stale, invalid, missing or implausible safety data, the VCU shall enter the interface-specific non-permissive/degraded behavior; no communication failure may increase torque, regen, HV or Track-mode authority.	Traffic matrix / TSN	Bench/HIL/network/ vehicle verification
ICD-VCU-022	Display link	DP/eDP/automotive SerDes	Frame rate	On stale, invalid, missing or implausible safety data, the VCU shall enter the interface-specific non-permissive/degraded behavior; no communication failure may increase torque, regen, HV or Track-mode authority.	Link CRC/monitoring for safety display path	Bench/HIL/network/ vehicle verification
ICD-VCU-023	Audio/haptic alert	I2S/TDM/IPC + actuator	Event	On stale, invalid, missing or implausible safety data, the VCU shall enter the interface-specific non-permissive/degraded behavior; no communication failure may increase torque, regen, HV or Track-mode authority.	Ack/monitor where safety allocated	Bench/HIL/network/ vehicle verification
ICD-VCU-024	Physical mode switch	Discrete/LIN/CAN	Event	On stale, invalid, missing or implausible safety data, the VCU shall enter the interface-specific non-permissive/degraded	Debounce/plausibility	Bench/HIL/network/ vehicle verification

ICD	Interface	Transport	Rate / Latency	Failure / Timeout Policy	Integrity / Security	Verification
				behavior; no communication failure may increase torque, regen, HV or Track-mode authority.		
ICD-VCU-025	Ignition/wake	Discrete/CAN PN/Ethernet TC10	Event	On stale, invalid, missing or implausible safety data, the VCU shall enter the interface-specific non-permissive/degraded behavior; no communication failure may increase torque, regen, HV or Track-mode authority.	Authorized source + debounce	Bench/HIL/network/ vehicle verification
ICD-VCU-026	PMIC/safety watchdog	SPI/hardwired	Sub-ms to ms	On stale, invalid, missing or implausible safety data, the VCU shall enter the interface-specific non-permissive/degraded behavior; no communication failure may increase torque, regen, HV or Track-mode authority.	Independent safety path	Bench/HIL/network/ vehicle verification
ICD-VCU-027	Secure element/HSM	Internal bus	On demand	On stale, invalid, missing or implausible safety data, the VCU shall enter the interface-specific non-permissive/degraded behavior; no communication failure may increase torque, regen, HV or Track-mode authority.	Privilege/secure storage	Bench/HIL/network/ vehicle verification
ICD-VCU-028	Manufacturing tool	DoIP/debug manufacturing channel	EOL only	On stale, invalid, missing or implausible safety data, the VCU shall enter the interface-specific non-permissive/degraded behavior; no communication failure may increase torque, regen, HV or Track-mode authority.	Authenticated station + signed manifest	Bench/HIL/network/ vehicle verification
ICD-VCU-029	Cloud/fleet telemetry	Telematics gateway	Seconds/minutes	On stale, invalid, missing or implausible safety data, the VCU shall enter the interface-specific non-permissive/degraded behavior; no communication failure may increase torque, regen, HV or Track-mode authority.	Privacy/ authentication; no direct torque	Bench/HIL/network/ vehicle verification
ICD-VCU-030	Charger/OBC state	CAN FD	20-100 ms	On stale, invalid, missing or implausible safety data, the VCU shall enter the interface-specific non-permissive/degraded behavior; no communication failure may increase torque, regen, HV or Track-mode authority.	E2E/timeout	Bench/HIL/network/ vehicle verification

Appendix F - Safety/Cyber Change Impact Triggers

Trigger	Change	Mandatory Re-Assessment
CI-01	Safety MCU or PMIC part change	Re-open MCU/PMIC safety assumptions, FMEDA/WCCA, timing, MCAL and fault-injection evidence.
CI-02	HPA SoC/hypervisor/OS major version	Re-run resource isolation, boot, memory, device assignment, cybersecurity and HPA-to-safety interface tests.
CI-03	BMS/inverter/ESC ICD change	Re-run torque/SOP/regen/HV timing, timeout, fault and degraded-state analysis.
CI-04	Driver-mode calibration increase	Re-run HARA impact, jerk/traction/thermal/SOP validation and market/Track-mode restrictions.
CI-05	CAN/Ethernet topology or bitrate	Re-run worst-case schedule, gateway latency, bus-off, VLAN/QoS and security analysis.

Trigger	Change	Mandatory Re-Assessment
CI-06	New OTA/update dependency	Re-run compatibility, rollback, vehicle-state activation and cybersecurity/update evidence.
CI-07	Power/thermal enclosure change	Re-run WCCA, transient, PI, thermal and EMC tests at maximum compute/display load.
CI-08	Manufacturing/service process change	Re-run station MSA, secure provisioning, variant binding, diagnostic permission and EOL proof tests.

15. Rev A2 System-Safety Gap Closure and HARA Addendum

The Rev A2 HARA adds hazards created specifically by central-compute consolidation: HPA/Android interference with motion authority, safety-domain brownout, cryptographic/network latency, Track-mode eligibility failure, critical-telltale loss, shared-memory contention, incompatible OTA/calibration and drive enable during charge/service states.

HARA ID	Malfunction	Potential Effect	Operational Situation	S	E	C	ASIL	Safety Goal
HARA-VCU-041	QM/HPA fault corrupts torque request or safety IPC	Unintended positive torque / hazardous acceleration	Driving with Android/HPA active	S3	E4	C3	D	SG-01
HARA-VCU-042	Safety MCU rail collapses during HPA load transient/cold crank	Loss of authoritative torque/HV supervision	Drive during severe LV event	S3	E3	C3	D	SG-01
HARA-VCU-043	CAN security processing causes stale torque command	Delayed/incorrect torque response	High load / security event	S3	E3	C2	C	SG-02
HARA-VCU-044	Track mode activated outside approved prerequisites	Excessive response / stability margin reduction	Public-road / degraded thermal/tyre state	S3	E3	C2	C	SG-05
HARA-VCU-045	Critical telltale lost after HPA/guest failure	Driver not informed of HV/brake/torque safety fault	Active driving	S3	E3	C2	C	SG-07
HARA-VCU-046	HPA memory/interconnect starvation delays gateway/service	Stale capability/state data influences motion request	Dense cockpit/network load	S3	E3	C2	C	SG-06
HARA-VCU-047	OTA/configuration activates incompatible calibration	Unsafe torque/mode/HV behavior	Post-update drive cycle	S3	E3	C3	D	SG-10
HARA-VCU-048	Charge/service state incorrectly permits drive torque	Vehicle moves while connected/serviced	Charging/service workshop	S3	E2	C3	C	SG-04

Safety-goal alignment. Rev A2 resolves the earlier torque-integrity ambiguity: SG-01 is an ASIL-D target; driver-mode hazards map to SG-05, infotainment/FFI hazards to SG-06, critical telltales to SG-07, network freshness to SG-09, update/configuration integrity to SG-10 and LV/thermal/platform power failures to SG-11.

16. Torque Safety Integrity, Decomposition and Safe-State Rationale

The VCU safety concept now separates the nominal vehicle-control feature set from the authoritative safety mechanisms required to prevent unintended torque. The safety MCU is the only central-compute element allowed to authorize propulsion torque after safety checks; HPA requests are advisory/untrusted.

Safety Function	Rev A2 Requirement	Primary Mechanism	Independent Evidence
Driver demand	ASIL-D target path for SG-01	Dual pedal plausibility, brake override, gear/HV state and bounded torque maps	Sensor FI + HIL + WCET
Torque monitor	Independent/different computation or decomposed safety mechanism	Command vs expected/delivered torque, inverter current/speed/acceleration plausibility	Dependent-failure analysis + dyno/HIL
Torque inhibit	Independent hardware/discrete or equivalent safety path	Safety MCU/PMIC -> inverter inhibit / local inverter limiter	Reaction-time fault injection
Local actuator protection	Always retained	Inverter overcurrent/SC/torque limits, ESC/brake wheel-slip authority	Supplier safety manual + vehicle integration
Safe state	Operational-context dependent	Immediate inhibit for unintended positive torque; controlled ramp/limp for selected loss-of-function faults	Scenario-based FSC/TSC validation

ISO 26262 decomposition credit is only taken when the independence and dependent-failure conditions are demonstrated. Two software tasks on one MCU/common data path are not automatically treated as independent safety channels.

17. Network Integrity, SecOC Schedulability and Cross-Domain Trust Boundary

Rev A2 rejects the idea that SecOC must always be removed from 10 ms torque messages. Message authenticity is threat-driven. Where the TARA requires it, the safety PDU uses both E2E and SecOC and the PDU/authenticator/freshness design is optimized so the deadline and CAN utilization still close.

Flow / Network	Protection	Rev A2 Rule	Rationale
Torque command 10 ms CAN FD	E2E mandatory; SecOC if TARA requires	HSM-accelerated auth/freshness; TARA-selected truncation; deadline/bus-load analysis	Do not blanket remove SecOC
Inverter status 10 ms CAN FD	E2E mandatory; authenticity per TARA	Age/counter/CRC; delivered torque/current plausibility	Safety MCU consumes directly
BMS SOP 10-20 ms CAN FD	E2E mandatory; authenticity per TARA	Stale/invalid can only reduce authority	No HPA bypass
ESC/brake 10 ms CAN FD	E2E mandatory; authenticity per TARA	ESC/ABS override has priority	Dual-bus where safety allocation requires
Ethernet service backbone	VLAN/ACL/rate limit + MACsec where TARA allocates	1GBASE-T1 baseline; 2.5/10G upgrade branch	Not sole torque-safety path
HPA->Safety MCU IPC	Whitelisted mailbox / safe IPC	Schema, range, counter, rate limit, freshness; no raw pointer/MMIO sharing	All HPA requests untrusted
Time sync	IEEE 802.1AS-2025 program reference for Ethernet time	Time quality + monotonicity; torque loop not dependent on Ethernet time	Flight recorder/service correlation
CAN roadmap	ISO11898-1:2024 compatible strategy	CAN FD baseline; CAN XL future branch assessed at next platform freeze	Do not mix without gateway policy
Interface	**Transport**	**Key Rev A2 Contract**	**Verification**
HPA -> Safety MCU	Safe IPC/mailbox	Schema, length, range, age, counter, rate limit; no direct actuator write	Malformed/flood/DMA test
Safety MCU -> inverter	CAN FD + hard inhibit	10 ms torque request, E2E; SecOC when TARA allocates; independent inhibit	CANoe/HIL/dyno
Safety MCU -> critical HMI	Safety warning path	Independent source-of-truth and bounded fallback presentation	Display/guest fault injection
VCU <-> charge state	CAN/Ethernet service	Drive inhibit while incompatible charging/service state is valid	Charge/drive HIL

18. Freedom-from-Interference, Common-Cause Failure and Cockpit Safety

A central ECU increases packaging efficiency but also creates new common-cause risks. Rev A2 explicitly analyzes shared enclosure, power connector, thermal event, water ingress, reset/clock, DDR/interconnect, Ethernet switch and EMC roots rather than assuming hypervisor isolation alone is sufficient.

Resource	Rev A2 Budget	Design Rule	Verification
Safety MCU CPU	<=70% worst-case utilization	Leave >=30% WCET/interrupt margin at max CAN/fault load	WCET + stress HIL
Safety MCU RAM/Flash	<=75% allocated	Keep diagnostic/update margin; no runtime heap in critical loops	Link map + runtime monitor
HPA CPU/GPU/NPU	<=75% sustained released workload	Reserve for UI spikes/update/background isolation	Target workload stress
HPA memory bandwidth	<=65% sustained released workload	Reserve >=35% to avoid DDR contention under multi-display/network bursts	SoC performance counters
HPA RAM	<=80% normal high-watermark	Bound each VM/cgroup; safety-support services preallocate bounded pools	Long-duration stress
DMA / IOMMU	Explicit allow-list	QM guest DMA cannot reach safety memory/MMIO	IOMMU fault injection
Cache / interconnect QoS	Priority/way/bandwidth partitioning where silicon supports	QM workload cannot starve safety-support/control IPC	Contention benchmark
Interrupt ownership	Static routing	Safety interrupts cannot be masked by Android guest	Fault injection
Power / reset	Independent roots	HPA reset/brownout does not reset safety MCU/PMIC/CAN	Brownout/reset test
Clock	Independent safety clock supervision	HPA PLL/clock fault cannot remove safety monitor timing	Clock fault injection

18.1 Critical telltale / warning boundary

Safety- or regulation-critical warnings cannot depend solely on a QM Android rendering chain. The system shall use a qualified safety-rendering mechanism, monitored safety overlay, independent telltale, or equivalent fallback according to the warning criticality and vehicle safety concept.

Cockpit Failure	Safety Concern	Rev A2 Response
Android guest crash	Loss of normal cluster/center display	Safety MCU retains motion control; fallback warning/telltale path indicates critical states
Display-link failure	Critical warning not visible	Detect link/display health; use independent redundant indicator or alternate display path as allocated
GPU/DDR overload	Late HMI while motion control healthy	Shed animations/IVI/background tasks; preserve warning service and safety gateway
Malicious HMI mode request	Unsafe Sport/Track/Valet change	HMI request is untrusted; safety MCU eligibility and signed calibration control actual state

19. Rev A2 TARA, OTA and Data-Governance Addendum

The TARA is expanded to cover the high-value cross-domain attack paths introduced by centralized cockpit consolidation: torque PDU spoofing, driver-mode policy abuse, HPA-to-safety IPC, software-supply-chain compromise, production debug, time manipulation, valet/geofence data, hypervisor management and cloud command misuse.

TARA ID	Asset	Threat Scenario	Impact	Risk	Key Controls	Owner
TARA-VCU-031	Torque PDU	Spoofed/authenticated-looking torque command from compromised gateway/IVI path	Hazardous acceleration	High	Default-deny route, safety gateway validation, E2E, TARA-allocated SecOC, inverter plausibility	Cyber/FuSa
TARA-VCU-032	Driver mode	Unauthorized Track/boost activation via HMI/profile manipulation	Unsafe dynamic response	High	Signed mode calibration, safety-MCU eligibility, owner/market policy	Cyber/Vehicle Dynamics
TARA-VCU-033	Safety IPC	Malicious or malformed HPA-to-MCU mailbox payload	Safety calculation corruption	High	Schema allow-list, range/rate/counter/age checks, memory isolation	Cyber/FuSa
TARA-VCU-034	OTA/SBOM	Compromised dependency or supplier package	Persistent malicious/unsafe ECU software	High	Signed SBOM/dependencies, PKI, anti-rollback, staged validation	Cyber/OTA
TARA-VCU-035	Debug port	Unauthorized production JTAG/SWD access	Key extraction/code modification	High	Disable/authenticate debug, tamper/audit	Cyber/HW
TARA-VCU-036	Time sync	Manipulated Ethernet time affects logs/service decisions	Forensic ambiguity / service error	Medium	Time quality, monotonic local safety time, no torque dependency	Cyber/Network
TARA-VCU-037	Valet/geofence profile	Owner/profile data theft or malicious cap change	Privacy / unsafe feature policy	Medium-High	Encrypted profiles, signed policy, RBAC	Cyber/Privacy
TARA-VCU-038	Hypervisor management	Guest escape / privileged service compromise	Cross-domain access	High	Safety architecture outside HPA, IOMMU, secure hypervisor config, minimal management plane	Cyber/Platform
TARA-VCU-039	Fleet telemetry	Cloud command channel attempts motion-authority path	Remote control risk	High	No cloud direct motion control; gateway separation; authenticated diagnostics	Cyber
TARA-VCU-040	Certificates/keys	Expired or weak lifecycle crypto blocks update/communication	Availability/security degradation	Medium	Crypto agility, rotation, expiry monitoring	Cyber/Lifecycle

19.1 OTA / SBOM / recovery

Function	Requirement	Implementation	Boundary
Secure boot	HSM roots on Safety MCU and HPA; measured/verified chain	Anti-rollback, signed boot/config	No unsigned recovery image
OTA	A/B or equivalent atomic strategy with signed dependency manifest	Update only in allowed vehicle state; power-loss recovery	ISO24089 / R156 process
SBOM	Machine-readable ECU software/component inventory	Link version -> vulnerability status -> affected VIN/variant	Supply-chain governance
Crypto agility	Algorithm/key-length abstraction and certificate rotation	Avoid hardcoded lifecycle crypto dead-end	OEM policy controls algorithms
Debug	Production debug disabled or authenticated	Physical service policy and tamper logging	No generic unlock
SecOC	TARA-driven on safety/security relevant PDUs	HSM acceleration; schedule verified	Independent from E2E
Ethernet	VLAN/ACL/rate limiting; MACsec where threat analysis allocates	HPA/IVI not permitted direct safety-network injection	Gateway default deny
IDS	CAN/Ethernet anomaly monitoring without blocking safety timing	Event forwarding to backend when available	Safety reaction takes priority

19.2 Driver and telemetry data governance

Data Domain	Principle	Control	Boundary
Driver profile / mode preference	Store minimum attributes required for feature	Encrypted at rest; owner/profile access controls	Market privacy/consent rules
Valet/geofence policy	Treat location and owner constraints as personal/security data	Purpose limitation and retention policy	Geofence not sole safety control
Flight recorder	Separate safety/event channels from entertainment history	Role-based service access; retention schedule	Legal/insurance rules vary
Cloud telemetry	Opt-in/contractual lawful basis where required	Data minimization, pseudonymization, backend ACL	No cloud dependency for safety
Voice/assistant/cockpit analytics	Outside safety authority	Dedicated privacy consent and delete/export workflow	No access to raw torque controls

20. Regulatory and Standards Hardening for 2026 Program Execution

The baseline now distinguishes published engineering standards, regulations whose applicable series must be frozen at homologation, and standards that are actively evolving. The compliance owner must perform a delta review rather than automatically adopting a newer edition into a frozen design.

Domain	Standard / Regulation	2026 Status / Note	VCU220 Evidence
Functional safety	ISO 26262:2018	Published baseline; third edition under development in 2026	Safety lifecycle/HARA/FSC/TSC/HW/SW safety case
Cybersecurity	ISO/SAE 21434:2021	Published	TARA, lifecycle cyber risk management
Software update	ISO 24089:2023	Published	Update engineering, dependency/rollback evidence
ASPICE	Automotive SPICE 4.0	Program process baseline	SYS.1/SYS.2/SYS.3 + HWE/SWE/SYS.4/5 handoff
Electrical loads	ISO 16750-2:2023	Current published edition	LV operating/transient test profile plus OEM requirements
Conducted transients	ISO 7637-2:2011	Published/confirmed; 12/24V	Bench transient immunity; OEM pulse profile still controls
EMC immunity	ISO 11452-1:2025 + relevant parts	Current general principles edition	Component immunity plan
Emissions	CISPR 25:2021	Published	On-board receiver protection / component emissions
CAN	ISO 11898-1:2024	Published; includes CAN Classic/FD/XL framework	Network architecture/future branch
DoIP	ISO 13400-2:2025	Current published transport/network edition	Diagnostics over IP
UDS	ISO 14229-1:2020	Published	Diagnostic services
EV electrical safety	ISO 6469-3:2021	Published; revision underway	HV state/interface vehicle safety evidence
Ethernet timing	IEEE 802.1AS-2025	Active; supersedes 2020	gPTP/time-sync architecture
Ethernet sleep/wake	OPEN Alliance TC10	Automotive Ethernet sleep/wake specifications	Partial networking/wake design
Cyber type approval	UN R155	Apply current series/amendments at market homologation	CSMS/type approval evidence
Software update type approval	UN R156	Apply current series/amendments at market homologation	SUMS/RXSWIN/update compliance
Vehicle EMC	UN R10	Apply applicable current series at homologation	Vehicle/component EMC type approval
EV powertrain safety	UN R100	Vehicle-level applicability	Rechargeable energy storage/electric powertrain interfaces
Braking interface	UN R13-H / applicable market braking regulation	Vehicle-level interface	Regen/friction blend must not compromise braking compliance
Steering interface	UN R79 / applicable market steering regulation	Vehicle-level interface	Driver-mode steering request remains inside EPS/regulatory boundary

20.1 Compliance change triggers

Trigger	Mandatory Action
ISO 26262 new edition before SOP compliance freeze	Safety-case delta review; no automatic re-baseline
UN R155/R156/R10 applicable-series update	Homologation applicability and evidence delta assessment
Network/diagnostic standard update	Supplier stack compatibility, gateway/ICD and DVP review
SoC/hypervisor/OS major release change	FFI, WCET/resource, cyber, update and regression impact analysis
Driver-mode market policy change	Calibration signature, HARA/vehicle dynamics and type-approval review

21. Rev A2 DFMEA and Project-Risk Hardening

The new failure analysis is focused on design mechanisms that can escape generic software testing: power-rail coupling, IOMMU mistakes, shared-memory starvation, torque-inhibit failure, partial mode application, crypto scheduling, safety-telltale loss, update dependency mismatch and lifecycle support gaps.

DFMEA ID	Block	Item	Failure Mode	Effect	S	Cause	O	Control	D	AP	Action	Owner
DFMEA-VCU-101	B01	LV front end	Buck-boost fails to support cold crank	Safety MCU reset during drive	10	Undersized converter/current limit	3	UV monitor	3	H	WCCA + low-voltage load test	Hardware
DFMEA-VCU-102	B01	Safety hold-up capacitor	Cap ESR/capacity degraded	Hold-up shorter than safety budget	9	Aging/temp/tolerance	4	Rail monitor / component derating	4	H	Energy/WCCA + EOL margin	Hardware
DFMEA-VCU-103	B02	Torque inhibit line	Stuck inactive	Cannot independently remove torque	10	GPIO/driver/wiring fault	2	Readback / inverter monitor	3	H	Diverse inhibit + fault injection	FuSa
DFMEA-VCU-104	B02	Torque monitor	Common algorithm defect with main arbiter	ASIL-D protection defeated	10	Insufficient independence	2	Review/test	5	H	Diverse monitor/decomposition analysis	FuSa
DFMEA-VCU-105	B03	IOMMU configuration	QM DMA region over-permissive	HPA corrupts safety-support buffer	10	Configuration error	2	Boot audit	4	H	Static allow-list + penetration/fault test	Platform
DFMEA-VCU-106	B03	HPA memory bandwidth	DDR starvation	Gateway/telltale latency increase	8	GPU/display burst	4	QoS/perf counters	4	H	Reserve + stress benchmark	Platform
DFMEA-VCU-107	B07	Track eligibility	Partial calibration applied before rejection	Torque/regen inconsistency	9	Non-atomic state transition	3	State machine	4	H	Two-phase commit mode apply	Software
DFMEA-VCU-108	B07	Boost timer	Mode cycling resets thermal debt	Repeated overtemperature/overpower	8	State reset defect	3	Thermal monitor	4	H	Persistent thermal debt accumulator	Software
DFMEA-VCU-109	B11	SecOC timing	Crypto causes missed 10ms deadline	Stale torque request	9	MAC/freshness overhead	4	Bus/CPU monitor	3	H	HSM + schedulability analysis	Network/Cyber
DFMEA-VCU-110	B11	Gateway ACL	IVI route accidentally permitted	Unauthorized torque traffic	10	Config error	2	Firewall audit	3	H	Default-deny signed config	Cyber
DFMEA-VCU-111	B12	Safety IPC mailbox	Malformed size/counter accepted	Corrupt safety input	10	Parser defect	3	Range/counter checks	3	H	Schema-generated parser + fuzzing	Software
DFMEA-VCU-112	B13	Safety telltale renderer	Android/guest crash removes warning	Driver unaware of safety fault	9	Shared display path	4	Renderer watchdog	4	H	Safety overlay/fallback path	Cockpit/FuSa
DFMEA-VCU-113	B14	Event recorder	NVM write interrupted	Missing forensic evidence	6	Power loss	4	Commit marker	4	M	RAM buffer + atomic flush	Diagnostics
DFMEA-VCU-114	B15	OTA dependency manifest	Incompatible safety calibration activated	Unsafe torque/mode behavior	10	Dependency/version error	2	Signature/version check	3	H	Atomic dependency graph verification	OTA
DFMEA-VCU-115	B16	Production debug	Debug left enabled	Key/code manipulation	10	Manufacturing config error	2	EOL security check	3	H	Authenticated debug + fuse/config verify	Cyber
DFMEA-VCU-116	B17	Common reset source	HPA fault resets safety MCU	Loss of motion supervision	10	Shared reset supervisor	2	Architecture review	4	H	Independent reset/PMIC	FuSa
DFMEA-VCU-117	B18	Thermal interface	Insufficient 130W heat rejection	SoC throttles/shuts down	8	Old 75W assumption	4	Thermal sensors	3	H	Liquid/conduction CFD + margin	Thermal
DFMEA-VCU-118	B18	PDN	HPA load step couples into safety rail	Safety MCU brownout/reset	10	Shared supply impedance	3	Rail monitors	3	H	Separate converters/PDN impedance target	PI
DFMEA-VCU-119	B19	Mode calibration binding	Wrong vehicle variant accepted	Unsafe performance profile	9	Provisioning mismatch	2	Hash/variant check	3	H	Signed VIN/powertrain binding	Manufacturing
DFMEA-VCU-120	B20	Hypervisor/OS EOL	No supported security patch	Cyber/service risk	7	Supplier lifecycle end	4	PCN/EOL tracking	4	M	Migration/last-time-support plan	Program

21.1 Priority program risks

Risk ID	Category	Risk	Cause	P	I	Score	Level	Mitigation	Owner	Gate	Status
RISK-VCU-046	Power/Thermal	130W cockpit-central workload exceeds enclosure/cooling capability	SoC/display feature growth	4	5	20	Critical	Freeze SoC workload + CFD early; liquid/conduction cooling branch; load shedding	Thermal Lead	SoC freeze	Open
RISK-VCU-047	FuSa	ASIL-D SG-01 not closed by allocated torque architecture	Old ASIL-C program wording	3	5	15	High	Update FSC/TSC; independent monitor/inhibit; decomposition analysis	Safety Lead	FSC/TSC	Open
RISK-VCU-048	Network/Cyber	SecOC/E2E overhead breaks 10ms CAN schedule	Security added late	3	5	15	High	HSM benchmark + PDU design + schedulability proof	Network/Cyber	ICD freeze	Open
RISK-VCU-049	Platform	Android/HPA memory	Shared DDR/GPU peak	4	4	16	High	QoS/quotas/memory	Platform Lead	B-sample	Open

Risk ID	Category	Risk	Cause	P	I	Score	Level	Mitigation	Owner	Gate	Status
		bandwidth starvation affects gateway/telltales						reserve/stress benchmark			
RISK-VCU-050	HMI	Critical warning lost with HPA guest failure	Single QM render path	3	5	15	High	Safety overlay/fallback indicator and watchdog	Cockpit/FuSa	DV	Open
RISK-VCU-051	OTA	Incompatible multi-domain update versions	Dependency graph not controlled	3	5	15	High	SBOM/ dependency manifest + atomic A/B + rollback	OTA Lead	SUMS release	Open
RISK-VCU-052	Driver Modes	Track/boost feature creates homologation/vehicle-dynamics conflict	Market policy and calibration drift	3	4	12	High	Market coding, driver confirmation, ESC envelope, legal review	Vehicle Dynamics	PV	Open
RISK-VCU-053	Lifecycle	Hypervisor/ OS/security component EOL before vehicle service life	Supplier roadmap mismatch	4	4	16	High	Lifecycle contract, migration path, PCN/EOL tracking	Program Lead	SOP+service	Open
RISK-VCU-054	Compliance	Standards/ regulation revision after Jan-2026 baseline causes late delta	2025/2026 updates	3	4	12	High	Standards watchlist and gate-based delta review	Compliance Lead	Homologation	Open
RISK-VCU-055	Common Cause	Single central ECU enclosure/power connector becomes systemic failure root	Physical consolidation	3	5	15	High	CCF analysis, independent safety rail, local actuator protections	System Architect	Architecture freeze	Open

22. Rev A2 Verification and Validation Architecture

Verification is expanded from requirement-level test planning to a multi-layered evidence campaign that stresses both safety and central-compute resource interactions. Every fault campaign must log the authoritative safety state, network age/integrity, power rails, HPA resource usage and local actuator acknowledgements.

Layer	Purpose	Representative Rev A2 Evidence
MIL/SIL	Control/model/calibration logic	Torque arbitration, regen blend, mode transitions, safe-state scenarios
Target bring-up	Power/boot/PMIC/memory/I/O	130 W load, cold crank, safety boot, watchdog/inhibit
Network bench	Scheduling/security/freshness	CAN FD E2E/SecOC, bus-off, replay, HPA route filtering
HIL	Closed-loop faults and timing	Pedal/BMS/ESC/inverter/charge state, VM/reset/resource faults
Dyno/brake rig	Delivered torque/regen performance	Torque step, gradient, regen blend, local limiter behavior
EMC/environmental	Physical robustness	ISO16750/7637, CISPR25/ISO11452, vibration/thermal
Cyber/OTA	Threat and recovery evidence	Fuzzing/penetration, debug lockdown, SBOM/update rollback/power loss
Vehicle DV/PV	Integrated behavior and homologation	Modes, braking/steering interfaces, telltales, charging interlock, regulations

Rev A2 Controlled Count	Value
SYS.1	58
SYS.2	115
HARA	48
Safety Goals	12
SYS.3	246
HSR	70
SW Safety	72
DFMEA	120
TARA	40
ICD	38
DVP	492
Validation	50
RTM	246